Architecture d’un système d’exploitation: Allocation mémoire

- Basé sur le cours de Marc Tchiboukdjian et les travaux de recherche de Sébastien Valat
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Enregistrement

- https://docs.google.com/forms/d/e/1FAIpQLScwu_BfoRir3osMhCeVg_Jdm1C7RPs2vPPeMZ2FqWvLuy6vjA/viewform?usp=pp_url

Supports de cours

- https://drive.google.com/drive/folders/16FnbMmbfreb2SJX0px-5ce5KFq0Pjd1M?usp=sharing

Bibliographie

- La programmation sous UNIX de Jean-Marie Rifflet
- Système d’exploitation d’Andrew Tanenbaum
- What Every Programmer Should Know About Memory de Ulrich Drepper
- Contribution à l'amélioration des méthodes d'optimisation de la gestion de la mémoire dans le cadre du Calcul Haute Performance de Sébastien Valat
- Framework de R&D http://mpc.hpcframework.com

Déroulement du cours

- Allocation mémoire en espace noyau
- Allocation mémoire en espace utilisateur
- Allocation mémoire en contexte HPC
- Mémoire unifiée sur GPU

La programmation sous UNIX de Jean-Marie Rifflet

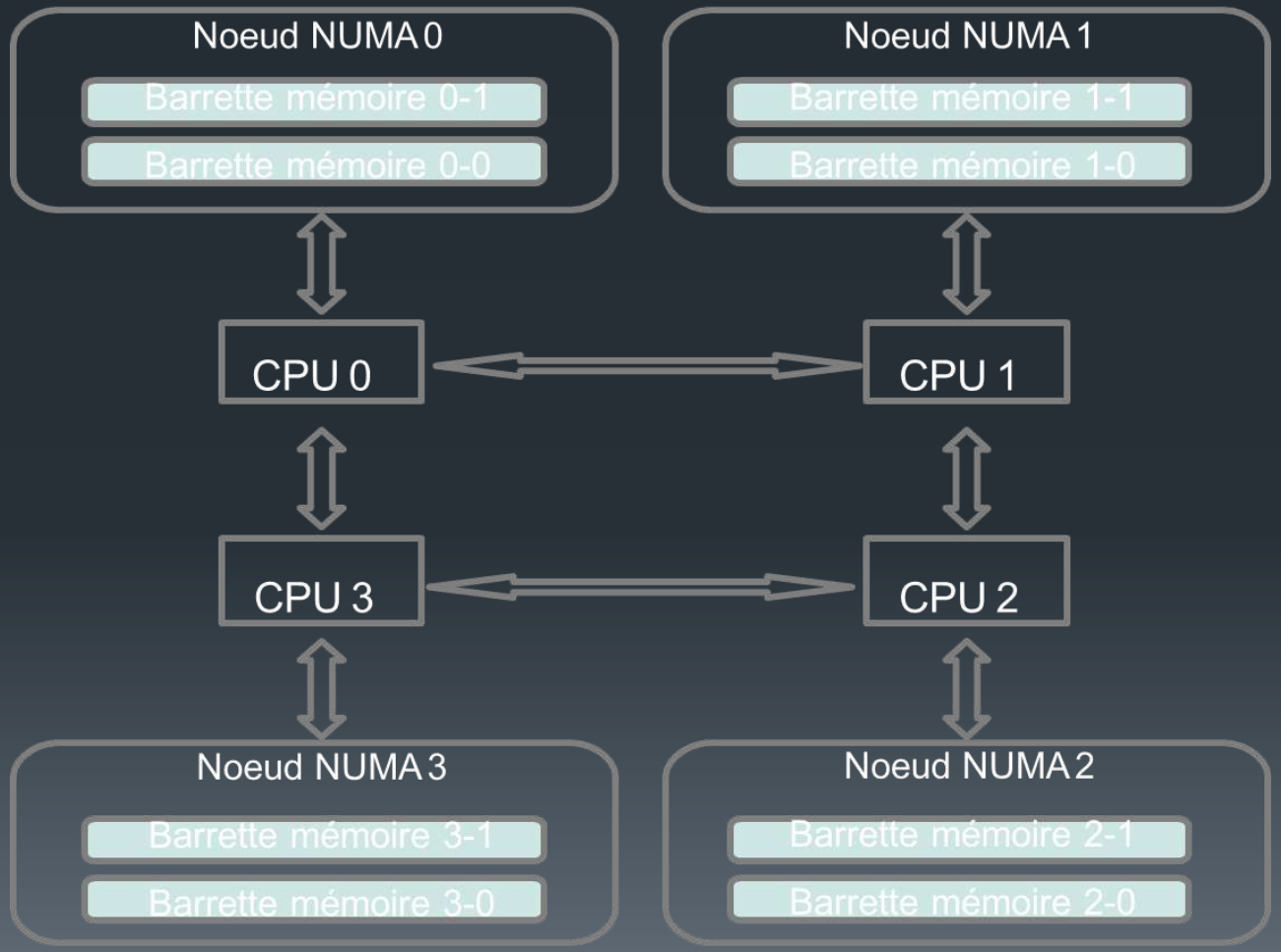
Système d’exploitation d’Andrew Tanenbaum

Allocation mémoire en espace noyau

Mémoire physique

- Un tableau de cases ou cellules mémoire
  - Chaque case : un nombre défini de bits
  - Chaque case a un numéro : une adresse
  - Mot mémoire : information contenue dans une case
- RAM pour Random Access Memory
- Taille sur un système actuel : de quelques Go (téléphone portable) à 1 To (un nœud d'un super-calculateur)

Mémoire physique



4 Nehalem EX processors

Mémoire vue par le programmeur

- Espace d'adressage logique
- Code du programme
  - Stocké dans la zone texte
- Variables allouées statiquement
  - Stockées dans la zone donnée
  - Durée de vie du programme
- Variables allouées automatiquement
  - Stockées dans la pile (stack)
  - Durée de vie liée à la fonction ou à un bloc
- Variables allouées manuellement
  - Stockées dans le tas (heap) En C avec malloc/free
  - Durée de vie gérée par le programmeur

Variables statiques et automatiques

- Variables statiques (ou globales)
  - Durée de vie du programme
  - En C :
    - variable globale (déclarée à l'extérieure de toute fonction)
    - variable préfixée par le mot clé static
- Variables automatiques (ou locales)
  - Durée de vie associée à leur portée (fonction ou bloc)
  - En C : variable locale

Mémoire vue par le programmeur

- Code et les données sont chargées en mémoire juste avec l'exécution du programme par le loader
- La pile est gérée
  - En C : par le compilateur
  - En Assembleur : par le programmeur
- Le tas est géré par le programmeur
  - Interface : malloc/free
  - Implémentation : libc (allocateur mémoire)

Malloc/Free

- Utilisation
  - #include <stdlib.h>
  - void *malloc(size_t size);
  - void free(void *ptr);
- Fait partie de l'espace utilisateur
  - L'implémentation de l'allocateur mémoire fait des appels systèmes
  - Groupe plusieurs appels à malloc (rapide) en un seul appel système (plus lent)
  - brk/sbrk : augmente la taille du tas
  - mmap : récupère de l'espace mémoire (libc : pour les grosses allocations)
- Critique pour les performances de certains codes (C,C++)
  - Parallélisme (plusieurs threads appellent malloc), Localité (NUMA)
  - D'autres implémentations que la libc existent (ex : Google TCMalloc)

Malloc/Free : problème de la fragmentation mémoire

- Après une série de malloc/free, la mémoire est fragmentée en parties utilisées et parties libérées

- Implémentation de l'allocateur mémoire
  - Choix de la zone où placer une nouvelle allocation
    - First fit / Best fit / Worse fit
- Compromis performance / consommation mémoire

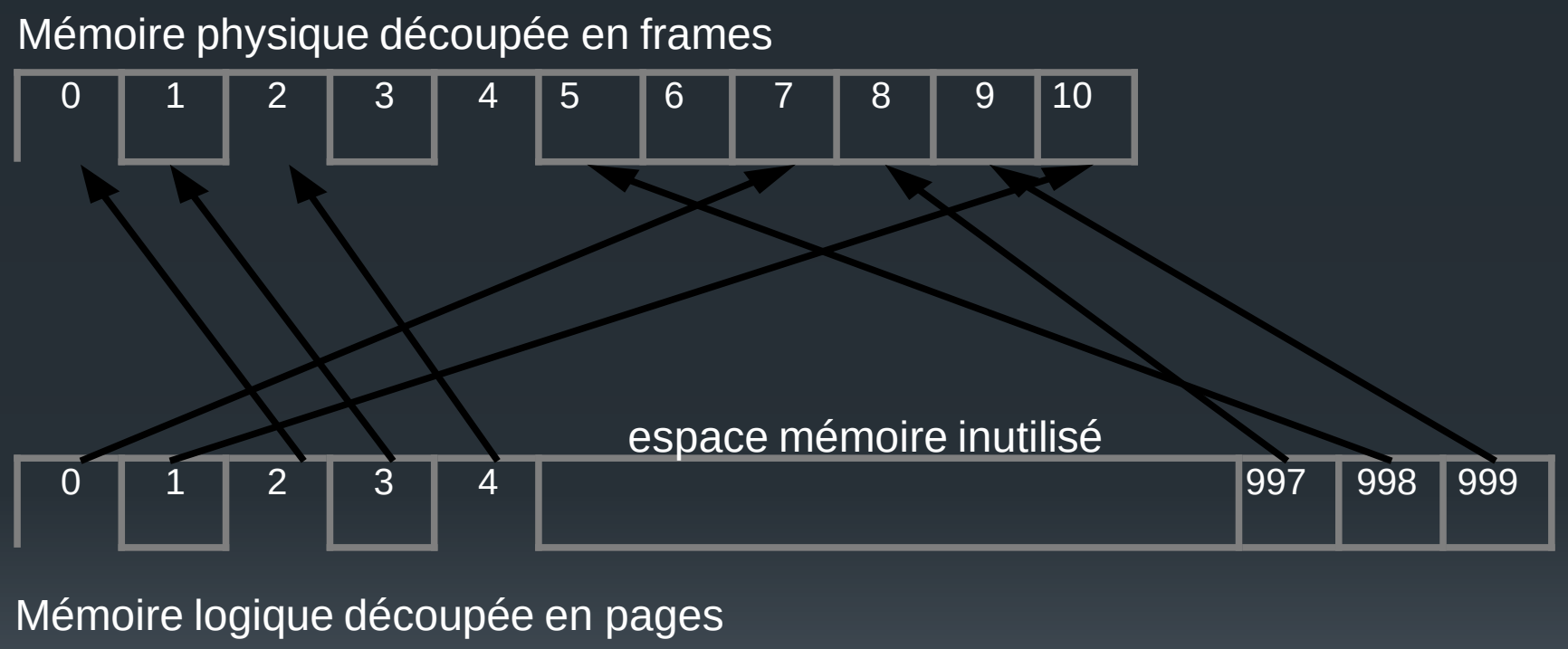
Problèmes posés par les adresses logiques

- Espace d'adresses logiques = abstraction de la mémoire
  - Simple pour le programmeur
  - Comment l'implémenter ?
- La mémoire logique est plus grande que la mémoire
  - physique. Comment fait on si on a moins de 2^{64} bits de mémoire ? (cas usuel)
- Multiprogrammation : plusieurs processus s'exécutent en « même temps »
  - Problème 1 : ils veulent utiliser la même adresse
  - Problème 2 : $\sum$ mémoires processus > mémoire principale

Solution : la mémoire virtuelle

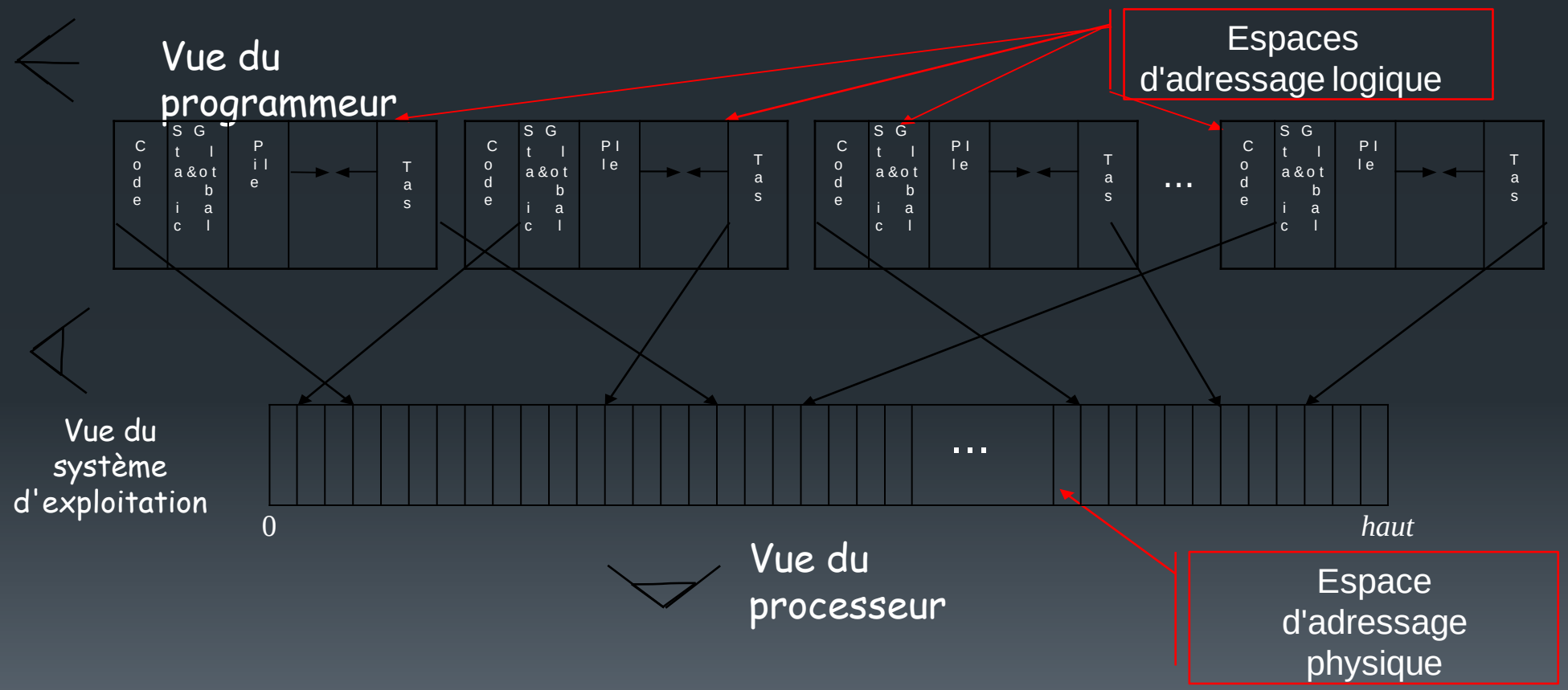
- Mémoire physique et mémoire logique sont découpées en utilisant la même taille de zone
  - Mémoire physique : en frames
  - Mémoire logique : en pages
  - Taille usuelle : 4KO (huge page 2MO)
- Adresse logique = un numéro de page + un déplacement

La mémoire virtuelle



La correspondance entre les pages et les frames est stockée dans la table des pages

Mémoire vue par le système d'exploitation



Gestion des frames et des pages

- Table des frames
  - Une case par frame Libre ou allouée
  - Si allouée, information sur le processus et la page
- Table des pages
  - Par processus
  - Fait partie de son contexte

Traduction des adresses logiques en adresses physiques : la table des pages

Implémentation de la table des pages système

- Registre de table de pages
  - Table des pages en mémoire centrale
  - Un registre contient l'adresse de la table
  - Implique des accès mémoire lors de la traduction des adresses (lent)
- Mémoire cache dédiée
  - TLB : Translation Look-aside Buffer

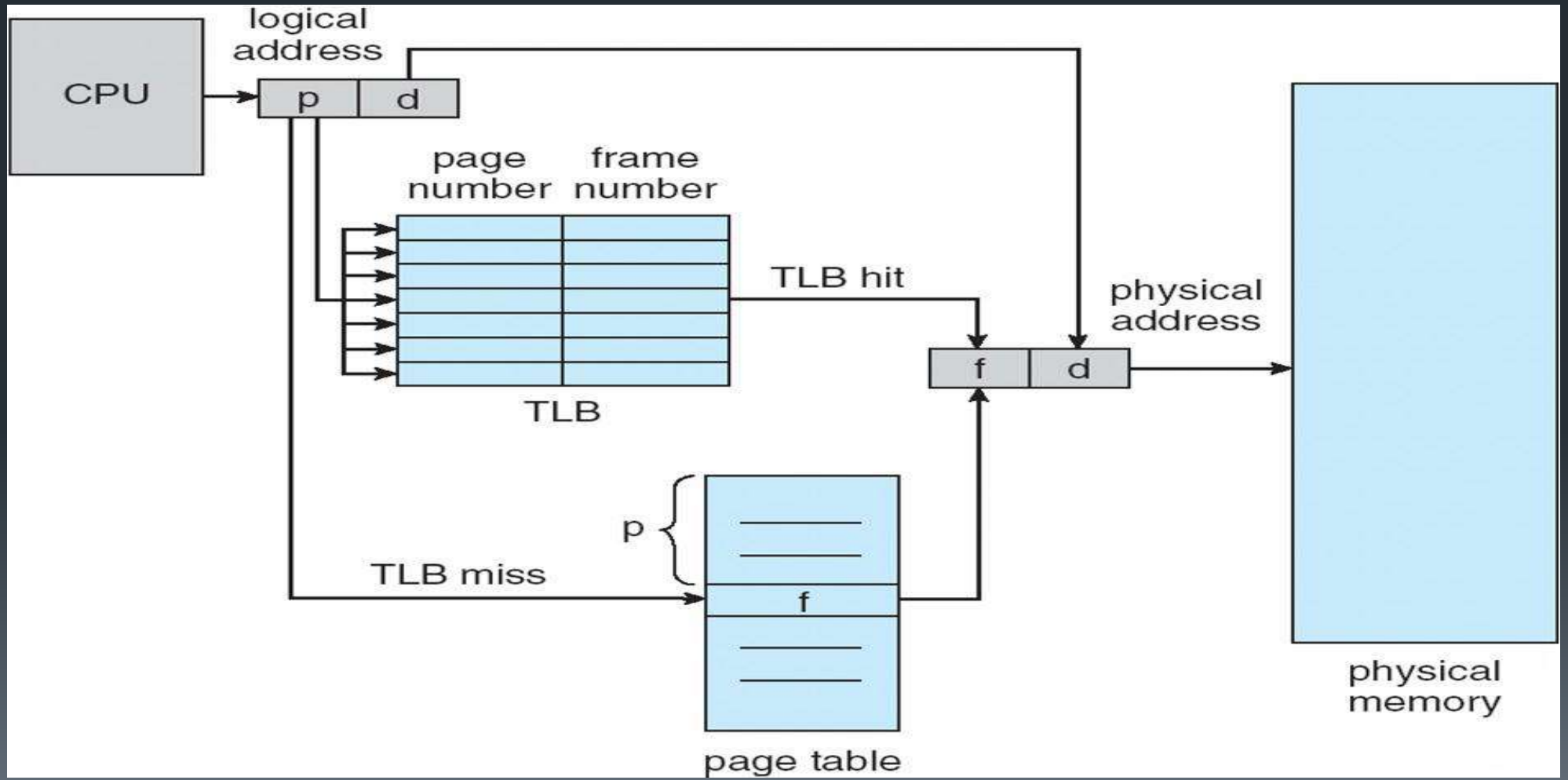
TLB : mémoire associative

- Mémoire associative : recherche très rapide en parallèle

Page#	Frame#

- Si le numéro de page est dans le TLB, on récupère le numéro de frame : très rapide car pas d'accès mémoire
- Sinon, récupérer le numéro dans la table des pages : lent car accès mémoire

Implémentation de la table des pages système avec TLB



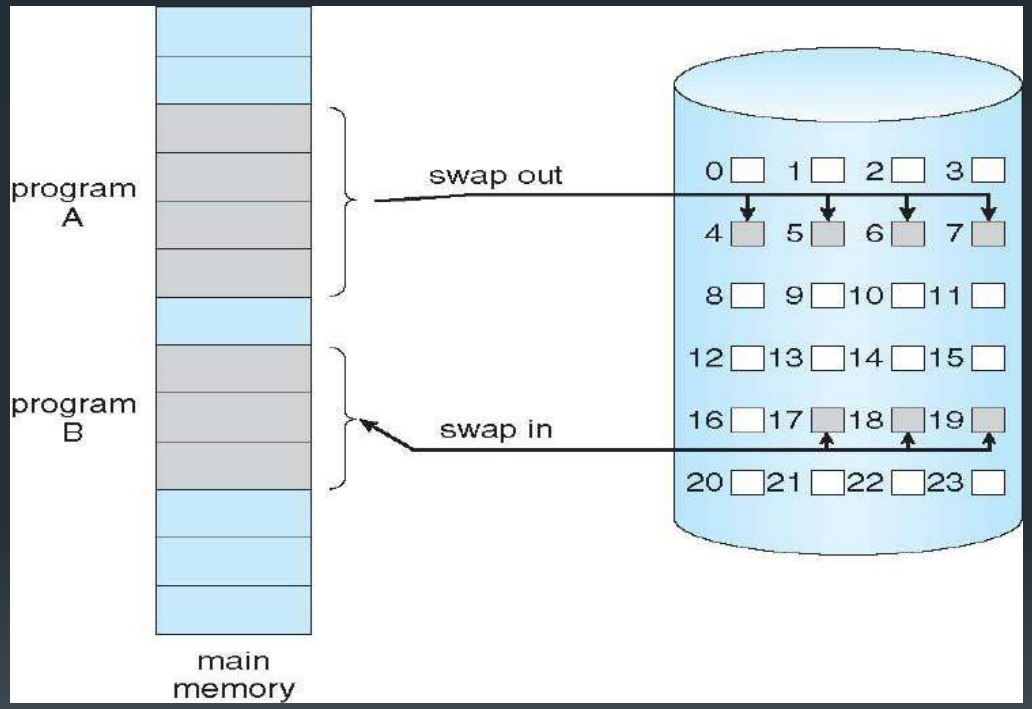
Fichier d'échange / Swap

- Que faire si le nombre de pages utilisés par tous les processus est supérieur au nombre de frames (taille de la mémoire physique) ?
- Les pages supplémentaires sont stockées sur le disque dans une partition spécifique ou un fichier : le swap
- Si #pages x 4 KO > taille(mémoire physique + swap), le système d'exploitation tue un processus qui consomme beaucoup de mémoire

Fichier d'échange / Swap

- Les processus ne sont chargés que partiellement
- Les pages non chargées sont stockées sur le disque
- Le chargement est paresseux (lazy) : une page est chargée lors du premier accès à une adresse contenue dans la page

Fichier d'échange / Swap

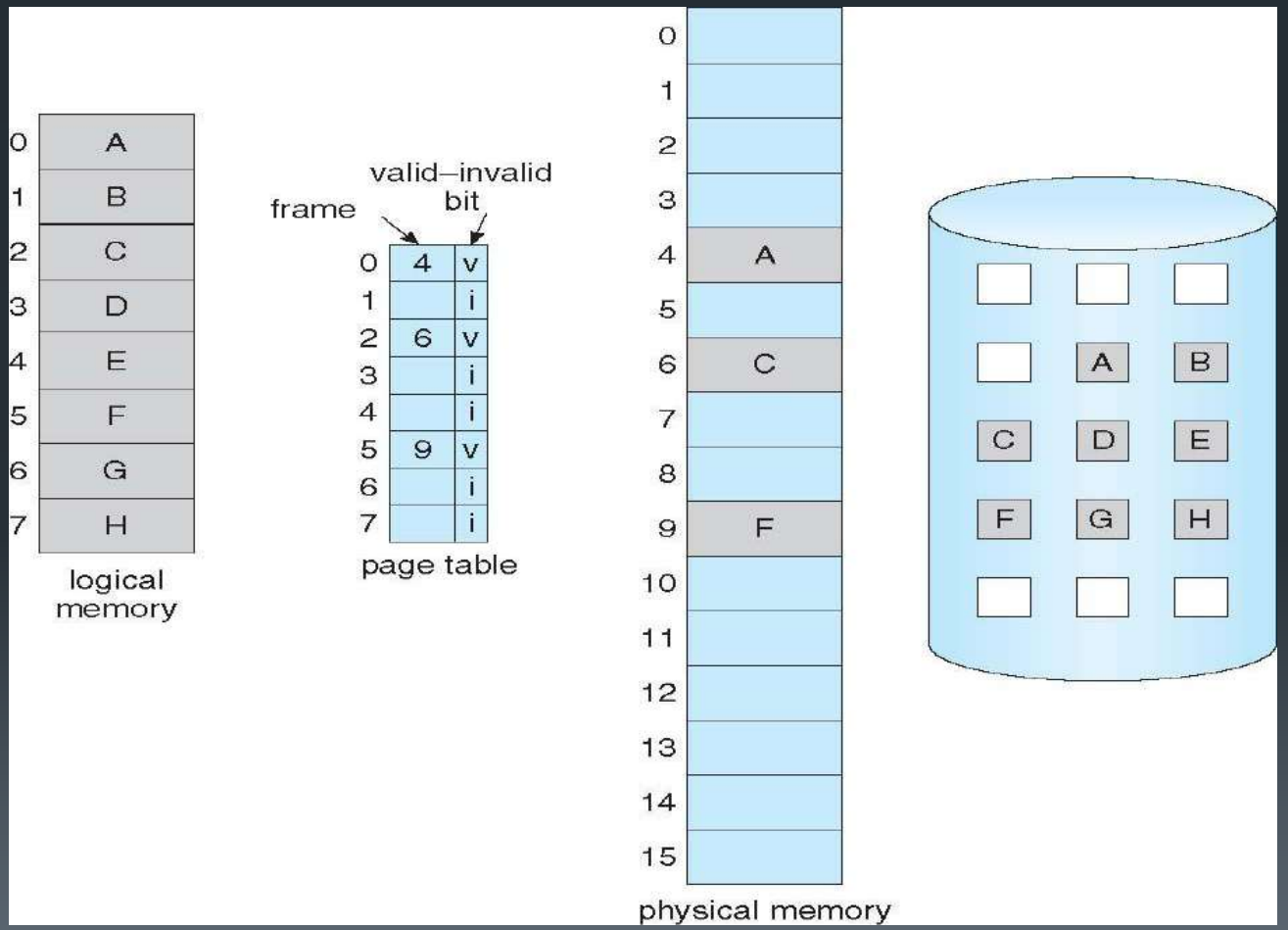


- Référence de page
  - Invalide
  - Page en mémoire : on continue l'exécution
  - Page sur disque : défaut de page (page fault)
    - On charge la page en mémoire
    - Si pas de place, remplacement de page

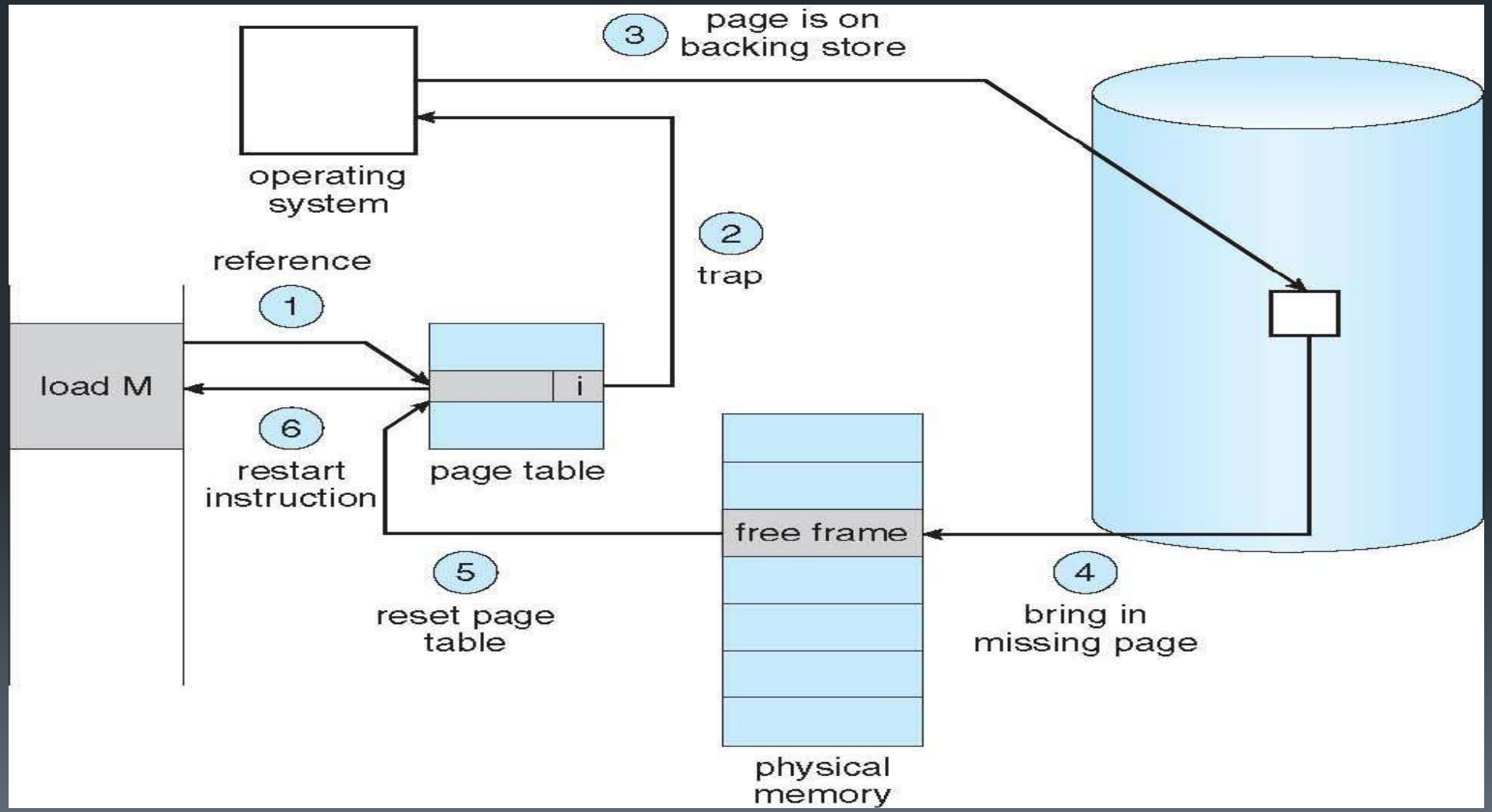
Bit de validité

- Comment déterminer si la page est en mémoire ou sur disque ?
- On rajoute un bit de validité dans la table des pages
- Si le bit de validité est à 1, la page est en mémoire à l'emplacement de la frame
- Si le bit de validité est à 0, la page est sur le disque : déclenchement d'un défaut de page

Bit de validité



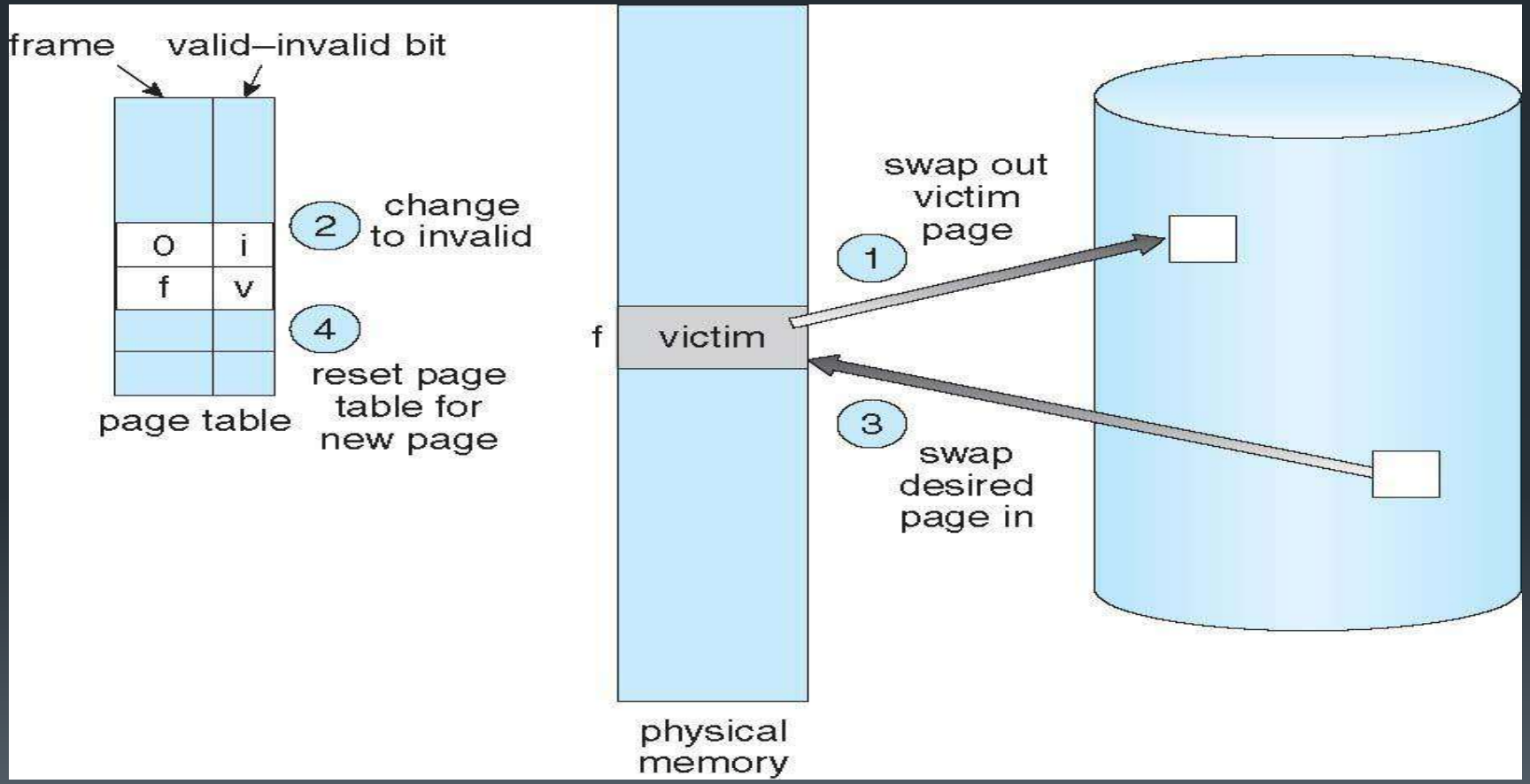
Gestion des défauts de page



Gestion des défauts de page

- Algorithme de préchargement
  - Quelles sont les pages à charger au lancement du processus ?
  - Cas extrême : aucune
  - Pagination à la demande pure
- Algorithme de remplacement
  - Si il n'y a pas de frame libre, laquelle des pages décharger ?
- Algorithme de répartition des frames entre processus
  - Combien de pages un processus a le droit de charger ?

Remplacement de page



Bit modifié (dirty)

- En plus du bit de validité, la table des pages comprend aussi un bit modifié
- Si la page a été modifié, le bit est à 1, on doit ré-écrire la page sur disque
- Si la page n'a pas été modifié, le bit est à 0, ce n'est pas la peine de ré-écrire la page sur disque

Algorithmes de remplacement

- Comment choisir la page victime ?
  - But : minimiser le nombre de défauts de page

- Algorithmes proposés
  - Random
  - FIFO (first in first out)
  - FIF (furthest in future)
  - LRU (least recently used)
  - Clock

Remplacement FIFO

- FIFO : first in first out
- Pages déchargées suivant l'ordre d'arrivée
- Gestion simple : une file d'attente FIFO

Pages adressées :

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Contenu des 3 frames :

f1	7	7	7	2	2	2	2	4	4	4	0	0	0	0	0	0	0	7	7	7
f2		0	0	0	0	3	3	3	2	2	2	2	2	1	1	1	1	1	0	0
f3			1	1	1	1	0	0	0	3	3	3	3	3	2	2	2	2	2	1

15 défauts de page

Remplacement FIF

- FIF : furtherst in future
- Remplacer la page qui sera accédée le plus tard
- Optimal mais nécessite de connaître le futur !

Pages adressées :

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Mémoire de 3 blocs :

b1	7	7	7	2	2	2	2	2	7
b2		0	0	0	0	4	0	0	0
b3			1	1	3	3	3	1	1

9 défauts de page

Remplacement LRU

- LRU : Least Recently Used
- Remplacer la page non utilisée depuis le plus longtemps

Pages adressées :

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Mémoire de 3 blocs :

b1	7	7	7	2	2	2	4	4	4	0	1	1	1
b2		0	0	0	0	0	0	0	3	3	3	0	0
b3			1	1	1	3	3	2	2	2	2	2	7

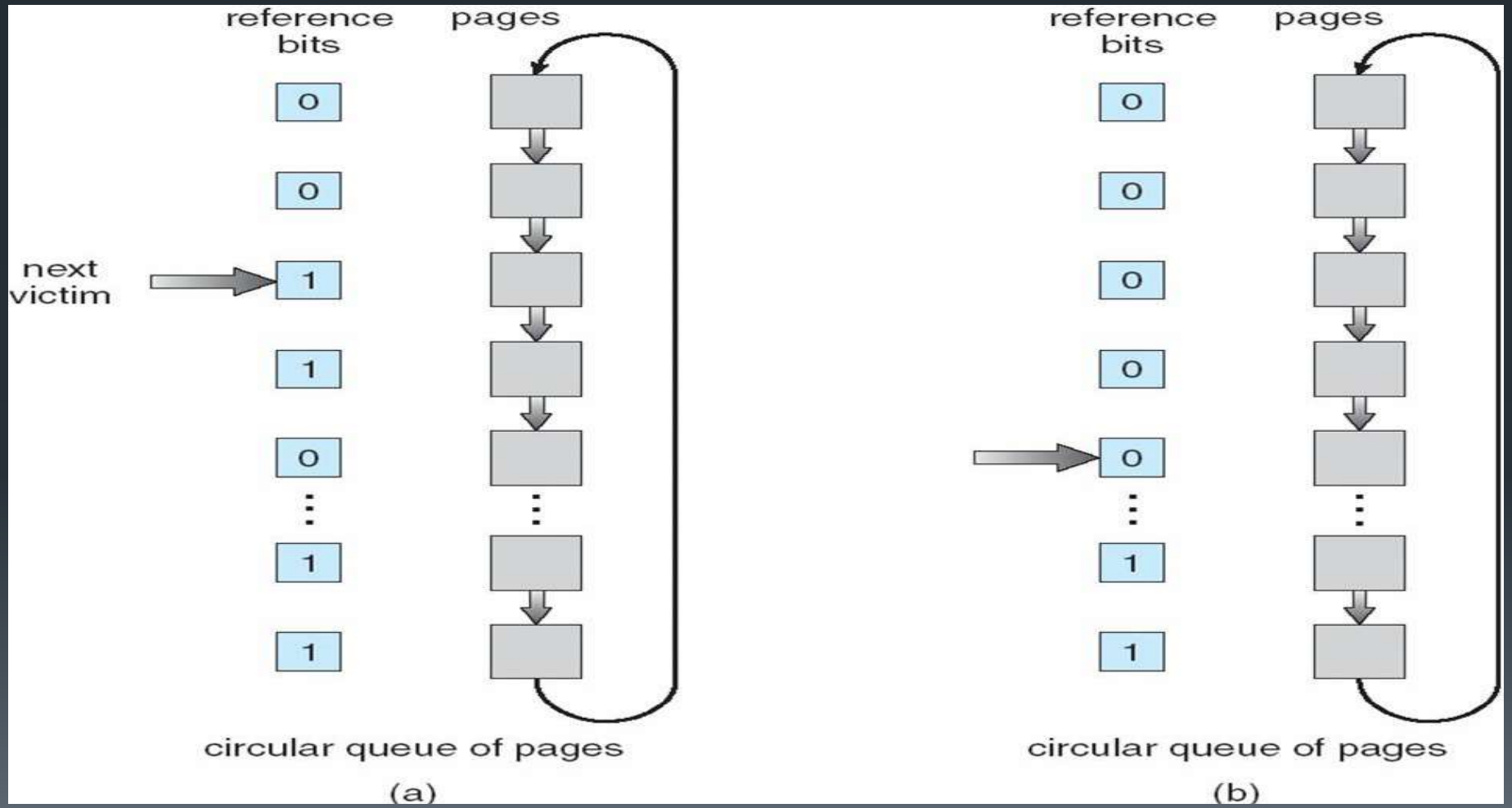
12 défauts de page

- Utilisé, considéré performant, implémentation coûteuse

Remplacement Clock

- Ou algorithme de la deuxième chance
- Bit de référence
  - Initialement page non utilisée : bit = 0
  - Utilisation de la page : bit = 1
- Deuxième chance
  - Liste circulaire FIFO
  - Si bit == 0, remplacement
  - Si bit == 1, bit ← 0, vers page suivante
- Approximation de LRU moins coûteux à implémenter

Remplacement Clock



Classement des algorithmes de remplacement de pages

- FIF
- LRU
- Clock
- FIFO
- Random
- Dépend des programmes

Répartition des frames entre les processus

- Comment partager les frames entre les processus ?
- Égalité entre processus
  - avantages / inconvénients ?
- Variations dynamiques

Gestion globale ou locale des frames

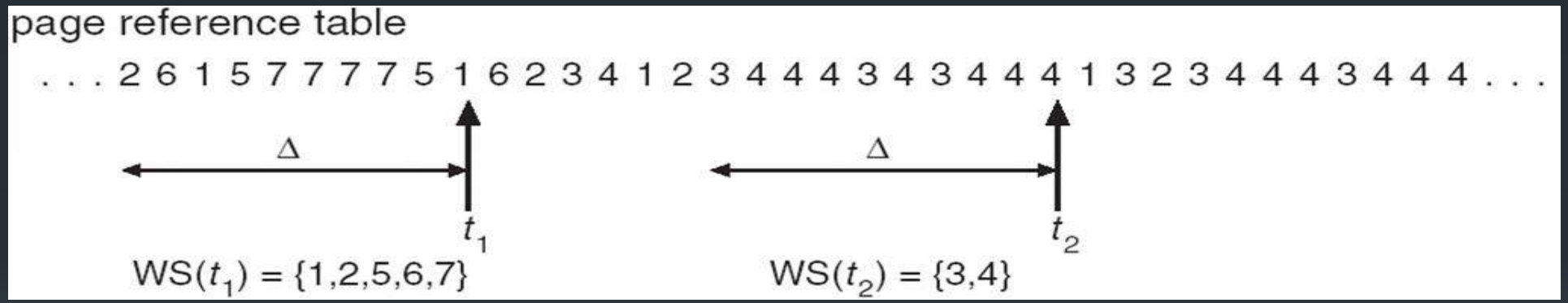
- Gestion globale
  - Quand une nouvelle frame est demandée par un processus, la frame allouée peut appartenir à un autre processus
  - Variation dynamique du nombre de frames par processus
- Gestion locale
  - Quand une page doit être déchargée, ne sont considérées que les frames allouées au processus
  - Le nombre de frames par processus reste constant

Problème du remplacement global

- Considérons la situation suivante
  - Un processus s'exécute et essaye d'accéder à des pages non chargées
  - Le chargement des pages implique des E/S disque, le processus est en attente et peut relâcher le processeur
  - Les autres processus n'ont pas assez de mémoire, ils génèrent aussi des défauts de page
  - Le processeur est inoccupé et aucun processus en mémoire n'est prêt, on va essayer de charger de nouveaux processus prêts
  - L'utilisation du processeur diminue, augmentation du nombre de processus en exécution
- Les processus passent plus de temps à charger des pages qu'à calculer : crash du système (trashing)

Remplacement local

- Comment décider du nombre de frames par processus ?
- Notion d'ensemble de travail (working set)
  - Ensemble des pages accédées dans le dernier intervalle de temps



- Algorithme du working set
  - si la somme des WS de tous les processus est supérieure à la mémoire physique
  - on suspend l'exécution d'un des processus et on décharge ses pages de la mémoire

Working Set

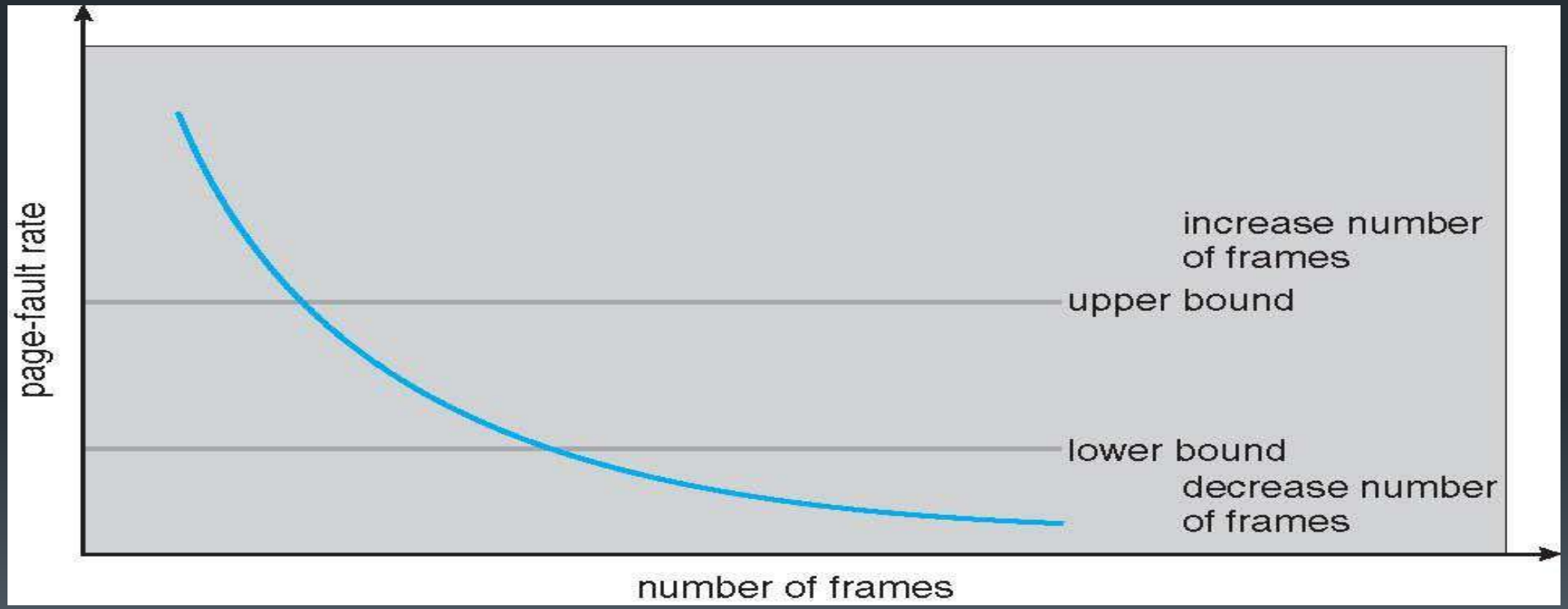

page-fault rate
increase number
of frames
upper bound
lower bound
decrease number
of frames
number of frames

Table des pages hiérarchique

- Taille d'un table de page de 4KO en 64bit
  - Taille d'une page : 2^12
  - Nombre de # de pages : 2^52
  - Taille de la table : 2^52 * 4O → énorme !

- Découper la table des pages en plusieurs niveaux

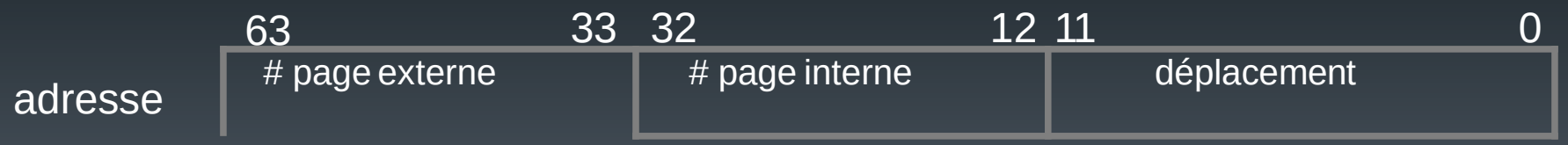
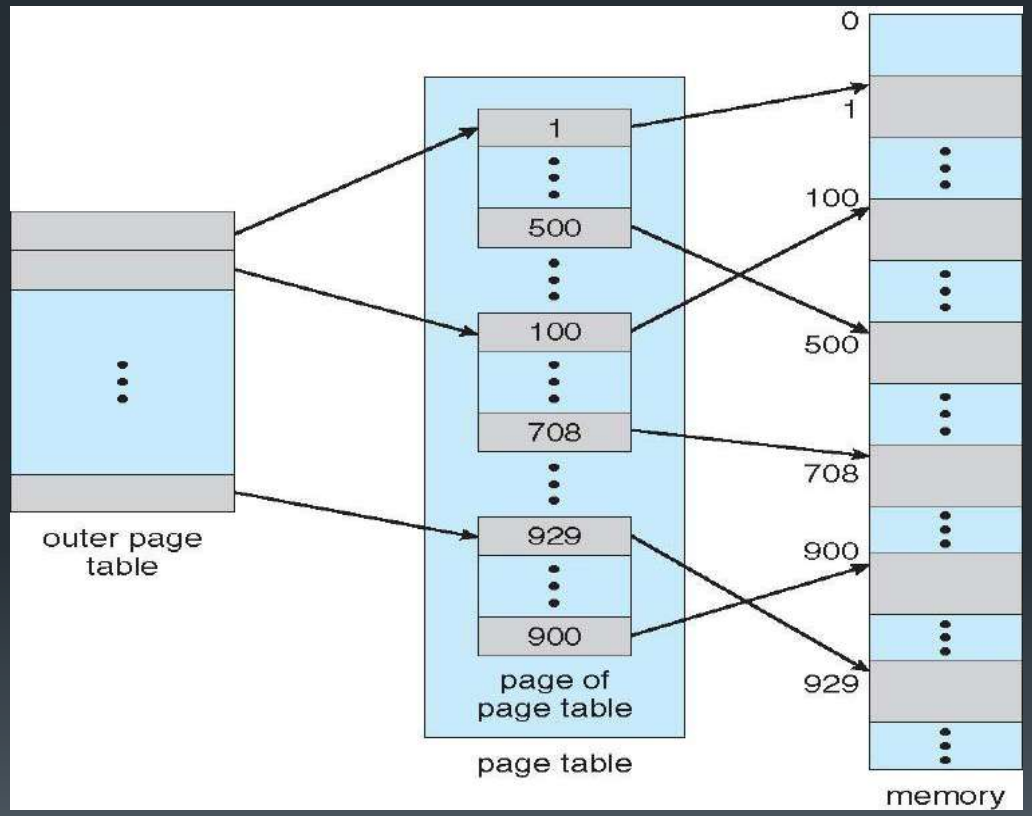
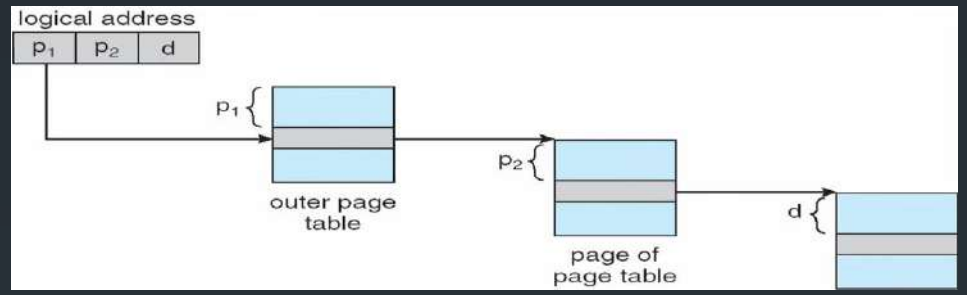


Table des pages hierarchiques

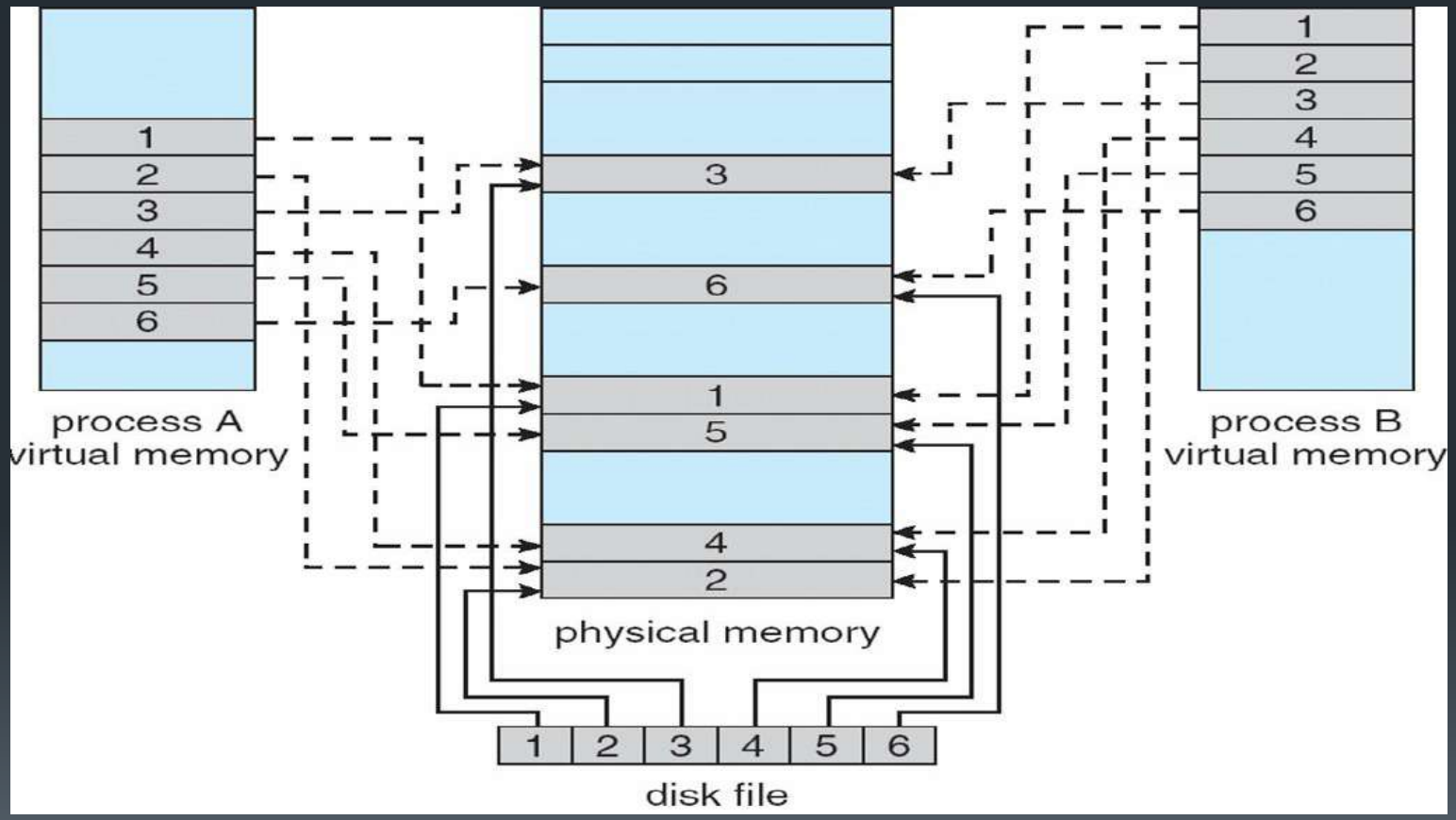




mmap

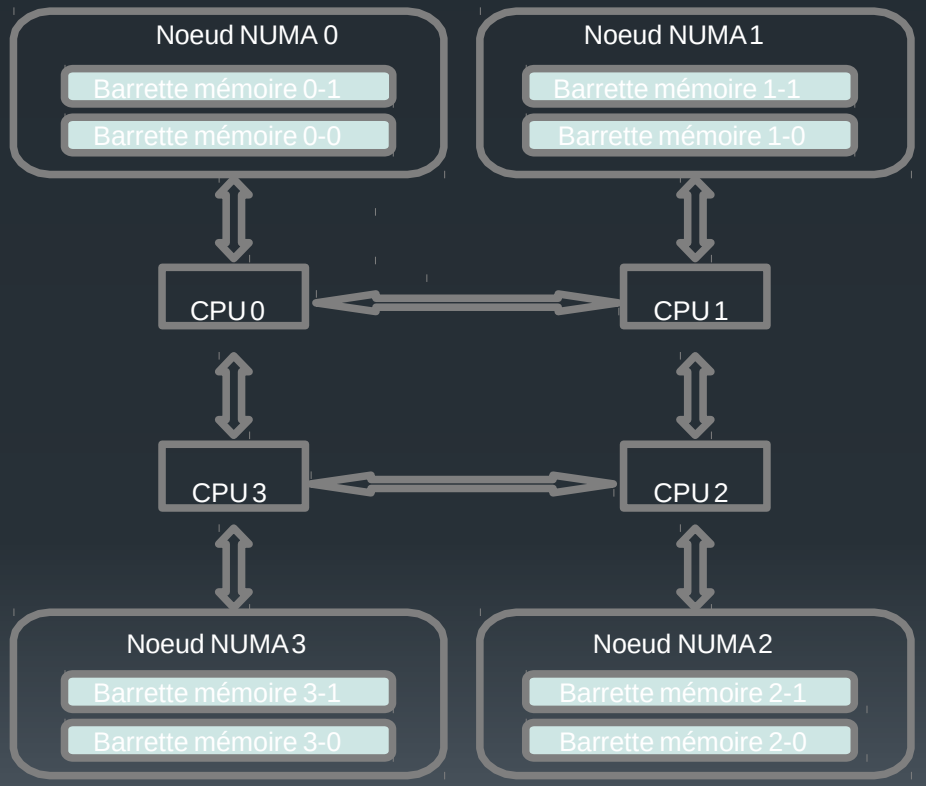
- void * mmap(void *start, size_t length, int prot , int flags, int fd, off_t offset);
  - mmap demande la projection en mémoire de length octets commençant à la position offset depuis un fichier indiqué par le descripteur fd
- remplace les E/S dans des fichiers par des opérations en mémoire : plus simple, plus rapide
- pagination à la demande : le fichier n'est pas intégralement chargé en mémoire
- permet de recouvrir les chargements par des calculs

mmap


1
2
3
4
5
6
process A
virtual memory
3
6
1
5
4
2
physical memory
1
2
3
4
5
6
process B
virtual memory
1
2
3
4
5
6
disk file

NUMA et la politique first touch

- Affinité mémoire : critique pour les performances
  - Bande passante
  - Latence
- First touch
  - premier accès à la page
  - Politique de linux par défaut
- Lorsqu'une page est accédée pour la première fois, on alloue une frame dans une barrette mémoire sur le même nœud NUMA que le thread qui a fait le défaut de page
- On peut modifier cette politique avec, par exemple, la libnuma
  - numactl --interleave



Mémoire virtuelle : conclusion

- 4 endroits où l'OS intervient
  1. Création de processus
     - décider de la taille de la table
     - créer la table des pages
  2. Exécution de processus
     - remise à zéro du TLB
  3. Défaut de page
     - déterminer l'adresse virtuelle qui fait défaut
     - swap entre page de remplacement et page demandée
  4. Terminaison
     - libérer la table des pages, les pages en mémoire

Issue de What Every Programmer Should Know About Memory de Ulrich Drepper

http://futuretech.blinkenlights.nl/misc/cpumemory.pdf

Allocation mémoire en espace utilisateur

Fonctionnement de malloc/free: Linux sans arena

- Approche centralisée
  - Verrou en entrée de fonction
  - Pas de distinction entre threads
  - Pas de notion de NUMA
  - First touch
  - Pas de lien avec l’ordonnanceur

Fonctionnement de malloc/free: Linux sans arena

Fonctionnement de malloc/free: Linux avec arena

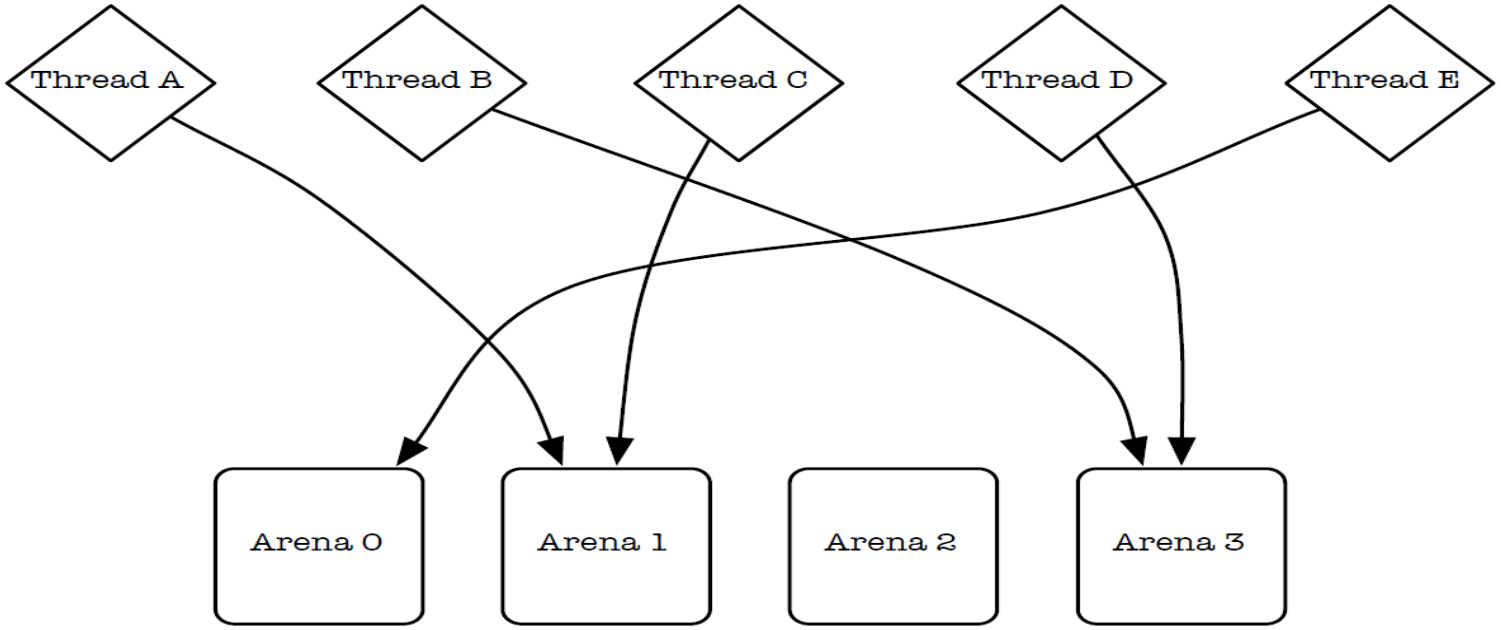


Figure 2: Larson and Krishnan (1998) hash thread identifiers in order to permanently assign threads to arenas. This is a pseudo-random process, so there is no guarantee that arenas will be equally utilized.

Fonctionnement de malloc/free: BSD

- Approche hiérarchique
  - Notion de thread
  - Verrous distribués
  - Pas de notion de NUMA
  - Pas de lien avec l’ordonnanceur

Fonctionnement de malloc/free: BSD

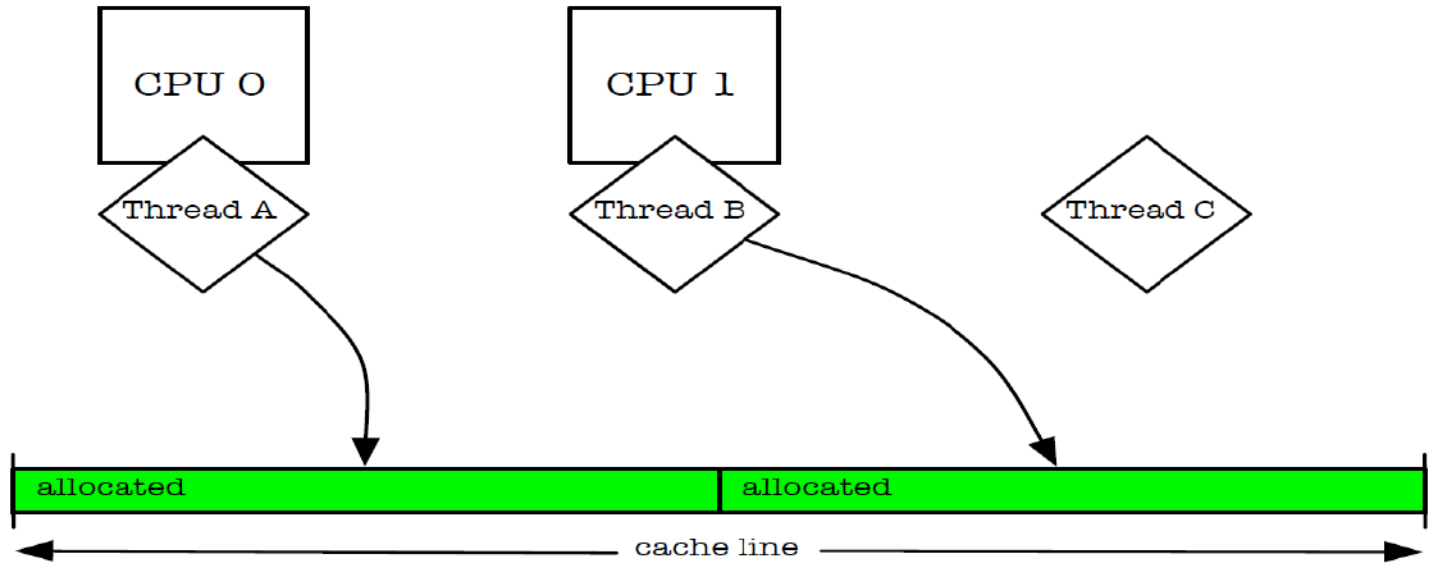


Figure 1: Two allocations that are used by separate threads share the same line in the physical memory cache (false cache sharing). If the threads concurrently modify the two allocations, then the processors must fight over ownership of the cache line.

Fonctionnement de malloc/free: BSD

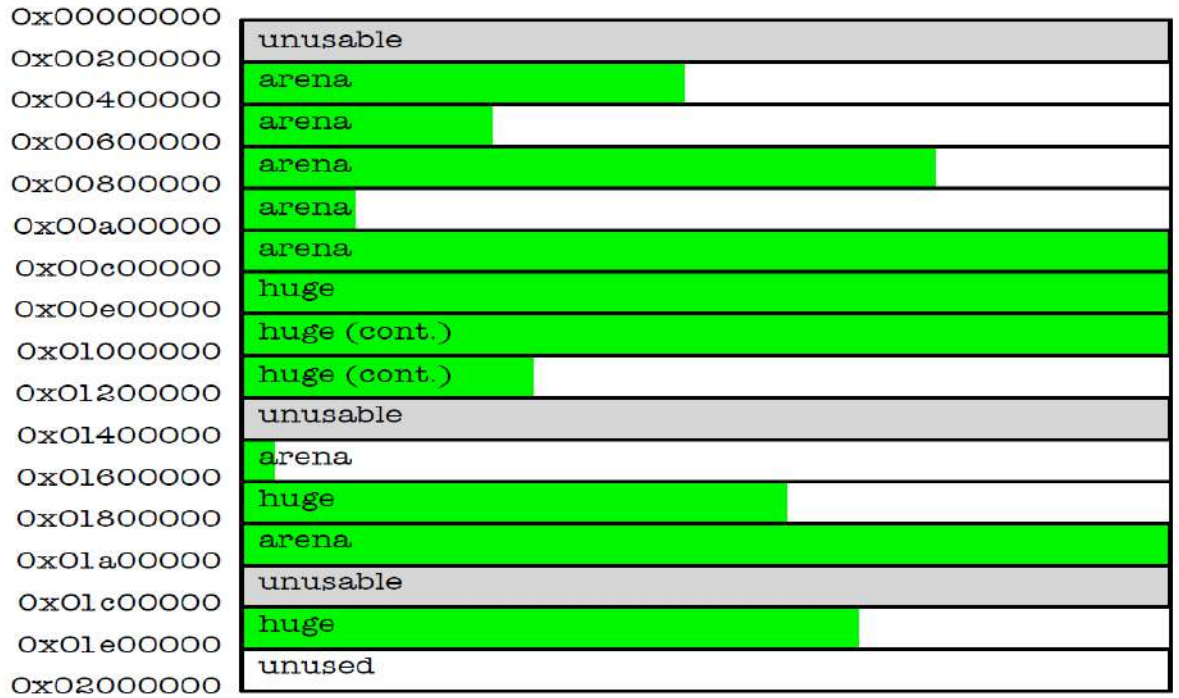


Figure 3: Chunks are always the same size, and start at chunk-aligned addresses. Arenas carve chunks into smaller allocations, but huge allocations are directly backed by one or more contiguous chunks.

Fonctionnement de malloc/free: BSD

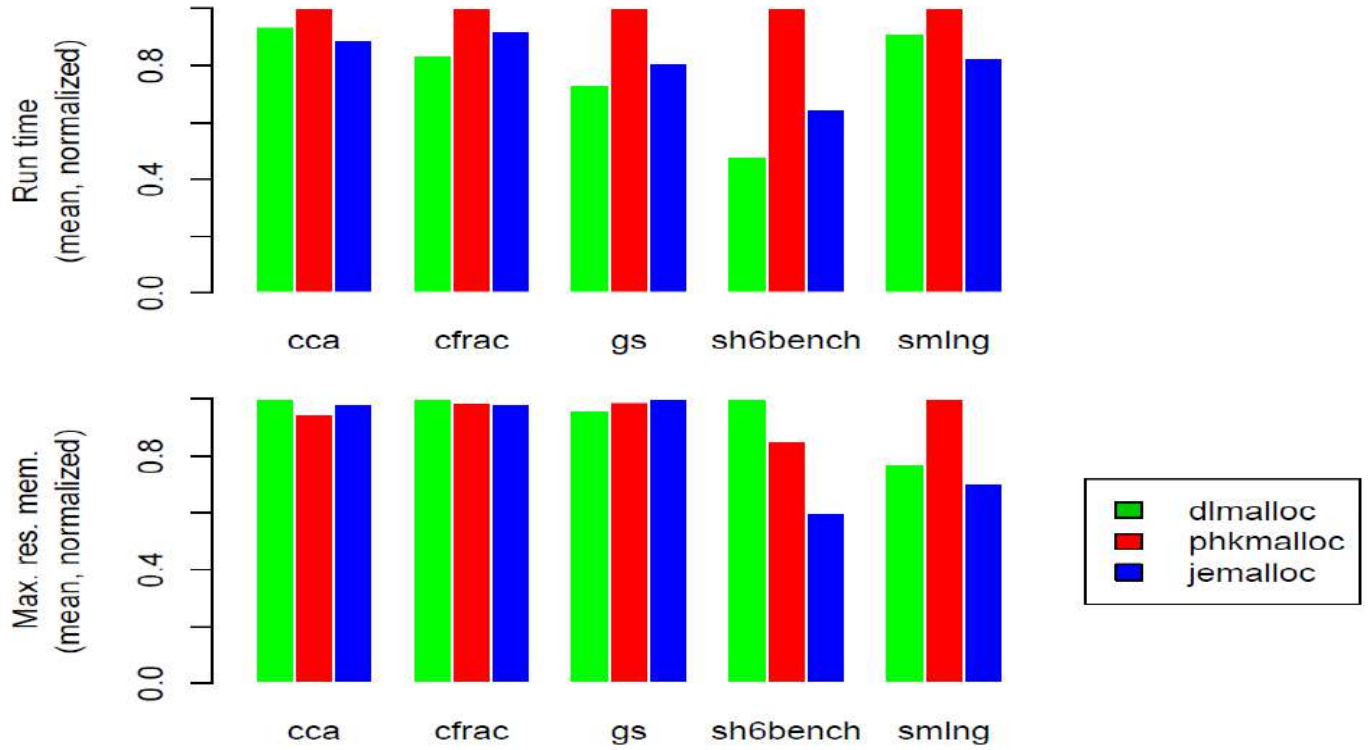


Figure 8: Scaled run time and maximum resident memory usage for five single-threaded programs. Each graph is linearly scaled such that the maximum value is 1.0.

Fonctionnement de malloc/free: BSD

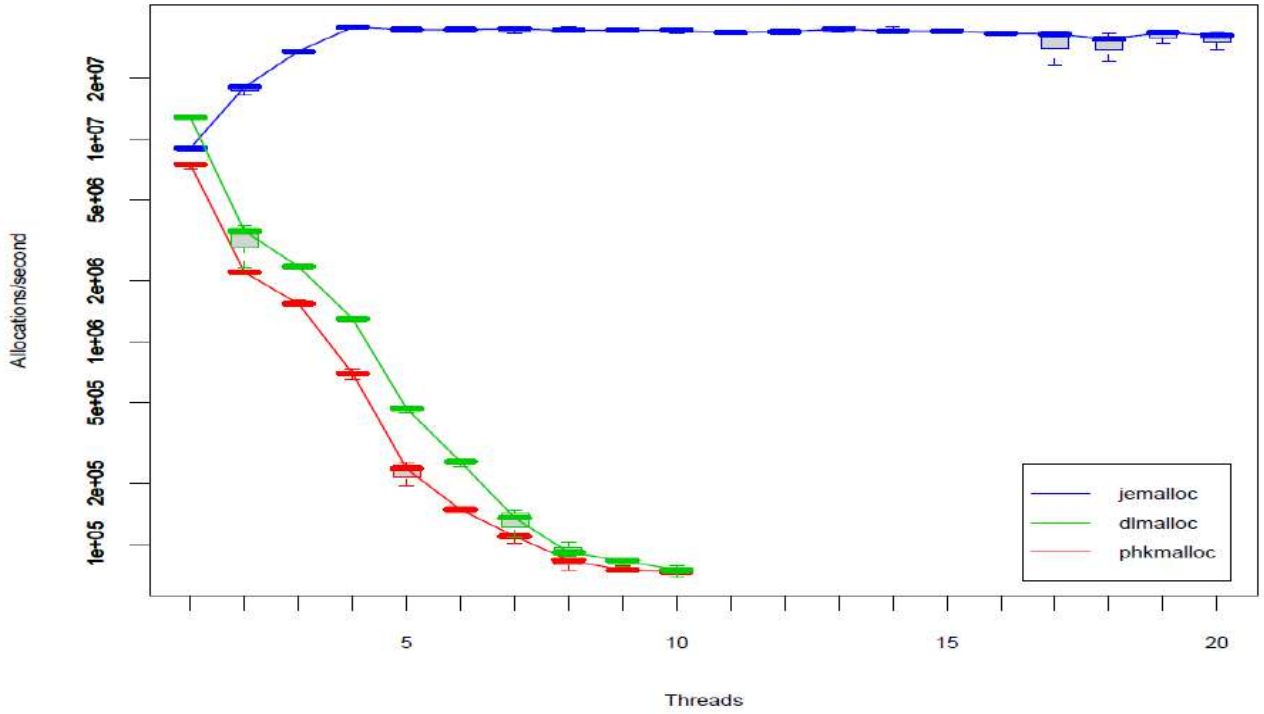


Figure 6: Allocator throughput, measured in allocations/second, for increasing numbers of threads. Each run performs a total of 40,000,000 allocation/deallocation cycles, divided equally among threads, creating one 512-byte object per cycle. All configurations are replicated three times, and the results are summarized by box plots, where the central lines denote the medians and the whiskers represent the most extreme values measured.

Fonctionnement de malloc/free

- Peu optimisé pour le contexte multithread
  - Support Linux des threads très limité (thread safe uniquement)
  - Non lié à l'ordonnanceur (migration de pages)
  - Pas de support OpenMP
- Pas de problème en MPI
  - Vision séquentielle
  - Pas de migration
  - Localité intrinsèque au modèle

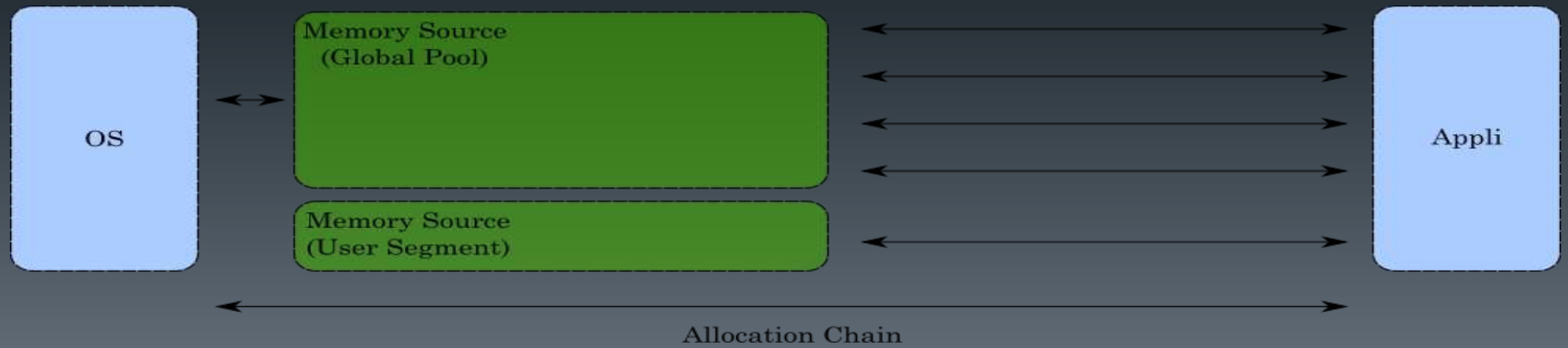
Issue des travaux de thèse de Sébastien Valat

https://tel.archives-ouvertes.fr/tel-01253537/document

Allocation mémoire en contexte HPC

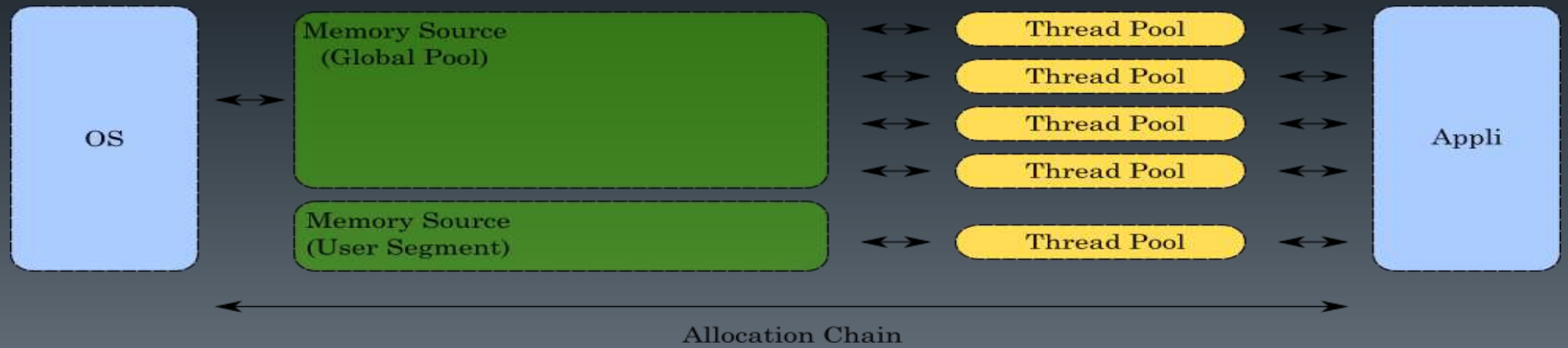
Memory Allocation in User Space

- User Space
  - Reduce the number of requests to the OS
- Drawbacks
  - Inefficient NUMA support
    - Data are allocated without NUMA knowledge
  - Inefficient multithread application support
    - Contention issue if threads target the same memory area



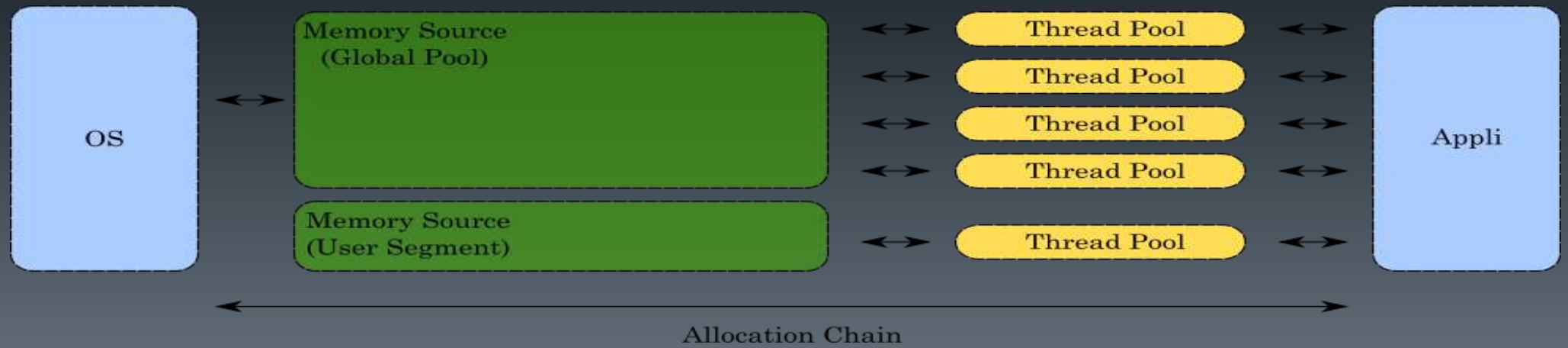
Data Locality in multithreaded context

- Data locality management with thread pools
  - Thread pools handle local management of allocations
    - List of free blocs (coming from macro-blocs splitting)
    - First level for algorithms (decision, splitting, fusion)
  - Exchange with Memory source via macro-blocs (> 2MB)
  - Ensure local memory accesses
    - Avoid false sharing
    - Easy when threads are bound, enable thread migration and keep data locality



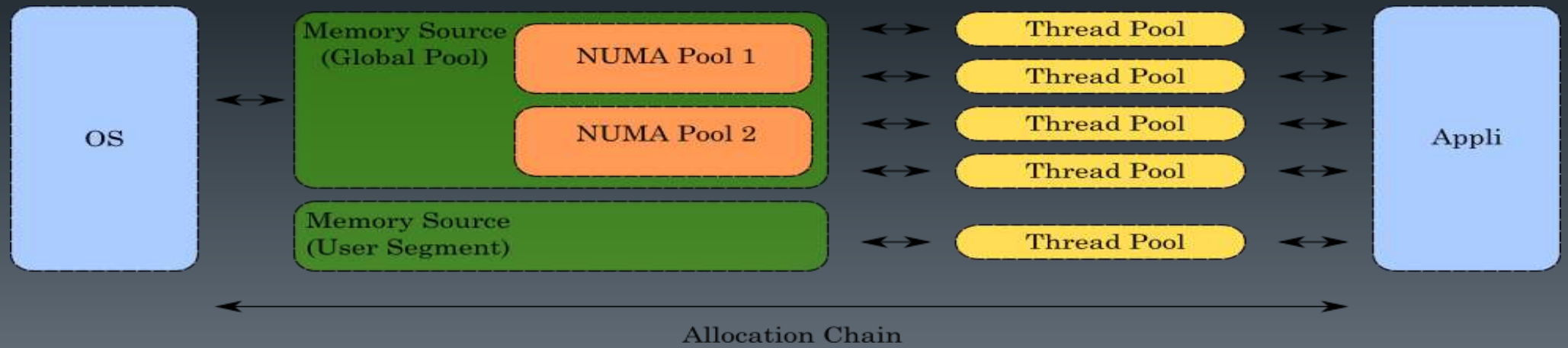
Data Locality in multithreaded context

- Drawbacks
  - Directly handle only allocation < 1 MB
    - Huge segment allocation are directly transferred to Memory Source
  - Multiple threads with huge segment allocation will cause contention of Memory Source
- Solution implemented in MPC allocator:
  - Do not tackle directly the Global Pool
    - Levels of NUMA pools to avoid contention and NUMA effects with huge segments



Data Locality in multithreaded context

- Keep data locality
  - Efficient NUMA support
  - Efficient highly multithread application support
  - How to handle thread migration
    - Link between thread scheduler and memory allocator
    - Next touch policy
  - Allow to handle concurrency more precisely on each NUMA level
  - Allow page “recycling” between threads in the same NUMA level



Huge segments issues

- Huge segment allocations still go through the OS
  - Performance loss due to system calls
  - For sizes > 1MB, allocation cost is driven by page-faulting
- MPC solution
  - The memory source is an allocator by itself
    - It keeps track of free macro-blocs for future reuse
      - If the segment is not too large
      - If the total amount of memory used for this macro-blocs cache is not tool large
    - Possibility to fuse adjacent macro-blocs to provide greater segments if necessary
  - Use *mremap* to resize segments if no matching segment is available
- Drawbacks
  - Greater latency when the system call is still necessary
    - Set up of macro-blocs buffers for each NUMA Pool to avoid system calls
  - Increase memory consumption
    - Possibility to adjust dynamically the amount of memory available for macro-blocs caches

AMR Code + MPC on Dual-Westmere (2*6 cores)

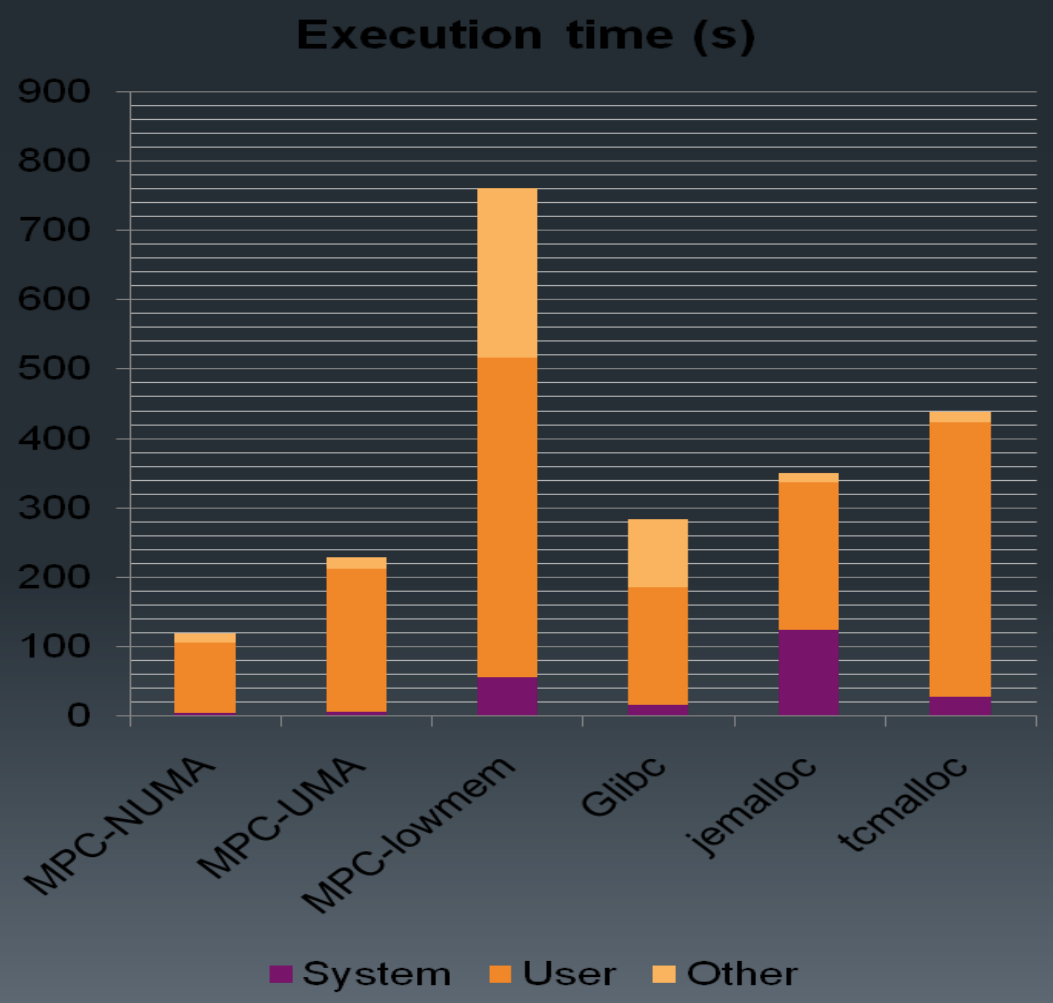
- Standard **4K pages**

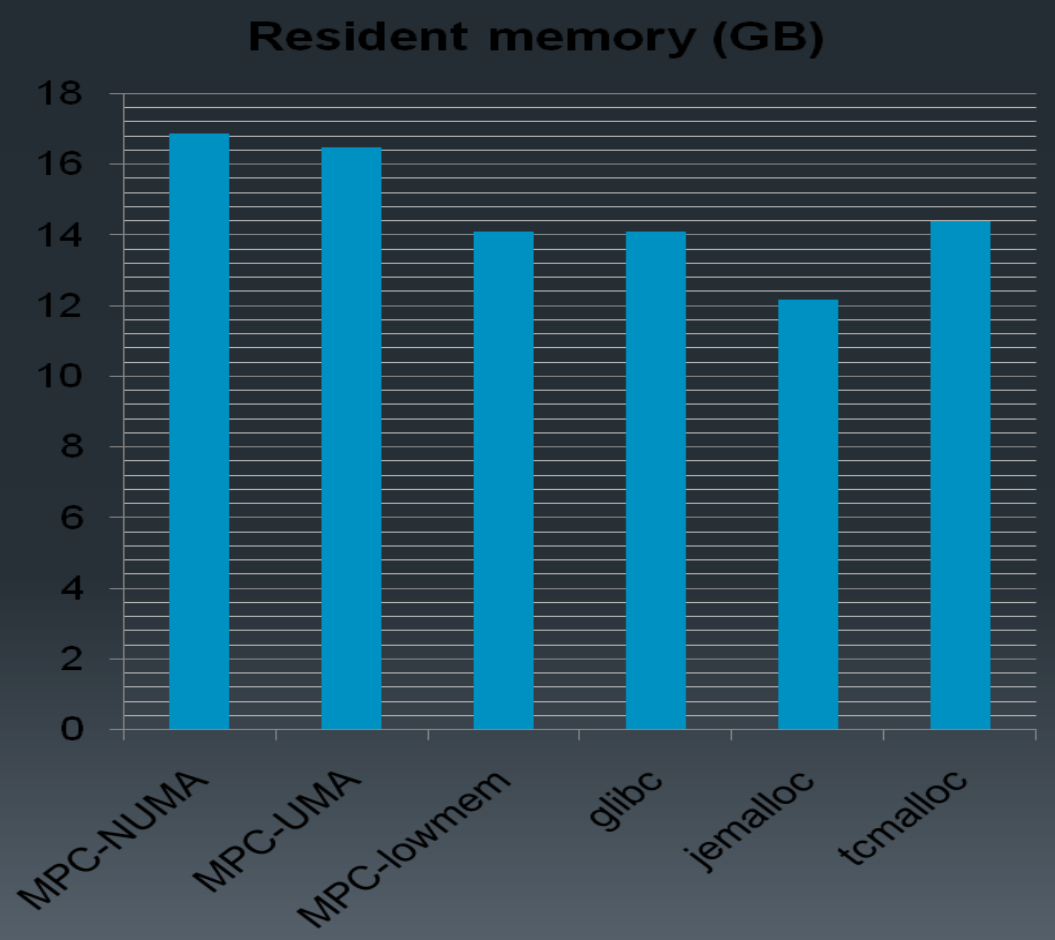
Allocator	Kernel	Total (s)	Sys. (s)	Mem. (GB)
MPC-NUMA	Std.	135.14	1.79	4.3
MPC-Lowmem	Std.	161.58	15.97	2.0
Jemalloc	Std.	143.05	14.53	1.9

- **Transparent Huge Pages**

Allocator	Kernel	Total (s)	Sys. (s)	Mem. (GB)
MPC-NUMA	Std.	137.89	1.86	6.2
MPC-Lowmem	Std.	196.51	28.24	3.9
Jemalloc	Std.	144.72	14.66	2.5

AMR Code + MPC on Nehalem-EX (128: 4*4*8 cores)

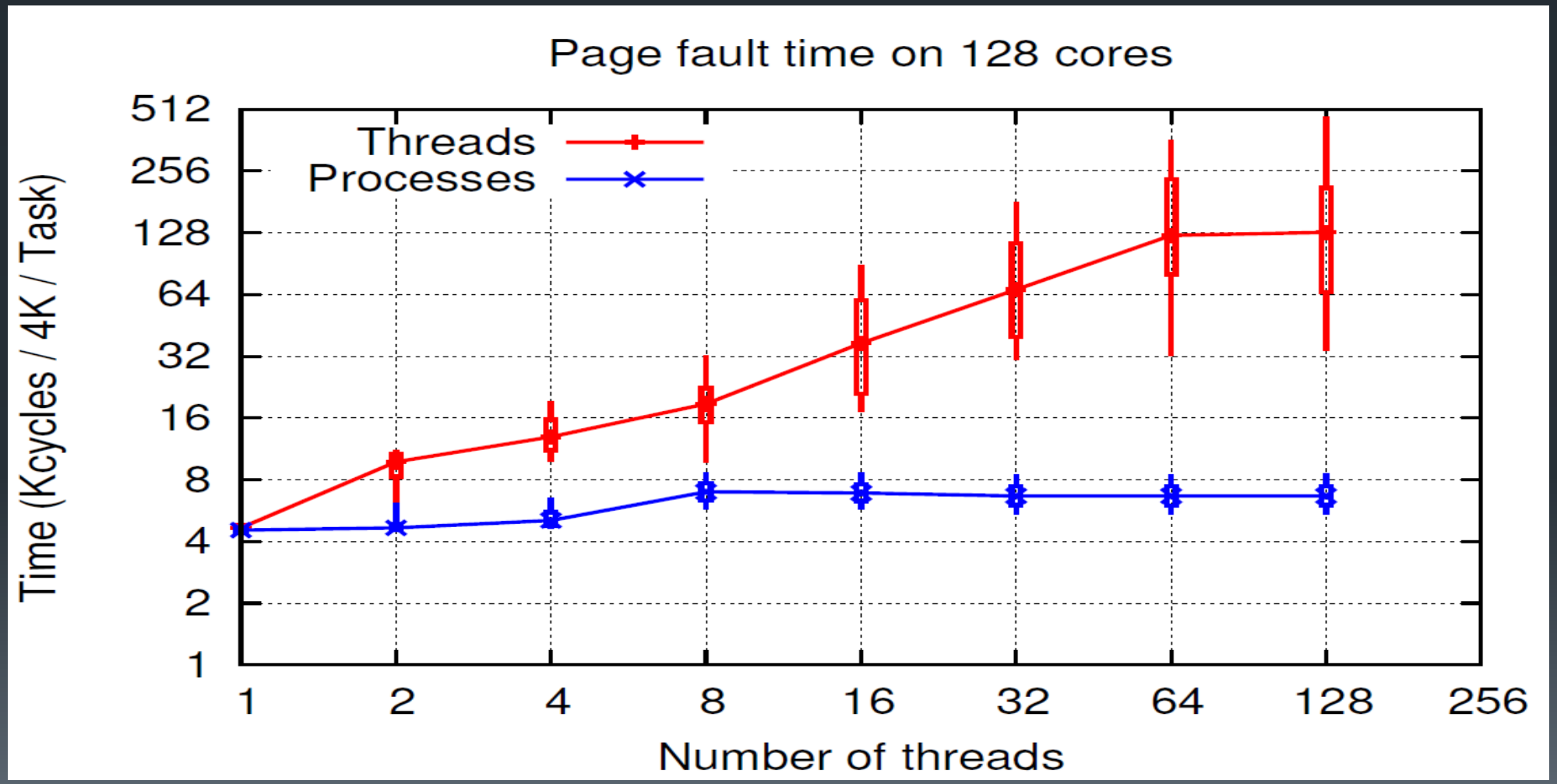




Memory allocation

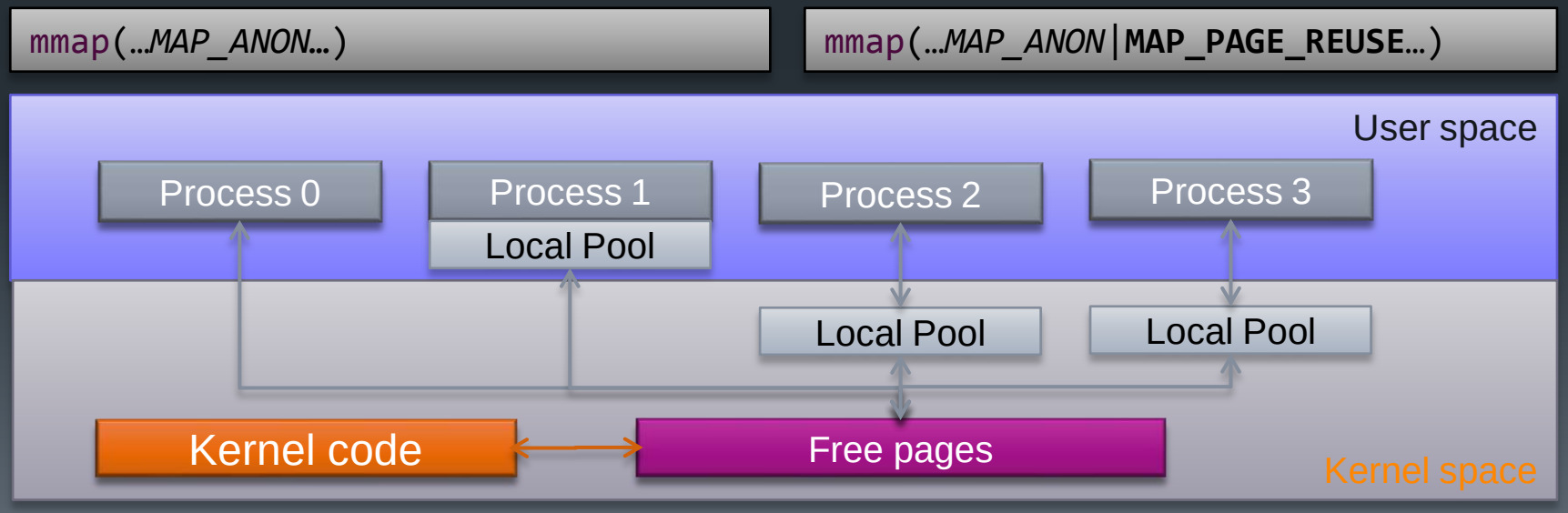
- Memory Semantic
  - Malloc only modify virtual memory space
  - Requests via *mmap* or *brk* system calls
- Lazy Memory Allocation: Page Fault
  - Physical pages are then provided upon a first touch policy via a page fault
  - Allocation cost is not limited to *malloc*
  - Cache usage is linked to physical pages allocation
- Reset memory pages
  - Security reason
  - Before providing page to the application
  - In kernel space

Page fault cost

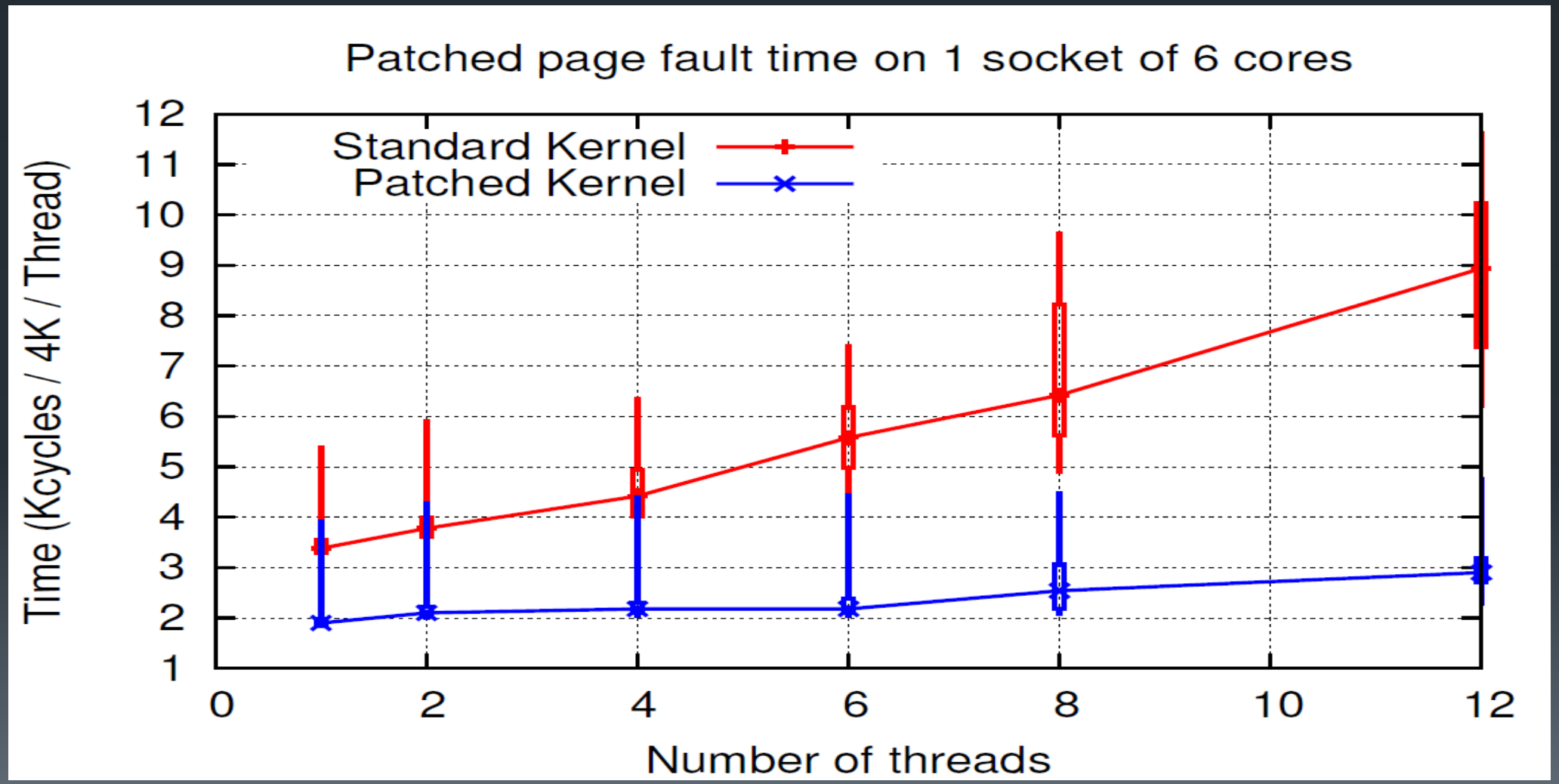


Page zeroing

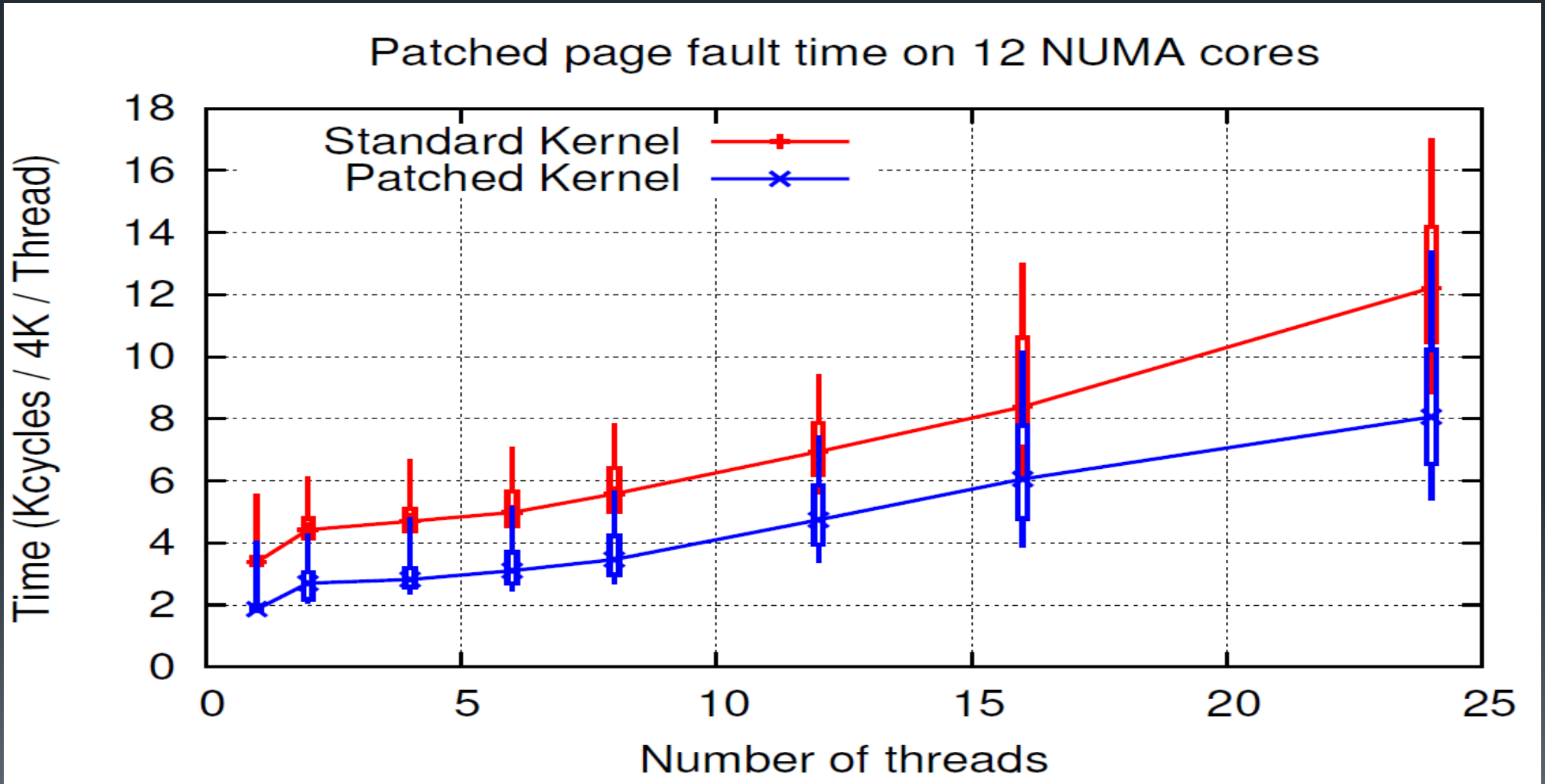
- Kernel Space
  - 40% of page fault execution time is due to zero-page.
  - Zero-page is useless from the application point of view.
  - Reuse dirty pages within the same process to avoid page zeroing.
  - Kernel patch using an extension to the mmap system call.



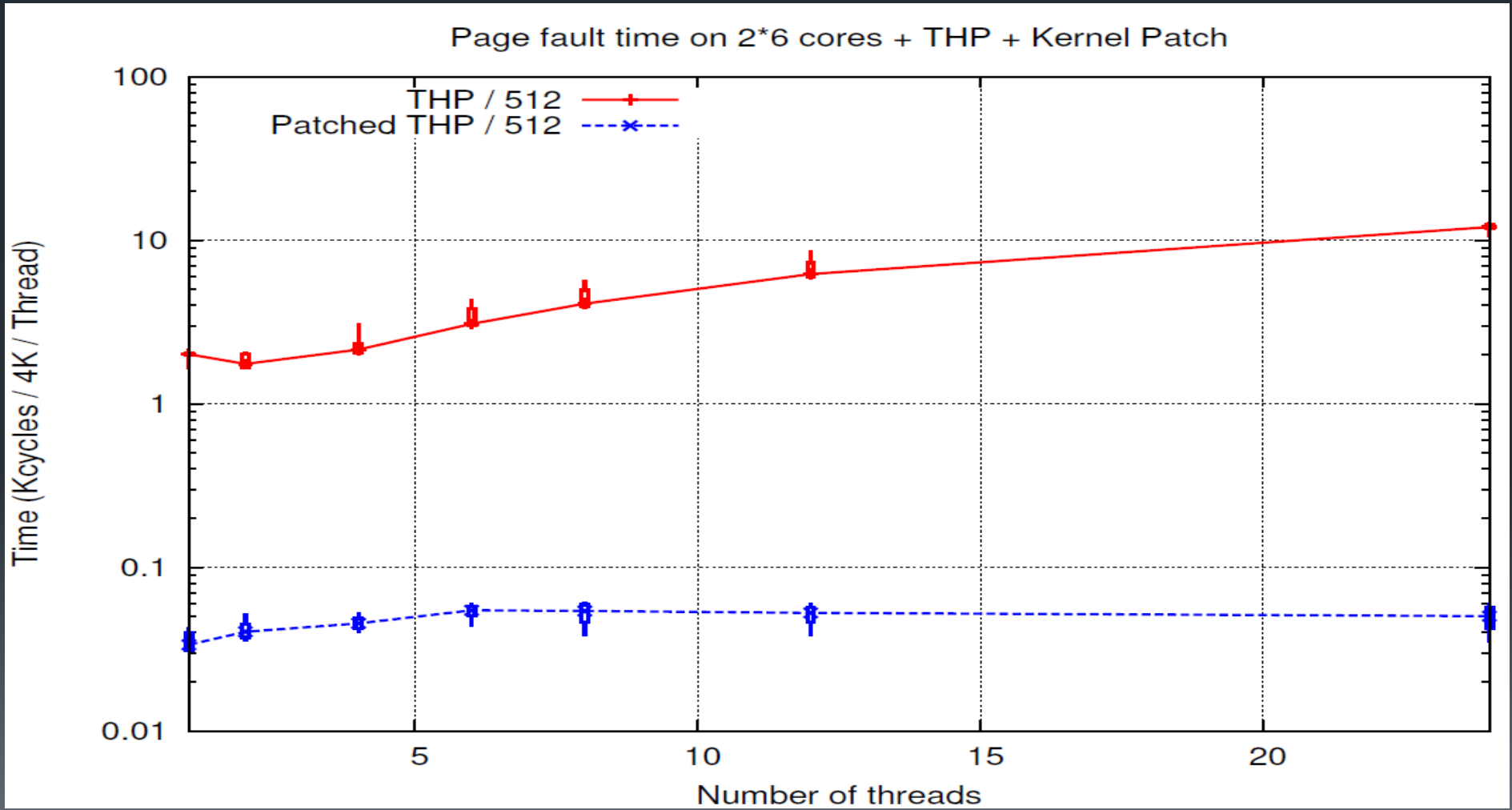
Page fault performance results



Page fault performance results



Page fault performance results



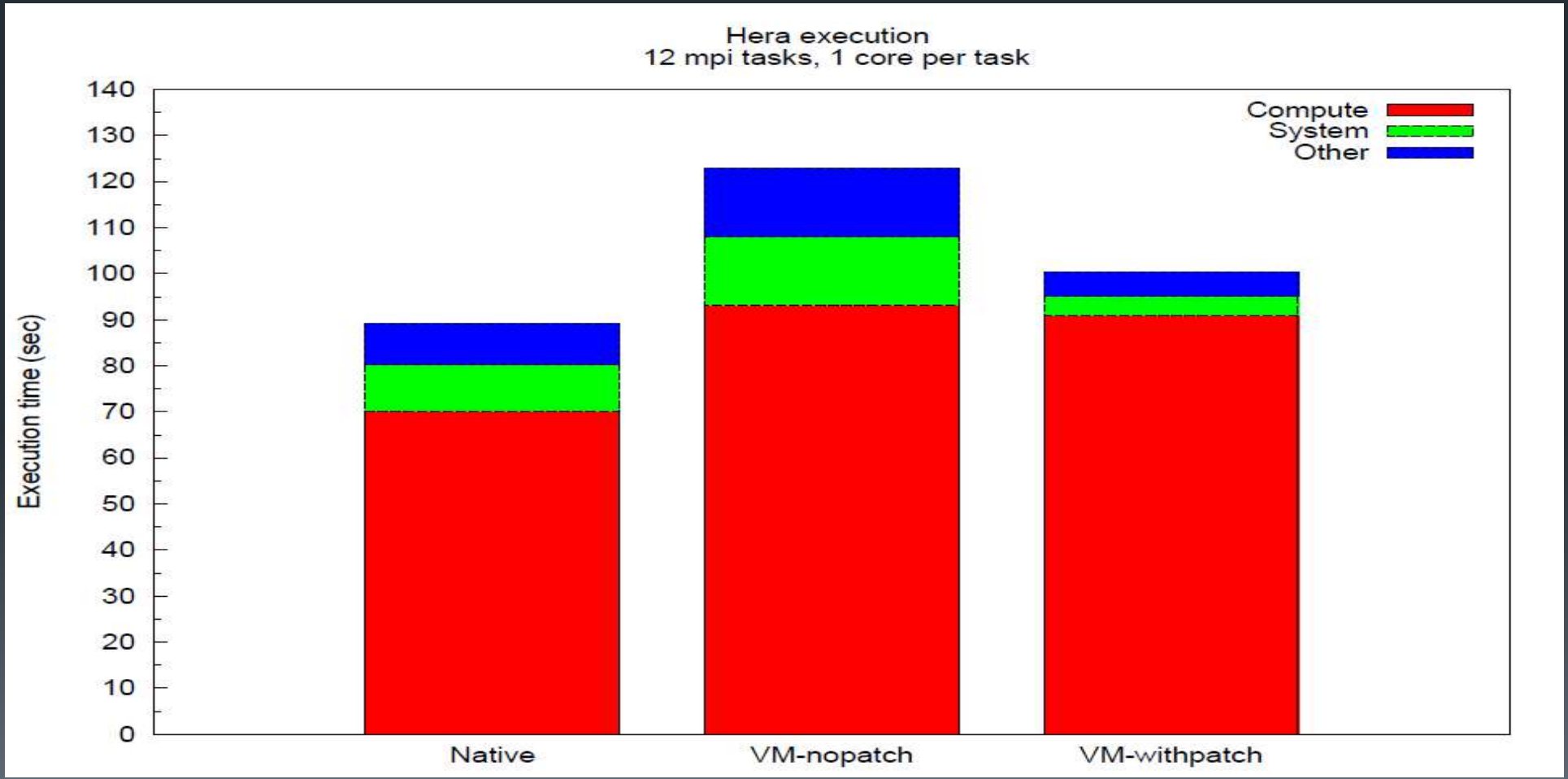
MPC on Dual-Westmere (2*6 cores)

- Kernel patch and standard **4K pages**

Allocator	Kernel	Total (s)	Sys. (s)	Mem. (GB)
MPC-NUMA	Std.	135.14	1.79	4.3
MPC-Lowmem	Std.	161.58	15.97	2.0
MPC-Lowmem	Patched	157.62	10.60	2.0
Jemalloc	Std.	143.05	14.53	1.9
Jemalloc	Patched	140.65	9.32	3.2

Allocator	Kernel	Total (s)	Sys. (s)	Mem. (GB)
MPC-NUMA	Std.	137.89	1.86	6.2
MPC-Lowmem	Std.	196.51	28.24	3.9
MPC-Lowmem	Patched	138.77	2.90	3.8
Jemalloc	Std.	144.72	14.66	2.5
Jemalloc	Patched	138.47	6.40	3.2

Results on Dual-Westmere (2*6 cores)



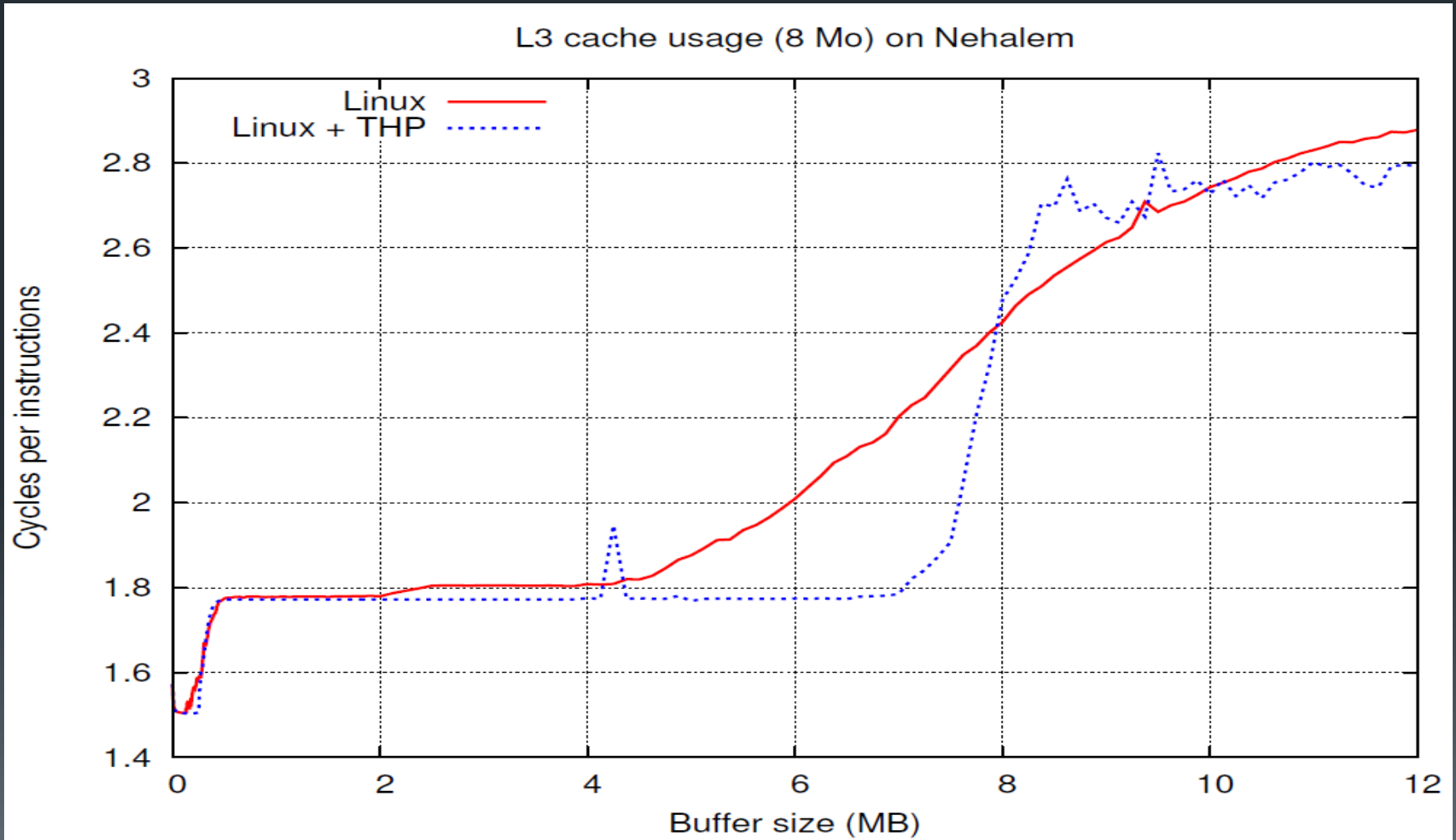
Physical page allocation

- Linux
  - Random page distribution
- Linux Transparent Huge Pages
  - Random page distribution
  - Huge pages
- OpenSolaris
  - Page coloring
  - Hash on virtual page addresses
  - Without PID
- FreeBSD
  - Page coloring
  - Hash on virtual page addresses
  - Without PID
  - Superpages

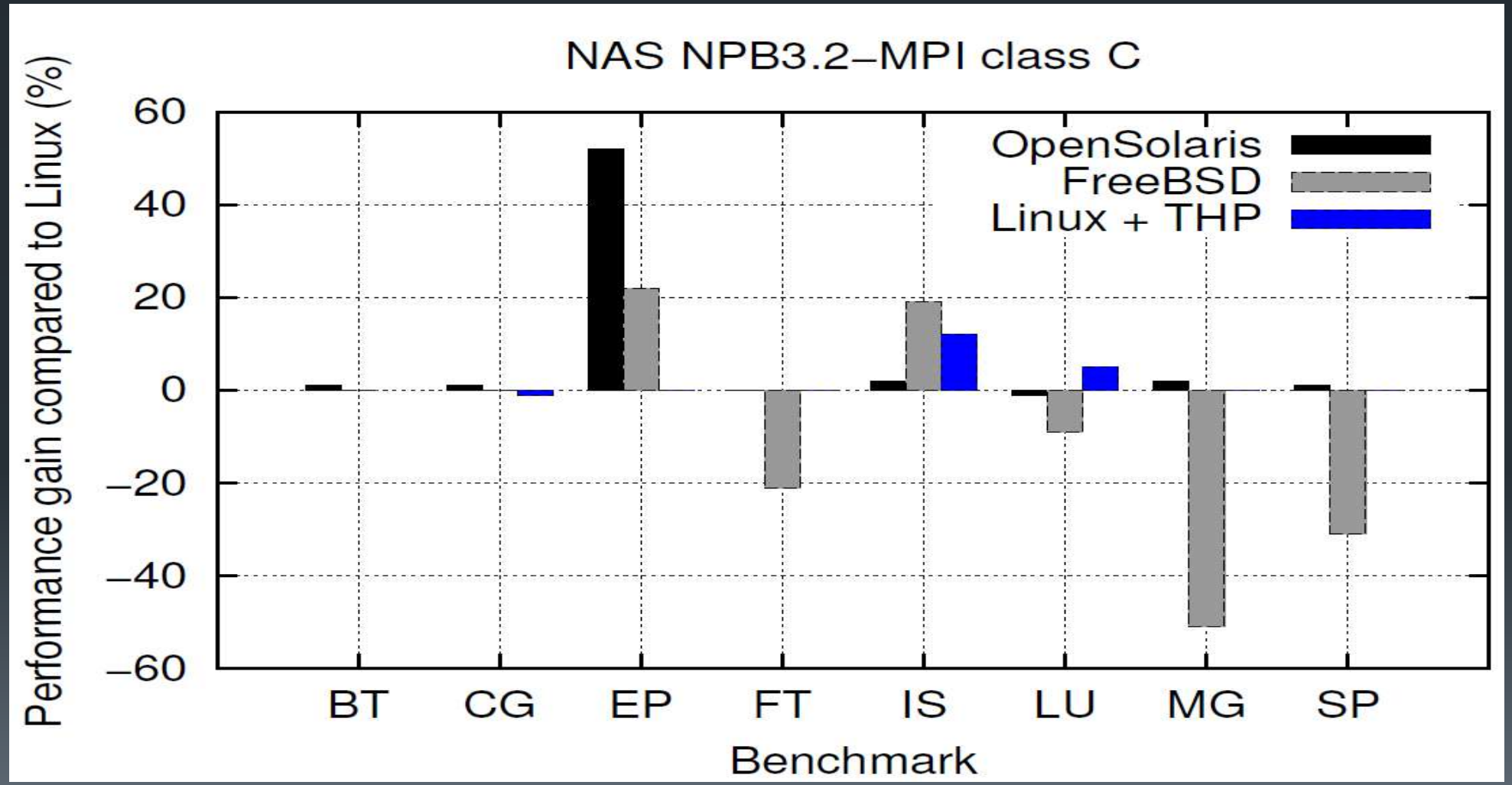
Physical page allocation

- Experimental setup
  - Processor Bi Intel Xeon-E5502 (quad core Nehalem)
  - Frequency 2.27 GHz
  - Cache L3 8 MB
  - Cache L2 256 KB
  - Cache L1 32 KB
  - Memory 24 GB
- Software stack
  - GCC 4.4.1
  - Identical software stack
  - Identical compilation options
- Native execution

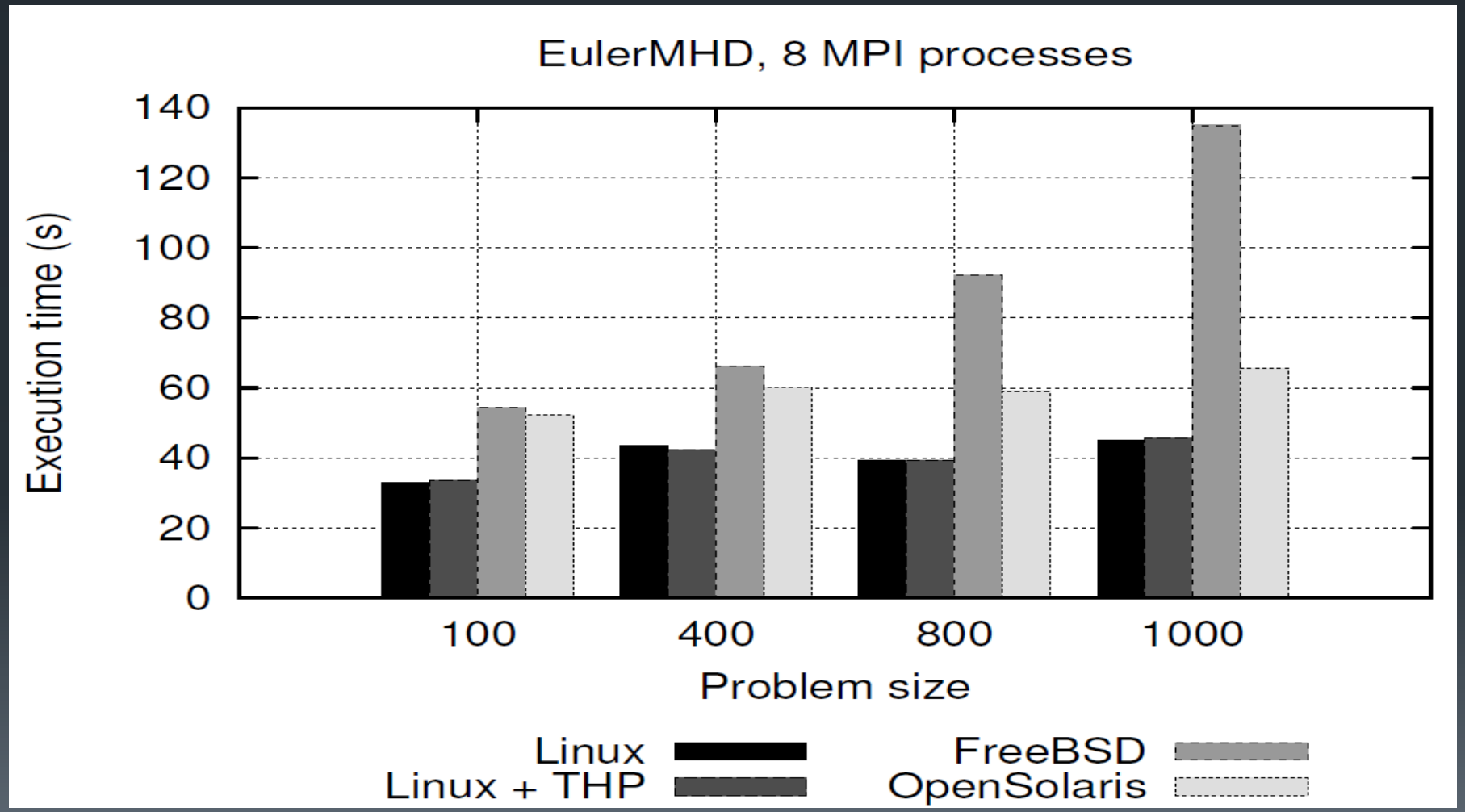
Cache usage


L3 cache usage (8 Mo) on Nehalem
Linux
Linux + THP
Cycles per instructions
Buffer size (MB)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
0
2
4
6
8
10
12

Impact of page allocation policy on benchmarks



Impact of page allocation policy on application



Conclusion

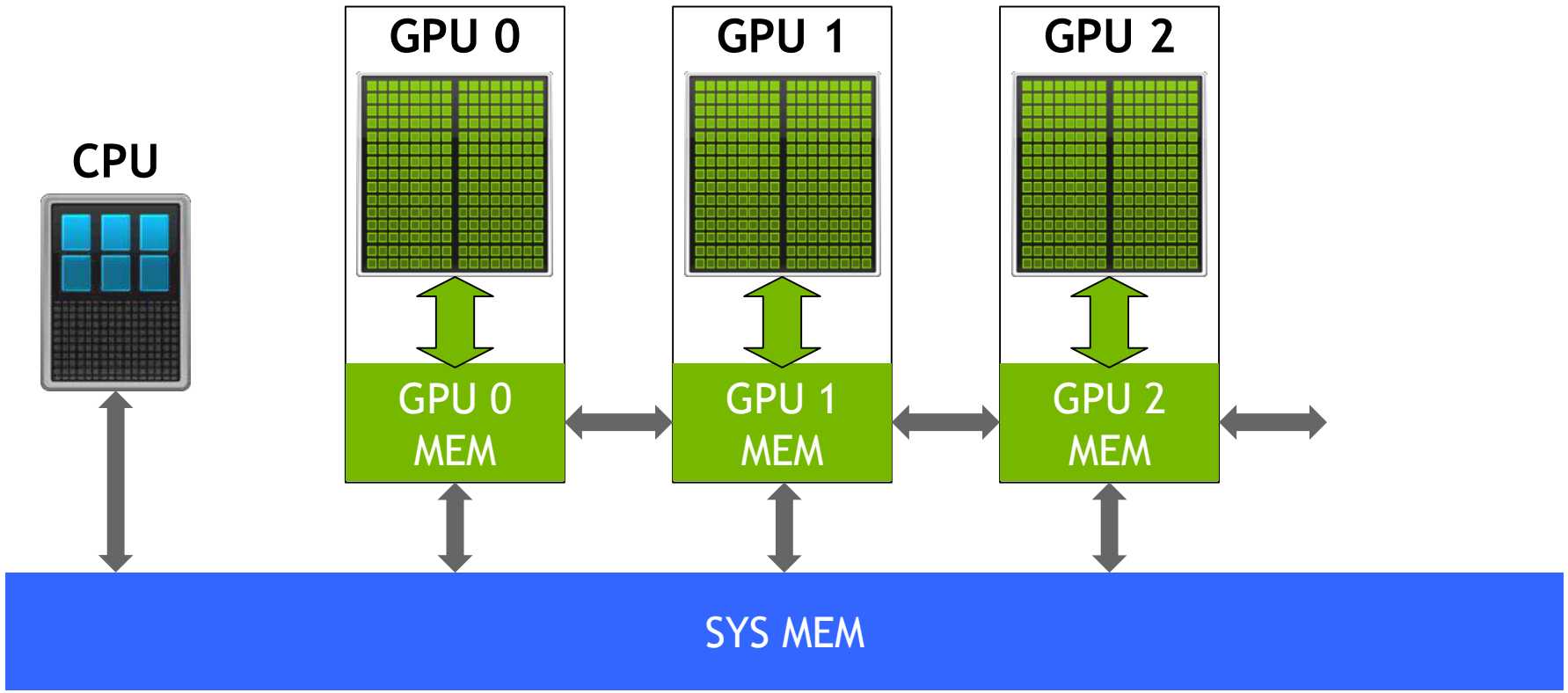
- Page zeroing
  - Kernel-space patch in Linux 2.6.32 and 2.6.36
  - Performance improvements
    - up to 45% on sequential page faults
    - up to 66% for 12 threads
  - Limitation for standard usage (outside HPC)
    - No support for SWAP
    - Need pool cleaning method

- Physical pages allocation
  - Huge impact on execution time
    - Up to 51% performance improvement and 91% performances decrease
  - POC: kernel module with contiguous physical pages allocation to improve cache usage

UNIFIED MEMORY ON PASCAL AND VOLTA

Nikolay Sakharnykh - May 10, 2017

HETEROGENEOUS ARCHITECTURES



UNIFIED MEMORY FUNDAMENTALS

Single Pointer

CPU code

```
void *data;
data = malloc(N);

cpu_func1(data, N);

cpu_func2(data, N);


cpu_func3(data, N);

free(data);
```

GPU code

```
void *data;
data = malloc(N);

cpu_func1(data, N);

gpu_func2<<<...>>>(data, N);
cudaDeviceSynchronize();

cpu_func3(data, N);

free(data);
```

UNIFIED MEMORY FUNDAMENTALS

Single Pointer

Explicit Memory Management

```
void *h_data, *d_data; h_data =
malloc(N); cudaMalloc(&d_data, N);
cpu_func1(h_data, N);
cudaMemcpy(d_data, h_data, N, ...)
gpu_func2<<<...>>>(data, N);

cudaMemcpy(h_data, d_data, N, ...)
cpu_func3(h_data, N);

free(h_data);
cudaFree(d_data);
```

Unified Memory

```
void *data;
data = malloc(N);

cpu_func1(data, N);

gpu_func2<<<...>>>(data, N);
cudaDeviceSynchronize();

cpu_func3(data, N);

free(data);
```

UNIFIED MEMORY FUNDAMENTALS

Deep Copy Nightmare

Explicit Memory Management

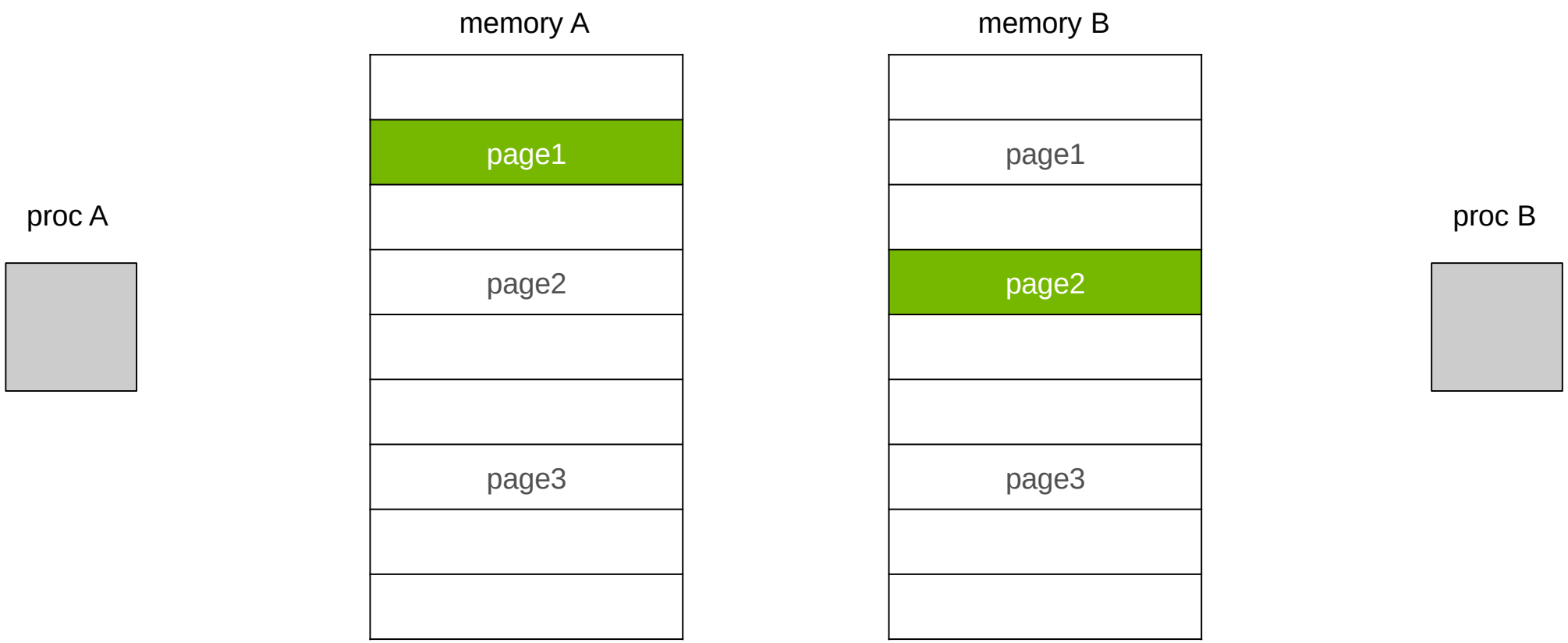
```
char **data;
data = (char**)malloc(N*sizeof(char*));
for (int i = 0; i < N; i++)
  data[i] = (char*)malloc(N);

char **d_data;
char **h_data = (char**)malloc(N*sizeof(char*));
for (int i = 0; i < N; i++) {
  cudaMalloc(&h_data2[i], N);
  cudaMemcpy(h_data2[i], h_data[i], N, ...);
}
cudaMalloc(&d_data, N*sizeof(char*));
cudaMemcpy(d_data, h_data2, N*sizeof(char*), ...);

gpu_func<<<...>>>(data, N);
```

Unified Memory

```
char **data;
data = (char**)malloc(N*sizeof(char*));
for (int i = 0; i < N; i++)
  data[i] = (char*)malloc(N);

gpu_func<<<...>>>(data, N);
```

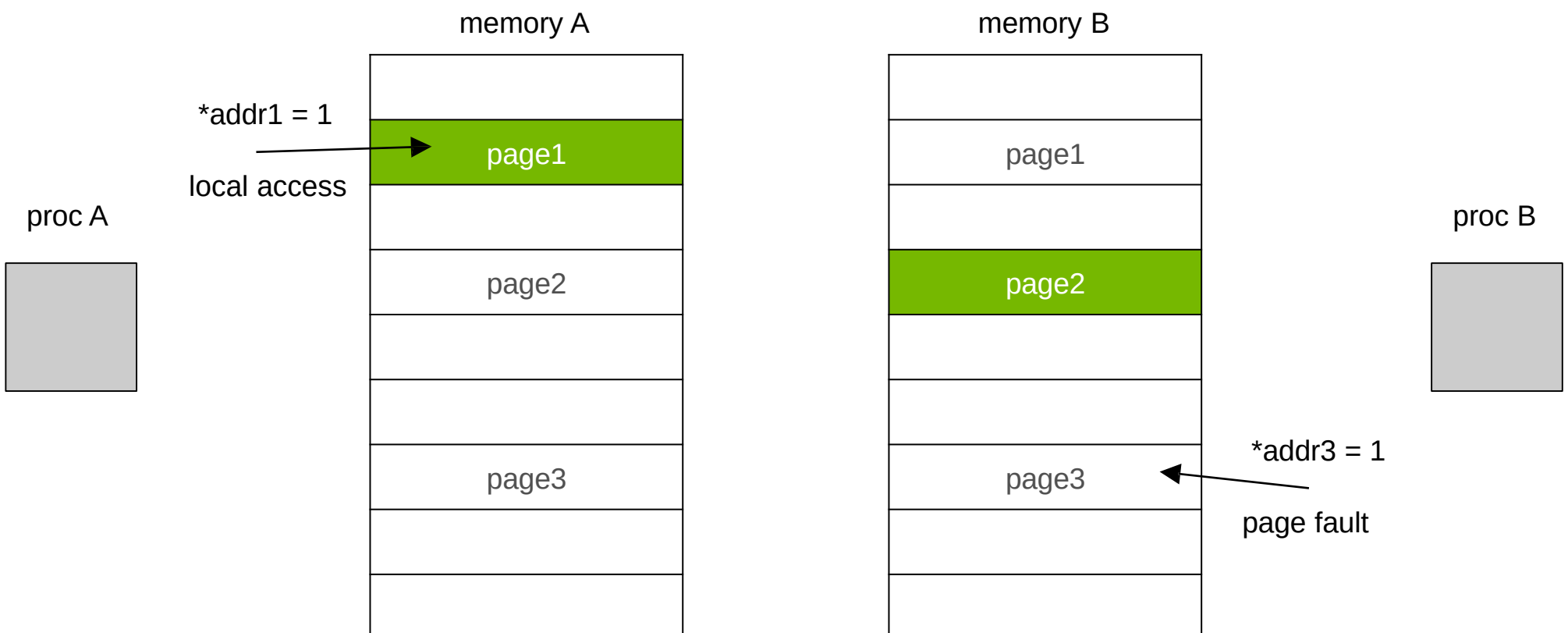
UNIFIED MEMORY FUNDAMENTALS

On-Demand Migration



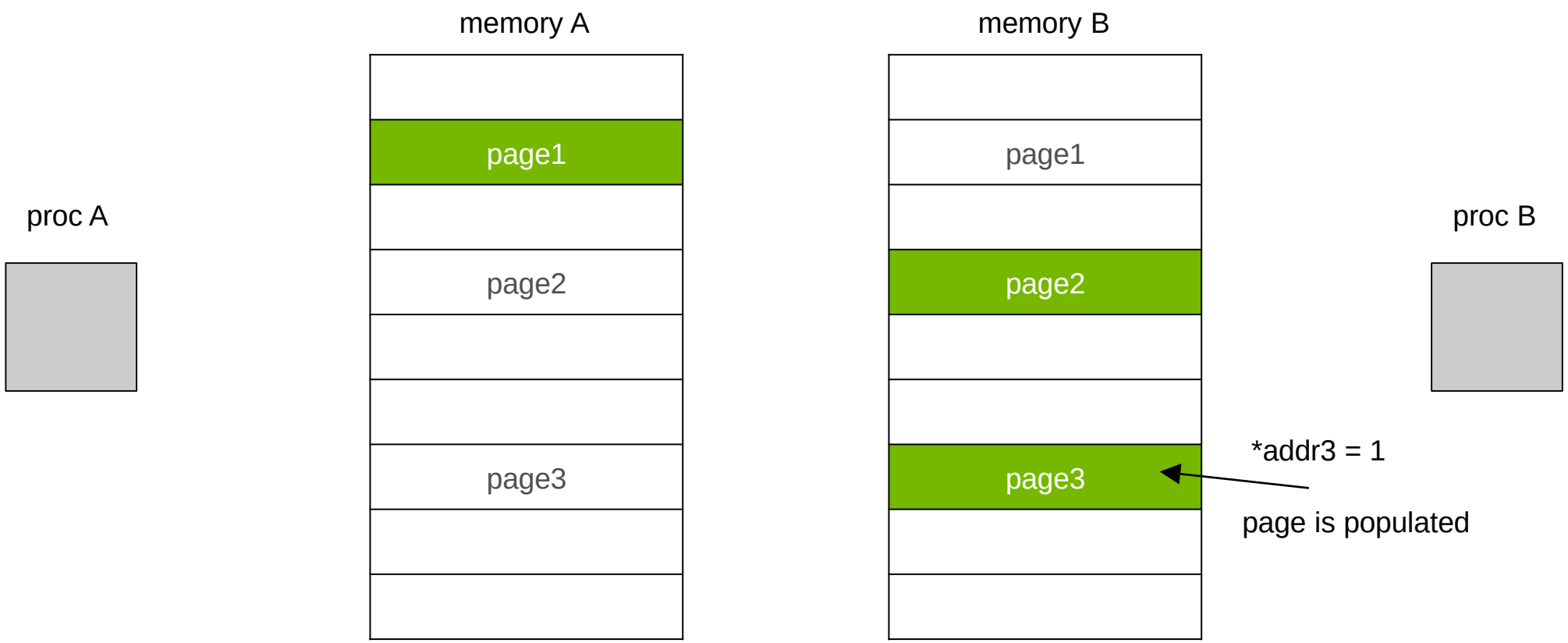
UNIFIED MEMORY FUNDAMENTALS

On-Demand Migration



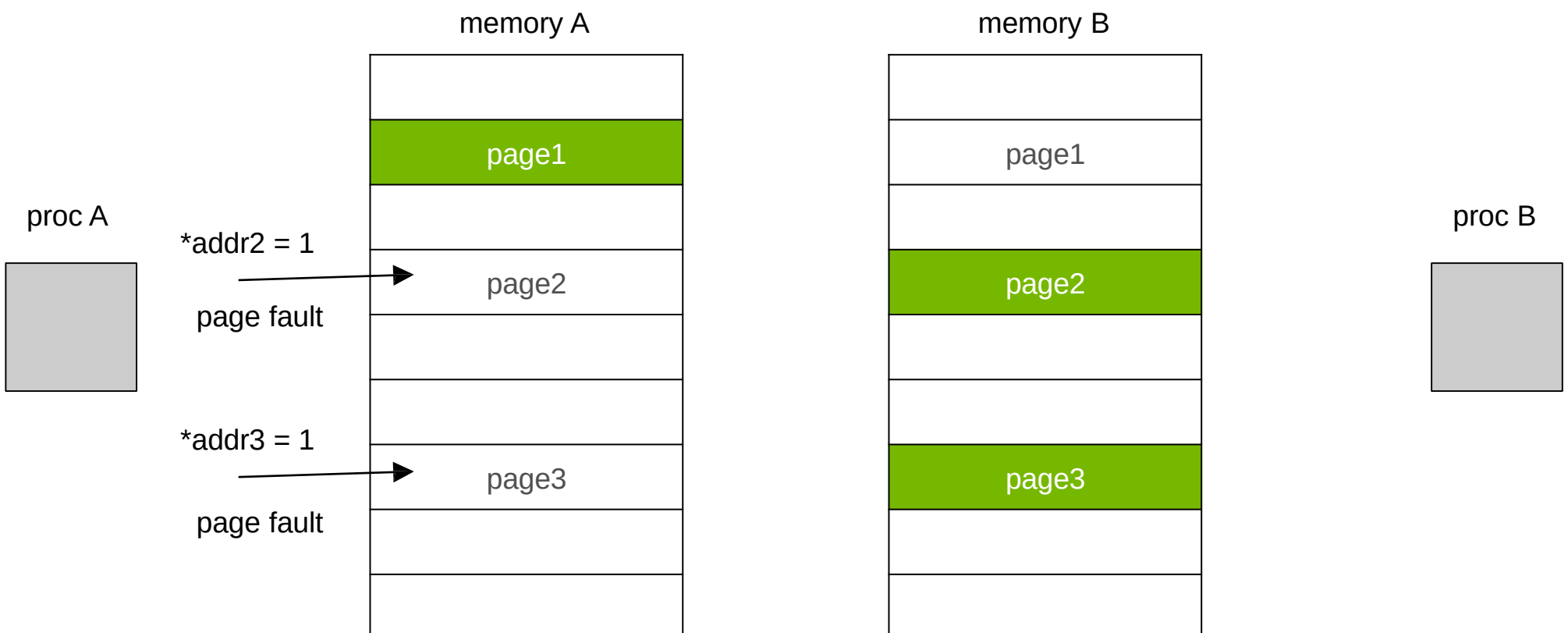
UNIFIED MEMORY FUNDAMENTALS

On-Demand Migration



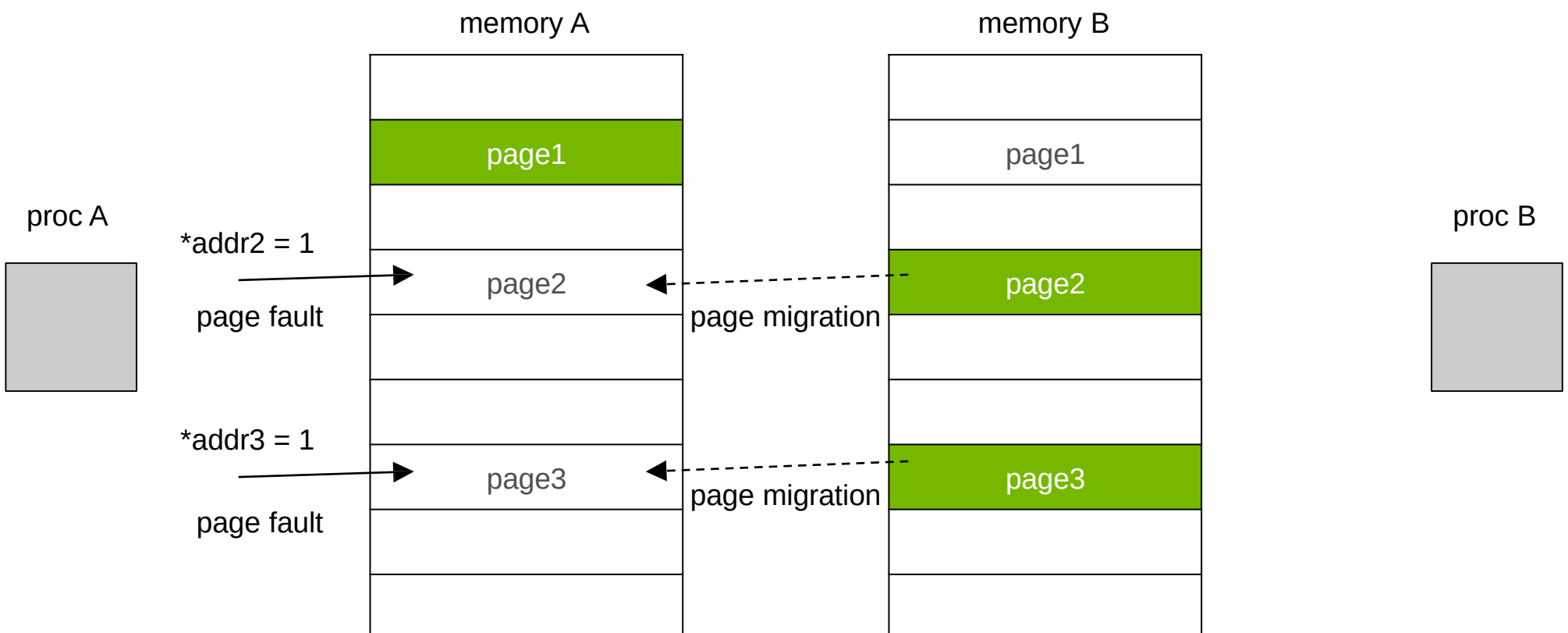
UNIFIED MEMORY FUNDAMENTALS

On-Demand Migration



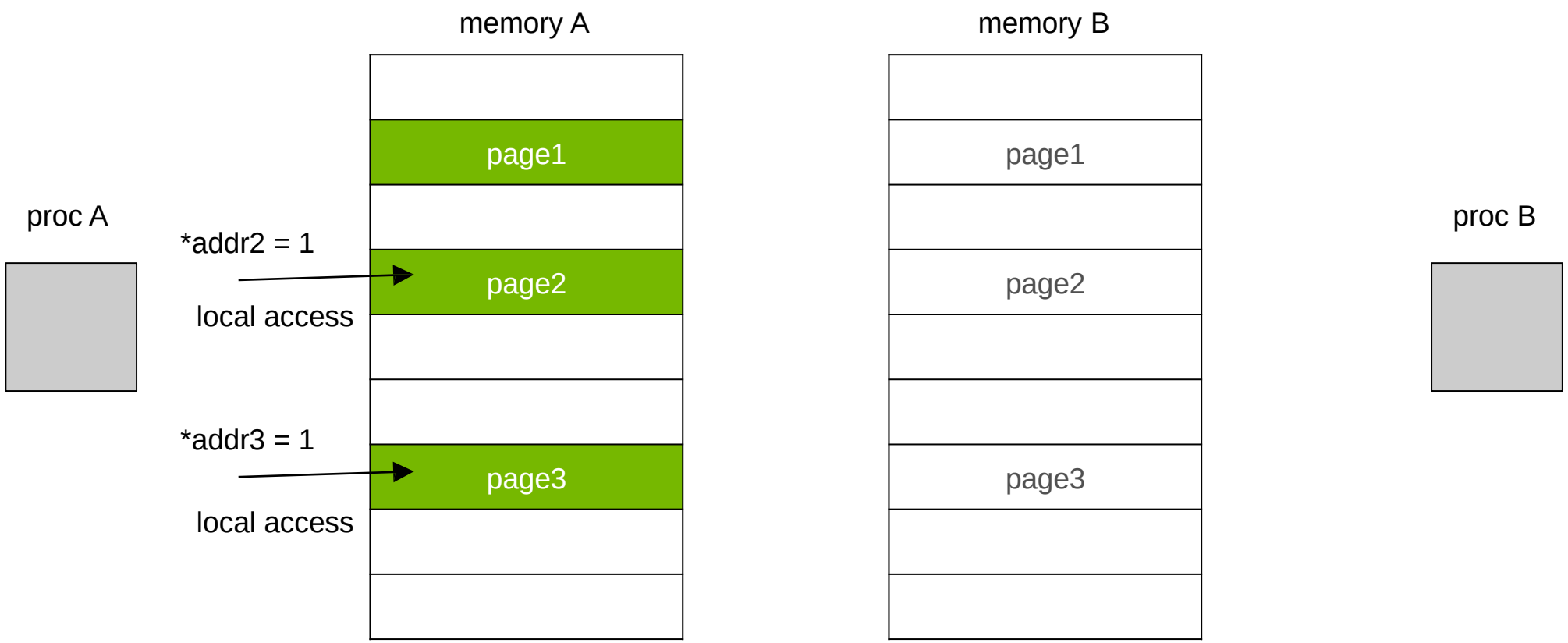
UNIFIED MEMORY FUNDAMENTALS

On-Demand Migration



UNIFIED MEMORY FUNDAMENTALS

On-Demand Migration



UNIFIED MEMORY FUNDAMENTALS

When Is This Helpful?

When it doesn't matter *how* data moves to a processor

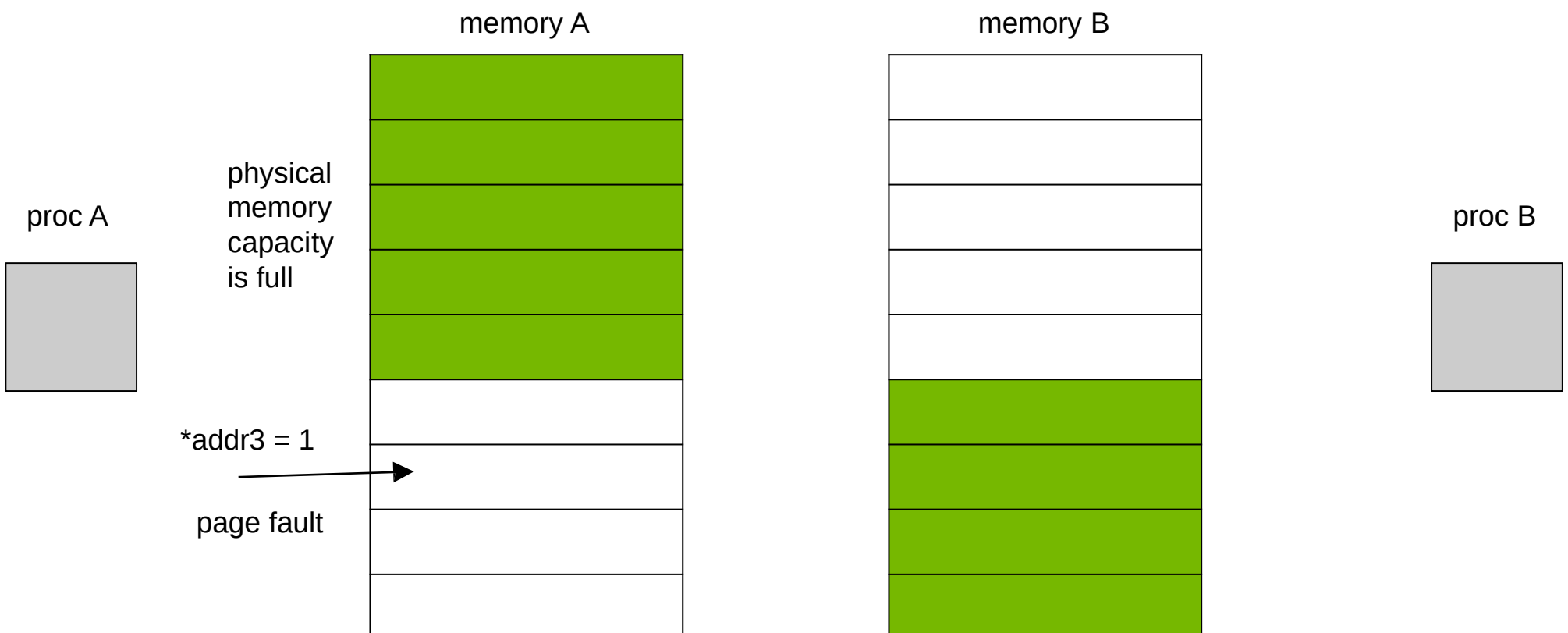
1) Quick and dirty algorithm prototyping
2) Iterative process with lots of data reuse, migration cost can be amortized
3) Simplify application debugging

When it's difficult to isolate the working set

1) Irregular or *dynamic* data structures, unpredictable access
2) Data partitioning between multiple processors

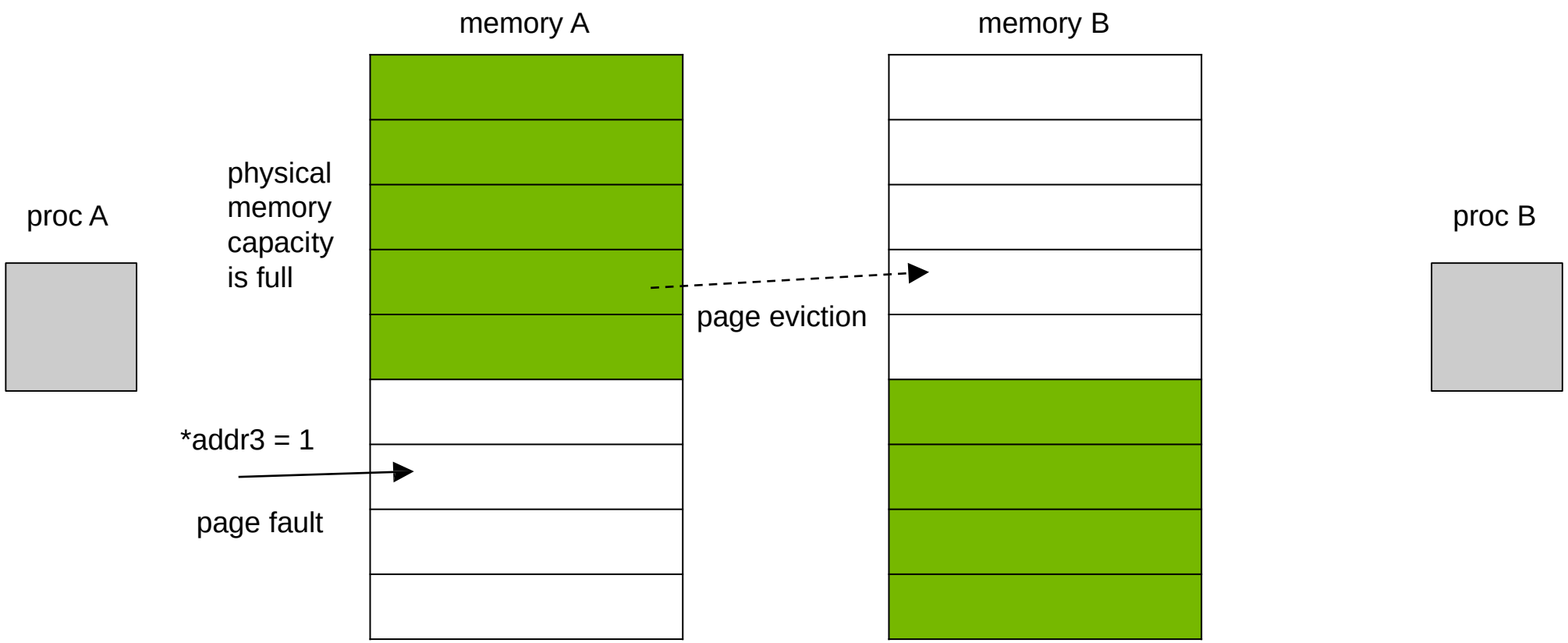
UNIFIED MEMORY FUNDAMENTALS

Memory Oversubscription



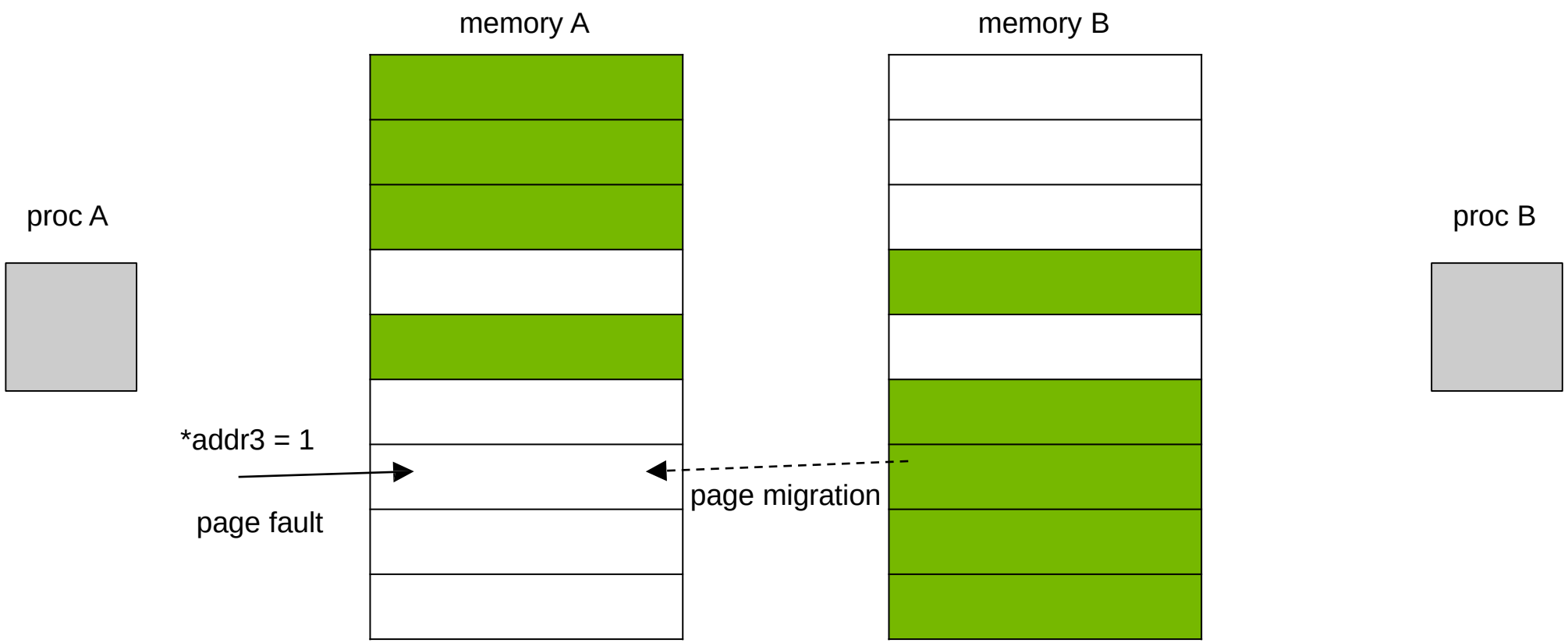
UNIFIED MEMORY FUNDAMENTALS

Memory Oversubscription



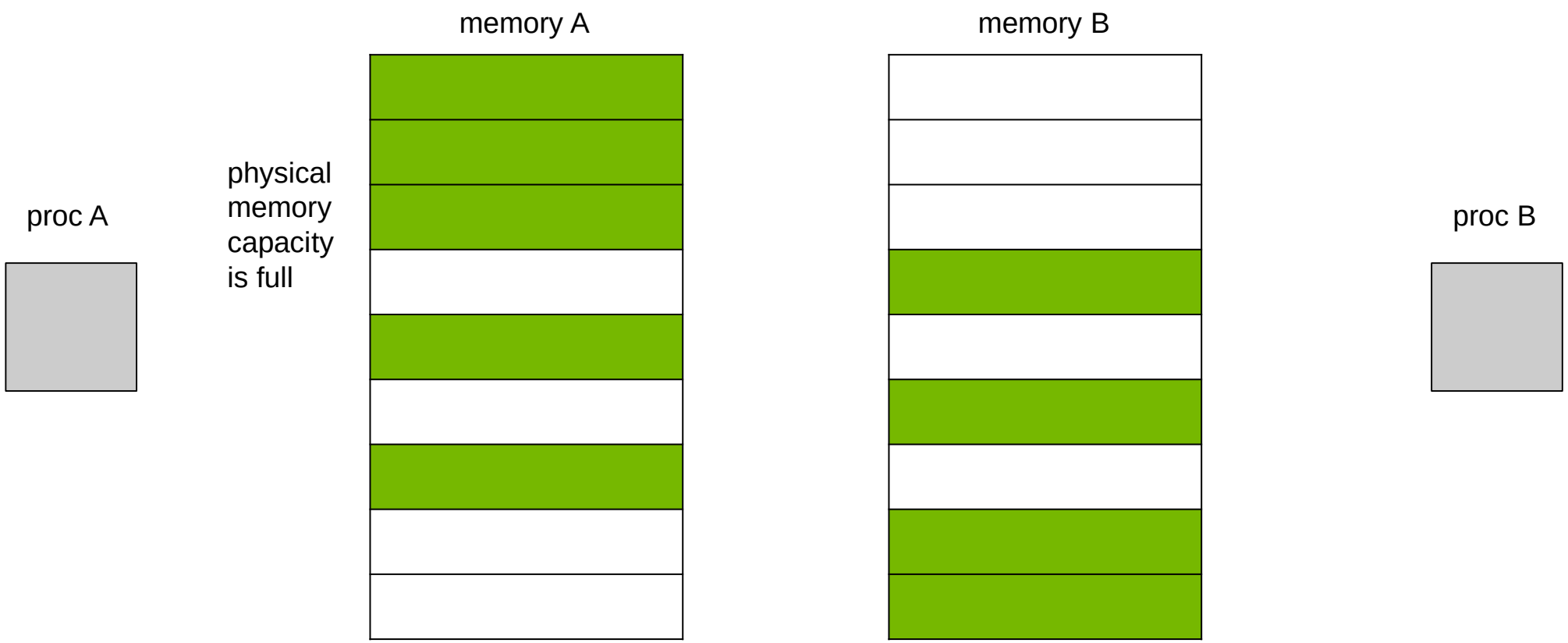
UNIFIED MEMORY FUNDAMENTALS

Memory Oversubscription



UNIFIED MEMORY FUNDAMENTALS

Memory Oversubscription



UNIFIED MEMORY FUNDAMENTALS

Memory Oversubscription Benefits

When you have **large** dataset and not enough physical memory

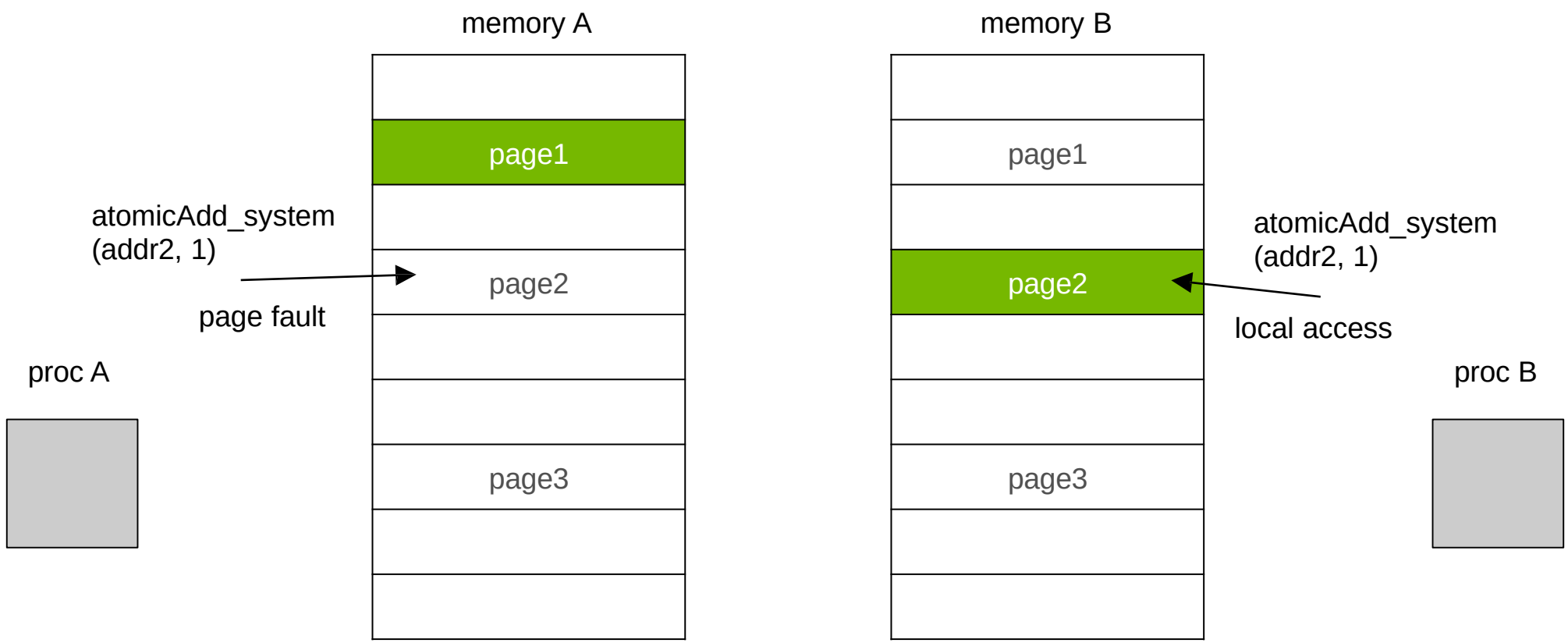
Moving pieces by hand is error-prone and requires tuning for memory size

Better to run slowly than get fail with out-of-memory error

You can actually get **high performance** with Unified Memory!

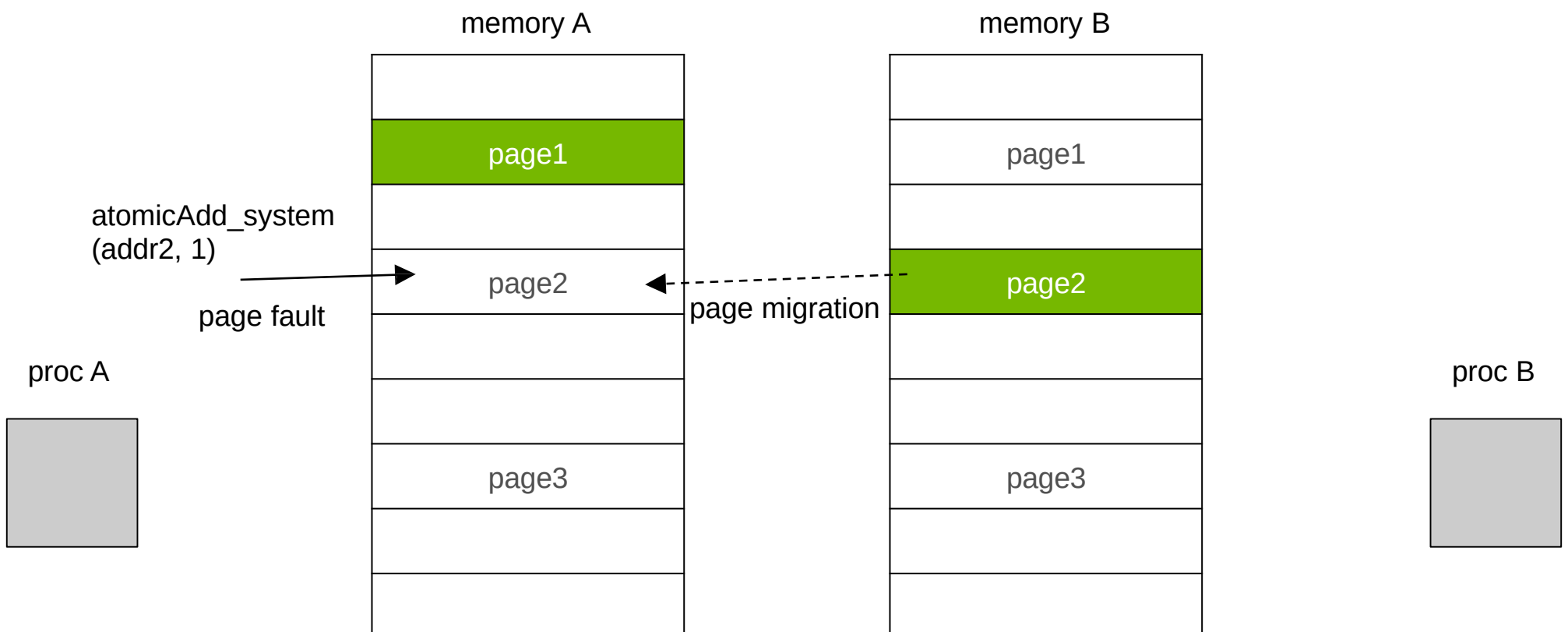
UNIFIED MEMORY FUNDAMENTALS

System-Wide Atomics with Exclusive Access



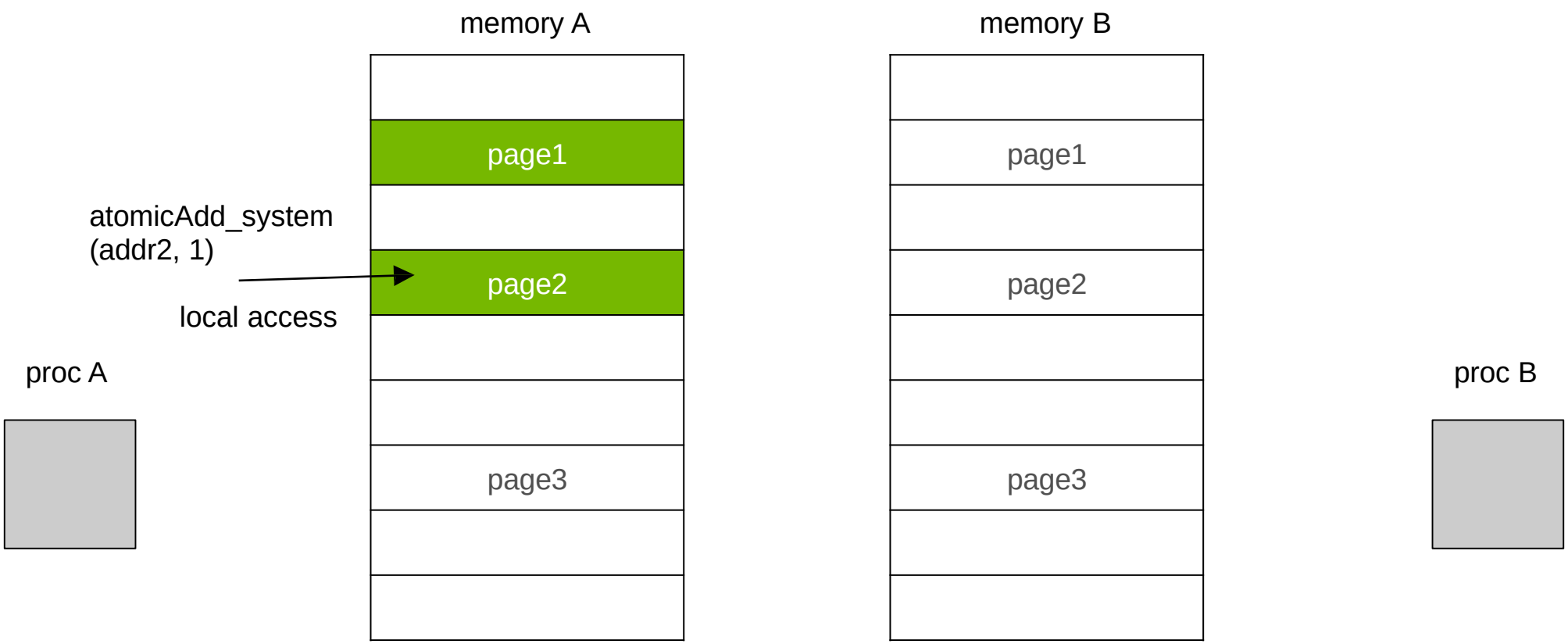
UNIFIED MEMORY FUNDAMENTALS

System-Wide Atomics with Exclusive Access



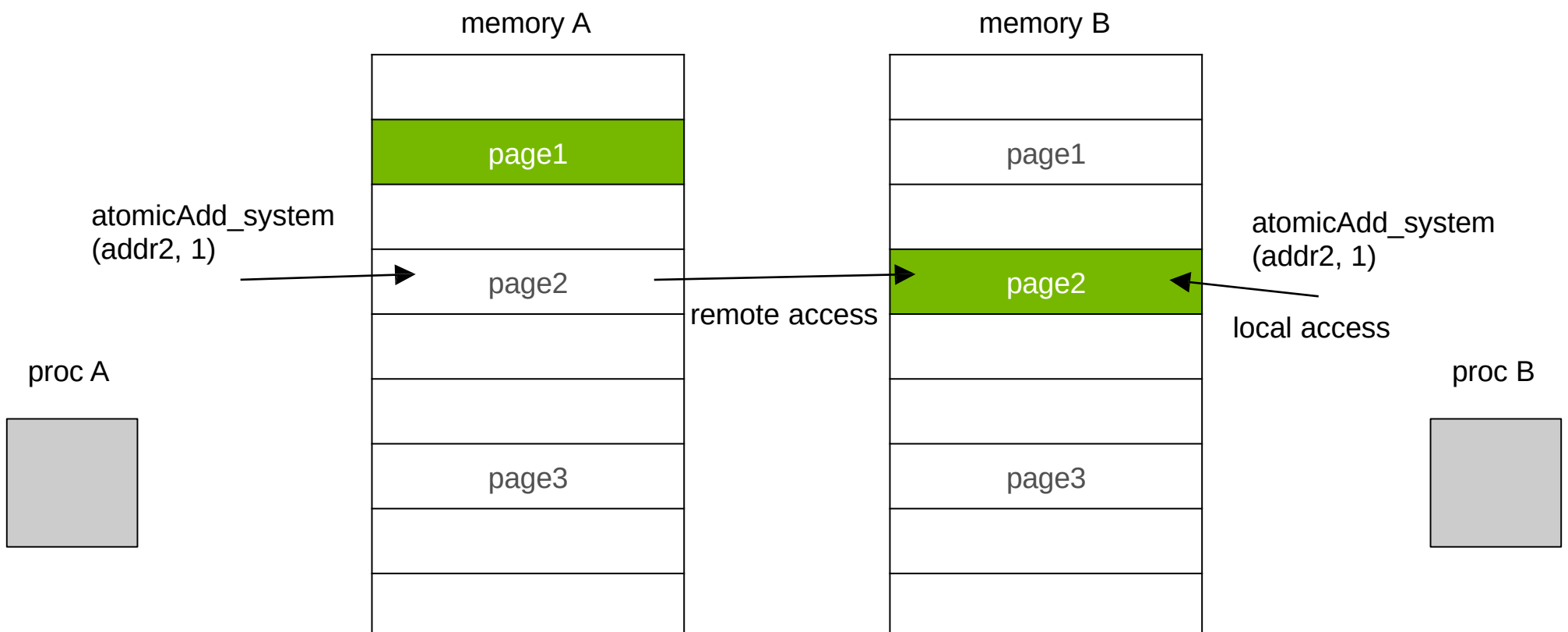
UNIFIED MEMORY FUNDAMENTALS

System-Wide Atomics with Exclusive Access



UNIFIED MEMORY FUNDAMENTALS

System-Wide Atomics over NVLINK*



*both processors need to support atomic operations

UNIFIED MEMORY FUNDAMENTALS

System-Wide Atomics

GPUs are very good at handling atomics from *thousands of threads*

Makes sense to utilize atomics between GPUs or between CPU and GPU

We will see this in action on a realistic example later on

AGENDA

Unified Memory Fundamentals

Under the Hood Details

Performance Analysis and Optimizations

Applications Deep Dive

UNIFIED MEMORY ALLOCATOR

Available Options

CUDA C: **cudaMallocManaged** is your most reliable way to opt in today

CUDA Fortran: **managed** attribute (per allocation)

OpenACC: **-ta=managed** compiler option (all dynamic allocations)

malloc support is coming on Pascal+ architectures (Linux only)

Note: you can write your own malloc hook to use cudaMallocManaged

HETEROGEENOUS MEMORY MANAGER

Work In Progress

Heterogeneous Memory Manager: a set of Linux kernel patches

Allows GPUs to access all system memory (malloc, stack, file system)

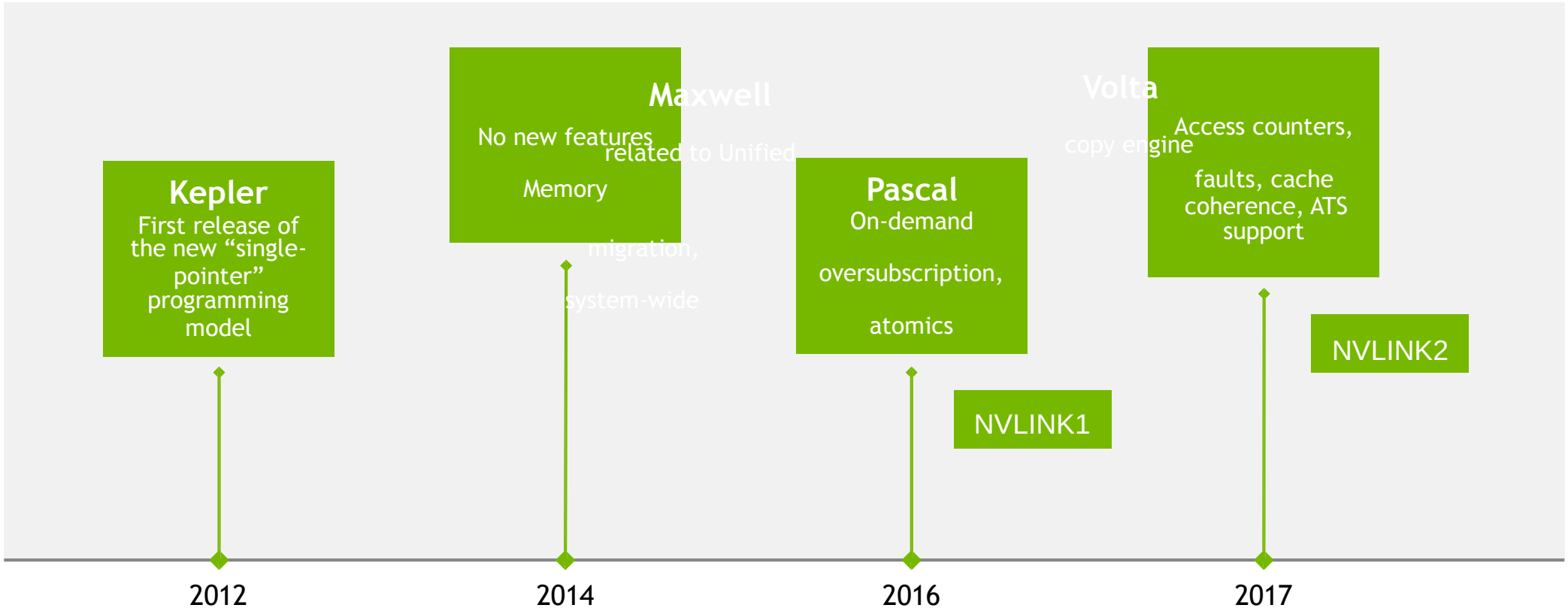
Page migration will be triggered the same way as for cudaMallocManaged

Ongoing testing and reviews, planning next phase of optimizations

More details on HMM today at 4:00 in Room 211B by John Hubbard

UNIFIED MEMORY

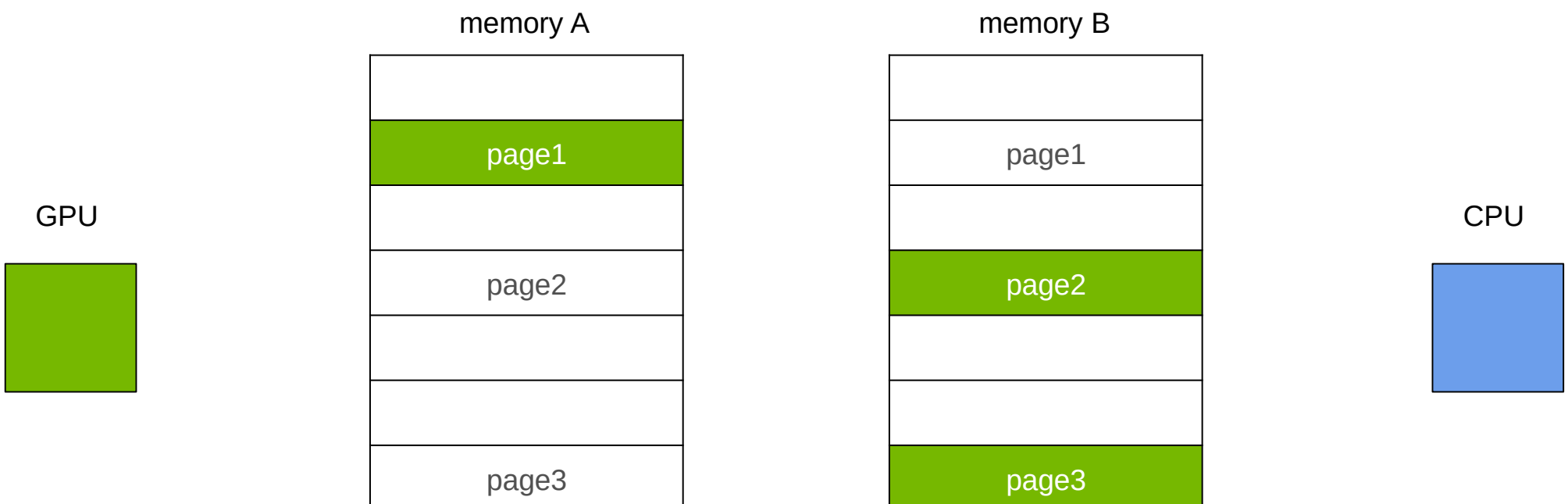
Evolution of GPU Architectures



UNIFIED MEMORY ON KEPLER

Available since CUDA 6

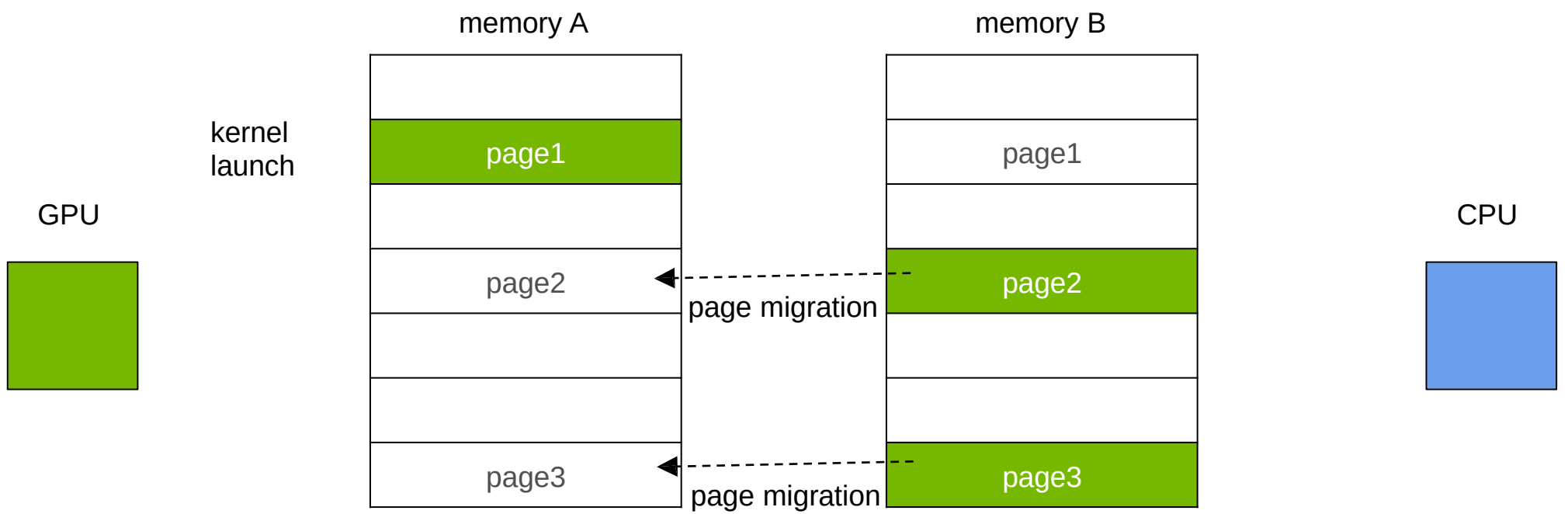
Kepler GPU: no page fault support, limited virtual space



UNIFIED MEMORY ON KEPLER

Available since CUDA 6

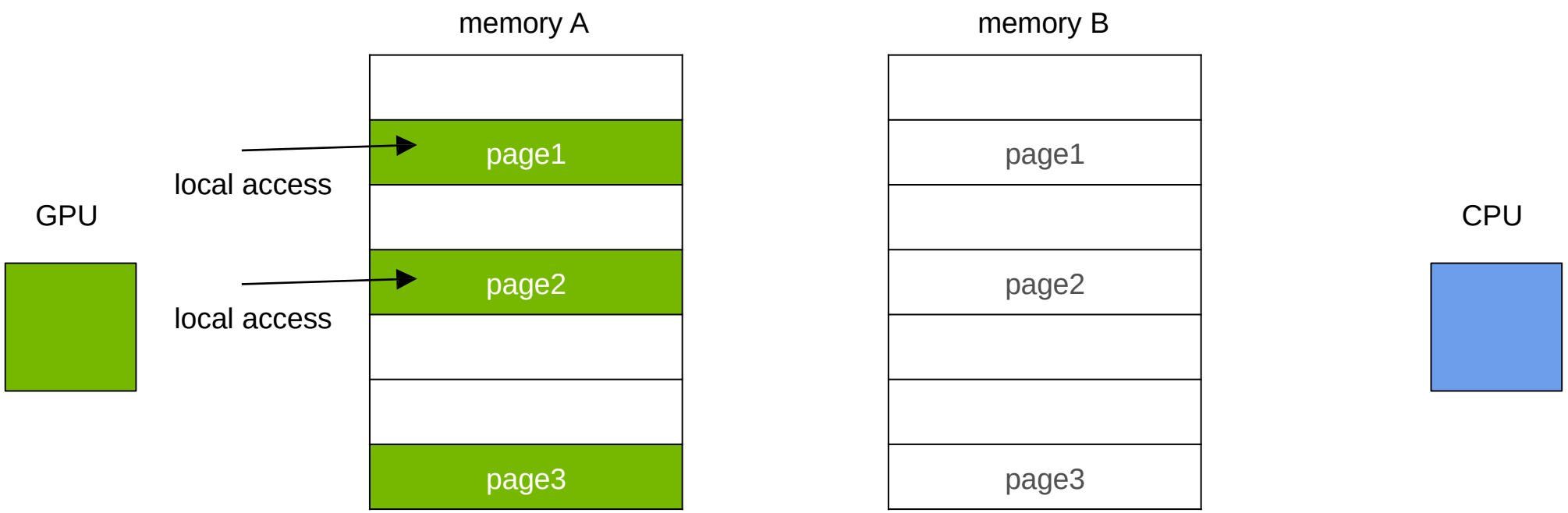
Bulk migration of **all pages** attached to current stream on kernel launch



UNIFIED MEMORY ON KEPLER

Available since CUDA 6

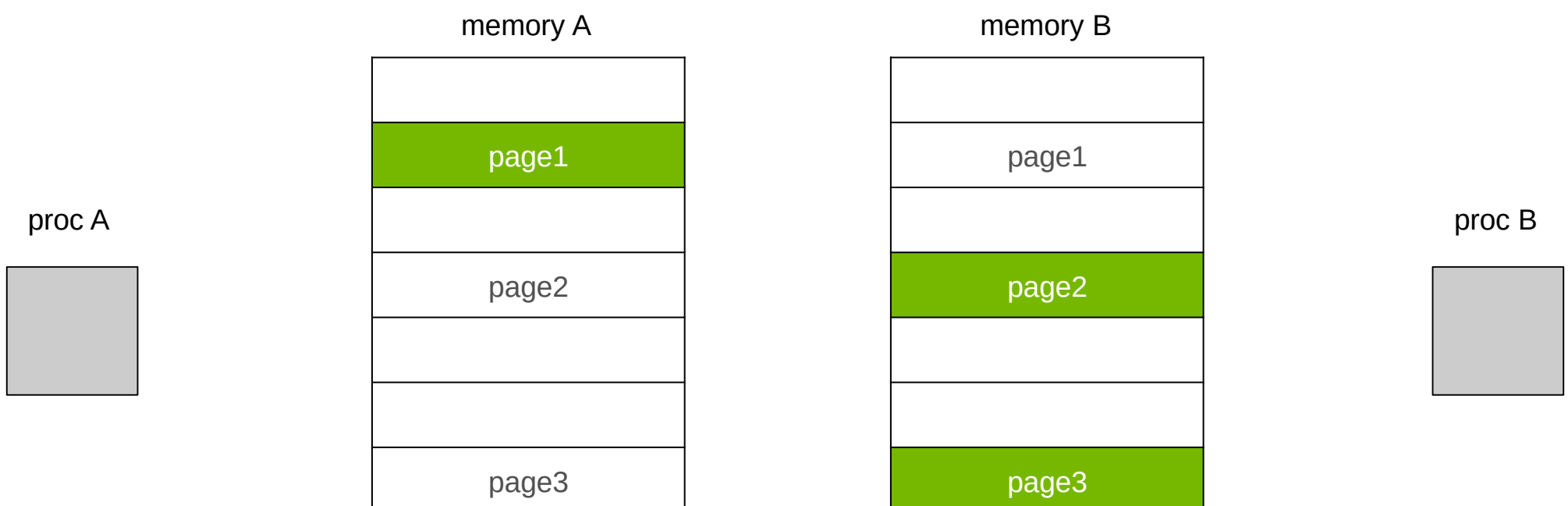
No on-demand migration for the GPU, no oversubscription, no system-wide atomics



UNIFIED MEMORY ON PASCAL

Available since CUDA 8

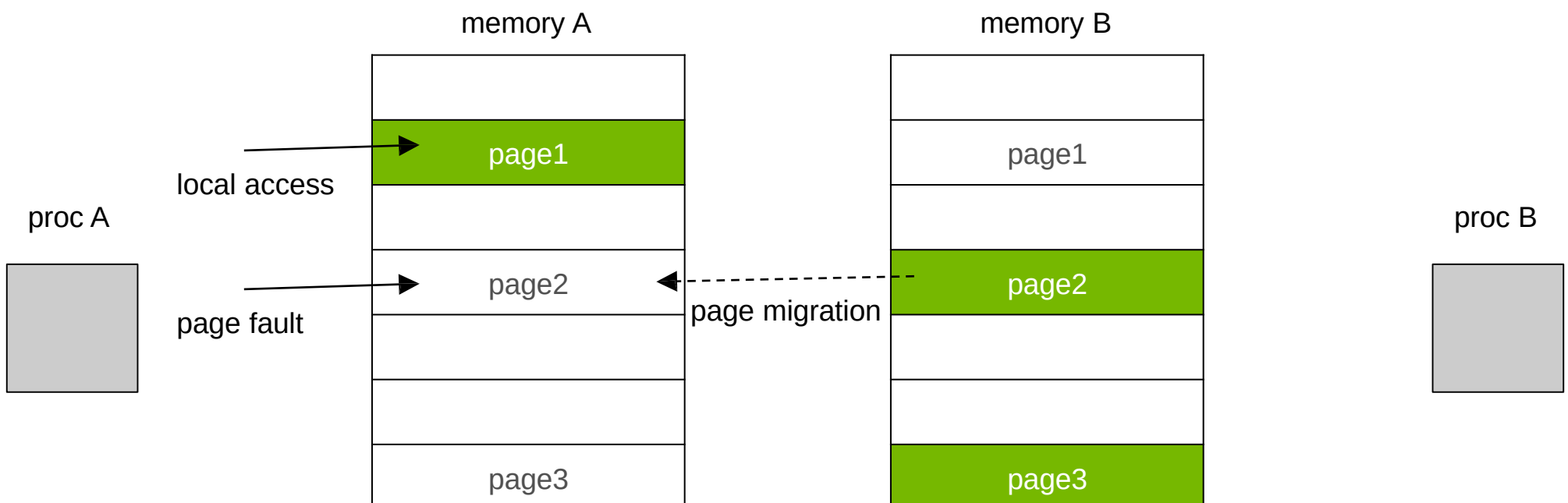
Pascal GPU: page fault support, extended virtual address space (48-bit)



UNIFIED MEMORY ON PASCAL

Available since CUDA 8

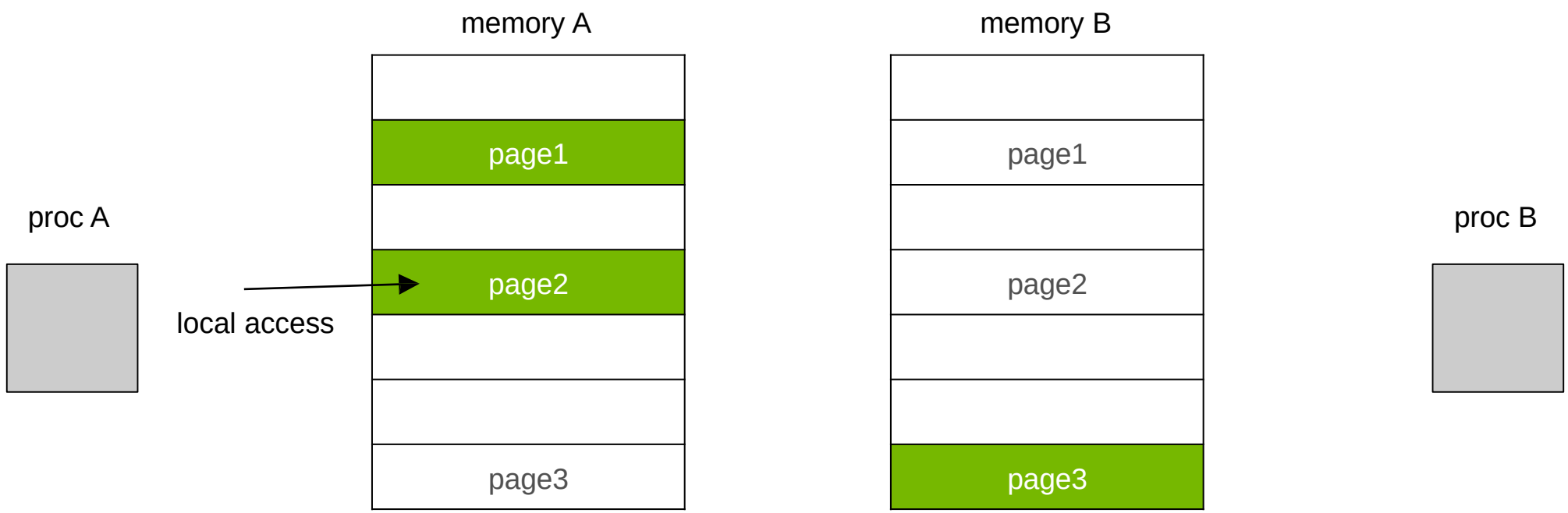
On-demand migration to accessing processor **on first touch**



UNIFIED MEMORY ON PASCAL

Available since CUDA 8

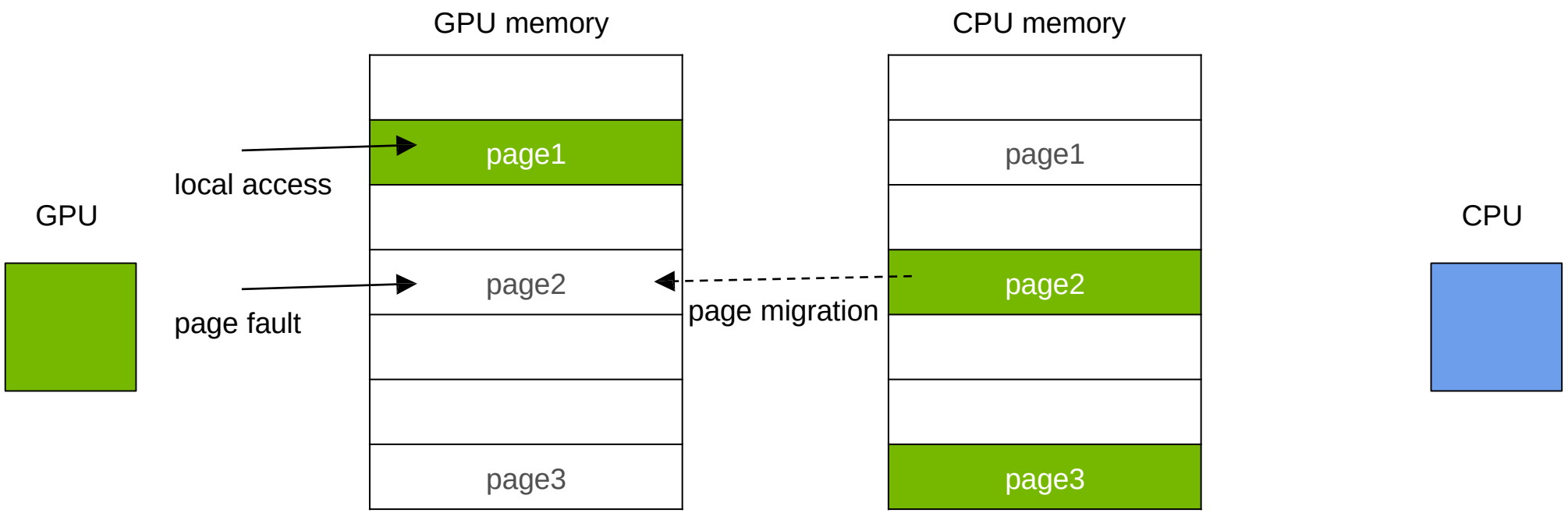
All features: on-demand migration, oversubscription, system-wide atomics



UNIFIED MEMORY ON VOLTA

Default model

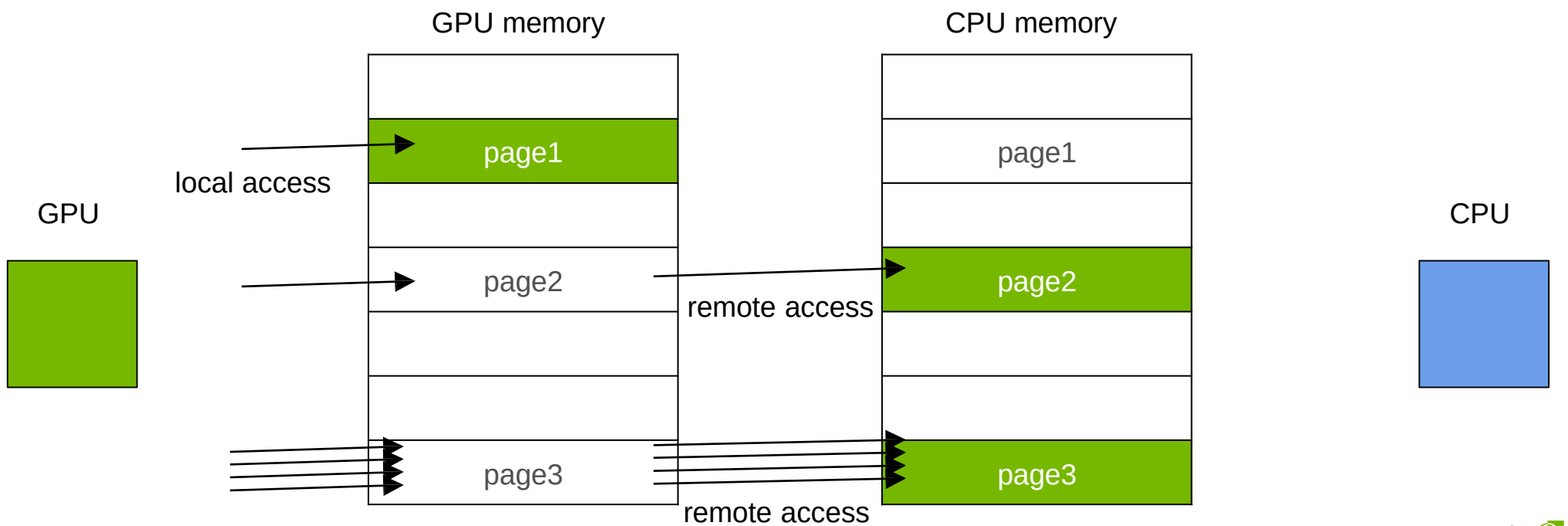
Volta GPU: uses fault on first touch for migration, **same as Pascal**



UNIFIED MEMORY ON VOLTA

New Feature: Access Counters

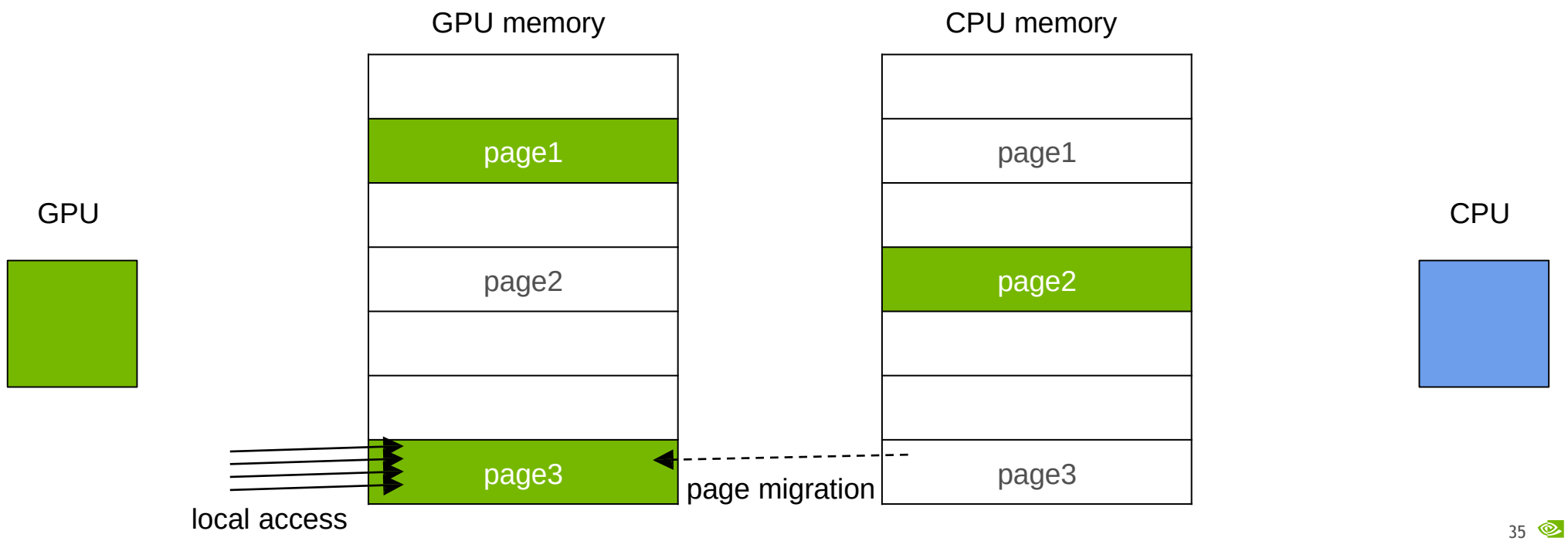
If memory is mapped to the GPU, migration can be triggered by access counters



UNIFIED MEMORY ON VOLTA

New Feature: Access Counters

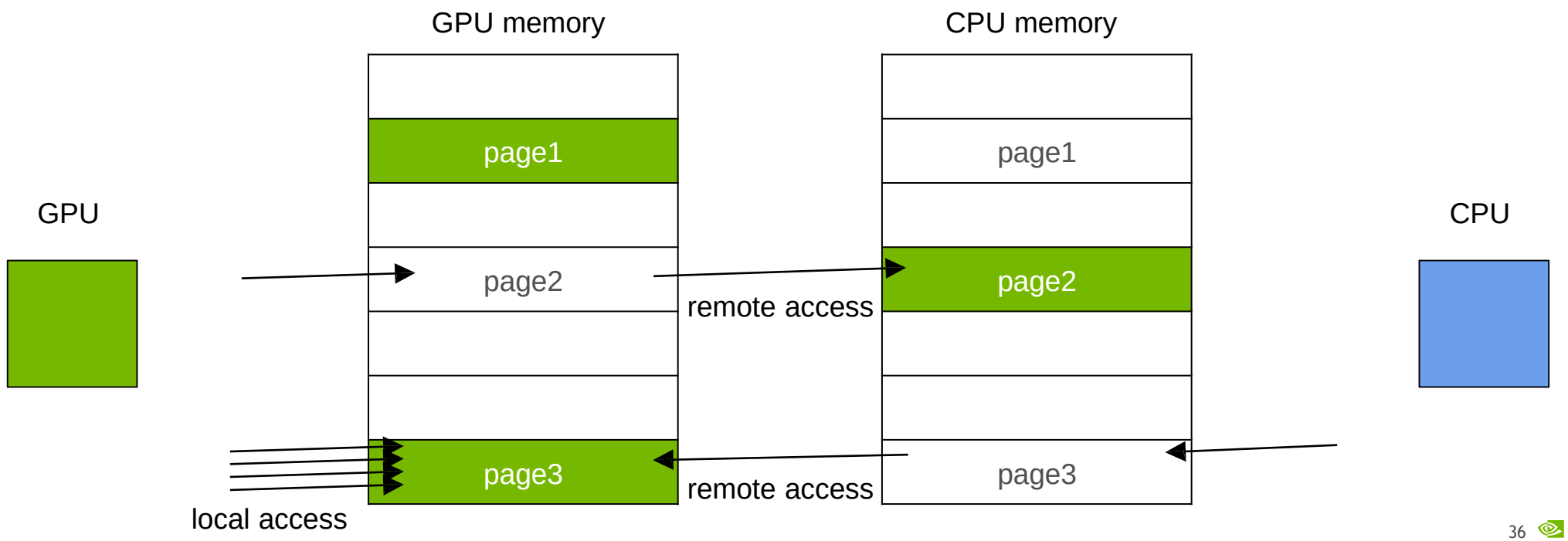
With access counters migration **only hot pages** will be moved to the GPU



UNIFIED MEMORY ON VOLTA+P9

NVLINK2: Cache Coherence

CPU can directly access and cache GPU memory; *native* CPU-GPU atomics



DRIVER HEURISTICS

Things You Didn’t Know Exist

The Unified Memory driver is doing intelligent things under the hood:

Prefetching: migrate pages proactively to reduce number of faults

Thrashing mitigation: heuristics to avoid frequent migration of shared pages

Eviction: what pages to evict when we need to make the room for new ones

You can’t control them but you can override most of these with hints

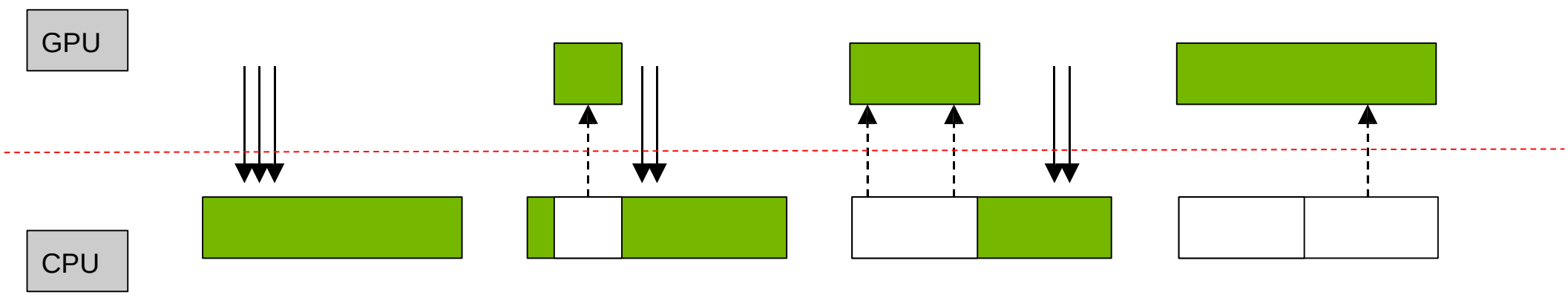
DRIVER PREFETCHING

Do Not Confuse with API-prefetching

GPU architecture supports different page sizes

Contiguous pages up to a larger page size are promoted to the larger size

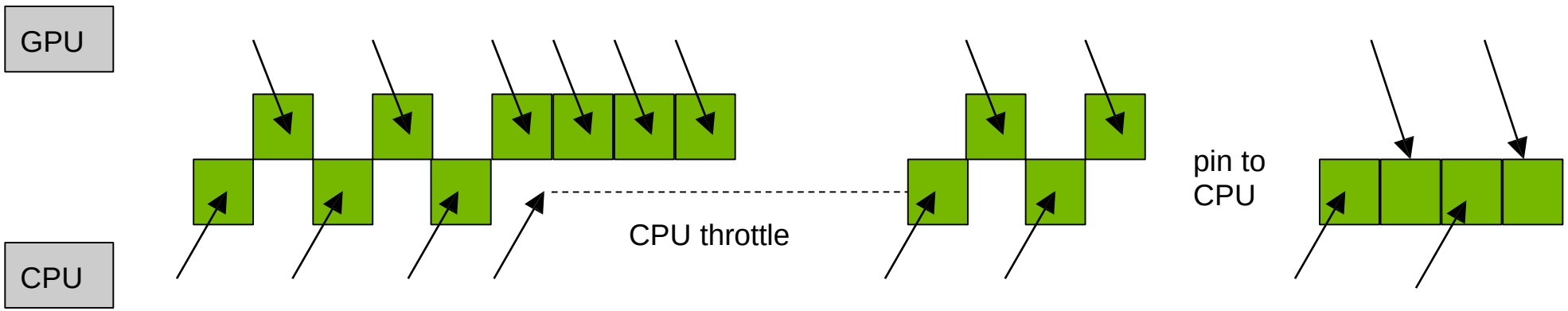
Driver prefetches whole regions if pages are accessed *densely*



ANTI-THRASHING POLICY

Frequent Access to Shared Data

Processors share the same page and frequently read or write to it



Pascal: when memory is pinned we lose any insight into access pattern

Volta: can use access counters information to find a better location

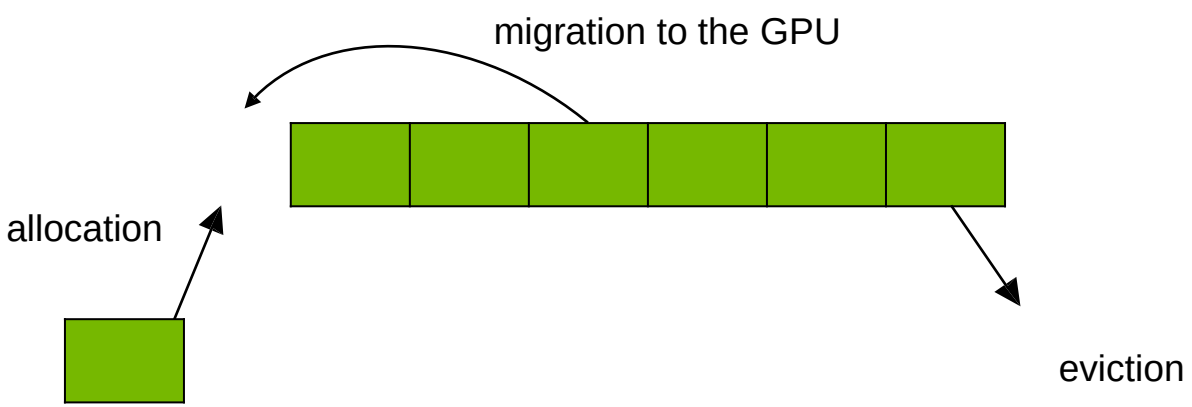
EVICTION ALGORITHM

What Pages Are Moving Out of the GPU

Driver keeps a single list of physical chunks of GPU memory

Chunks from the front of the list are evicted first (LRU)

A chunk is considered “in use” when it is fully-populated or migrated



AGENDA

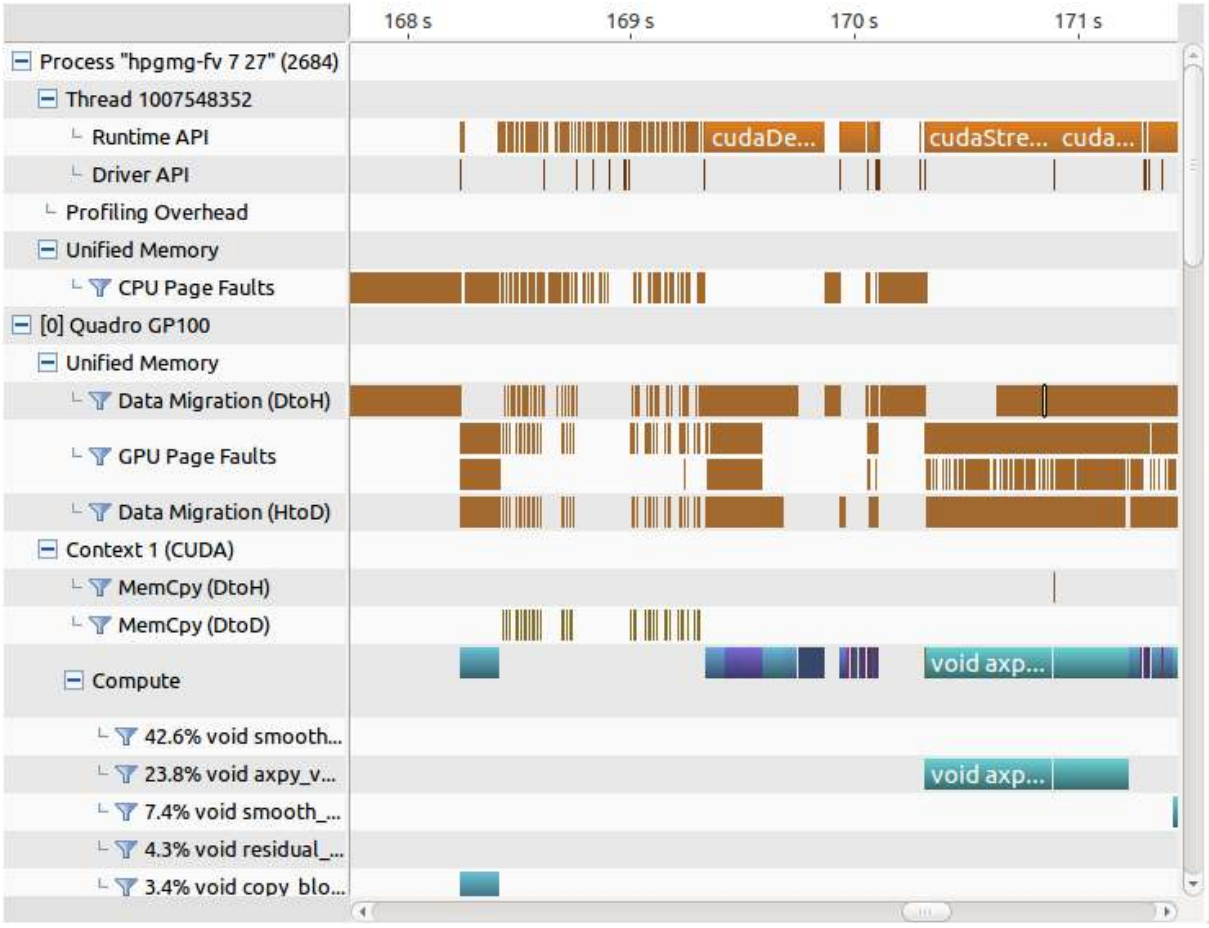
Unified Memory Fundamentals

Under the Hood Details

Performance Analysis and Optimizations

Applications Deep Dive

PROFILER: INSPECT



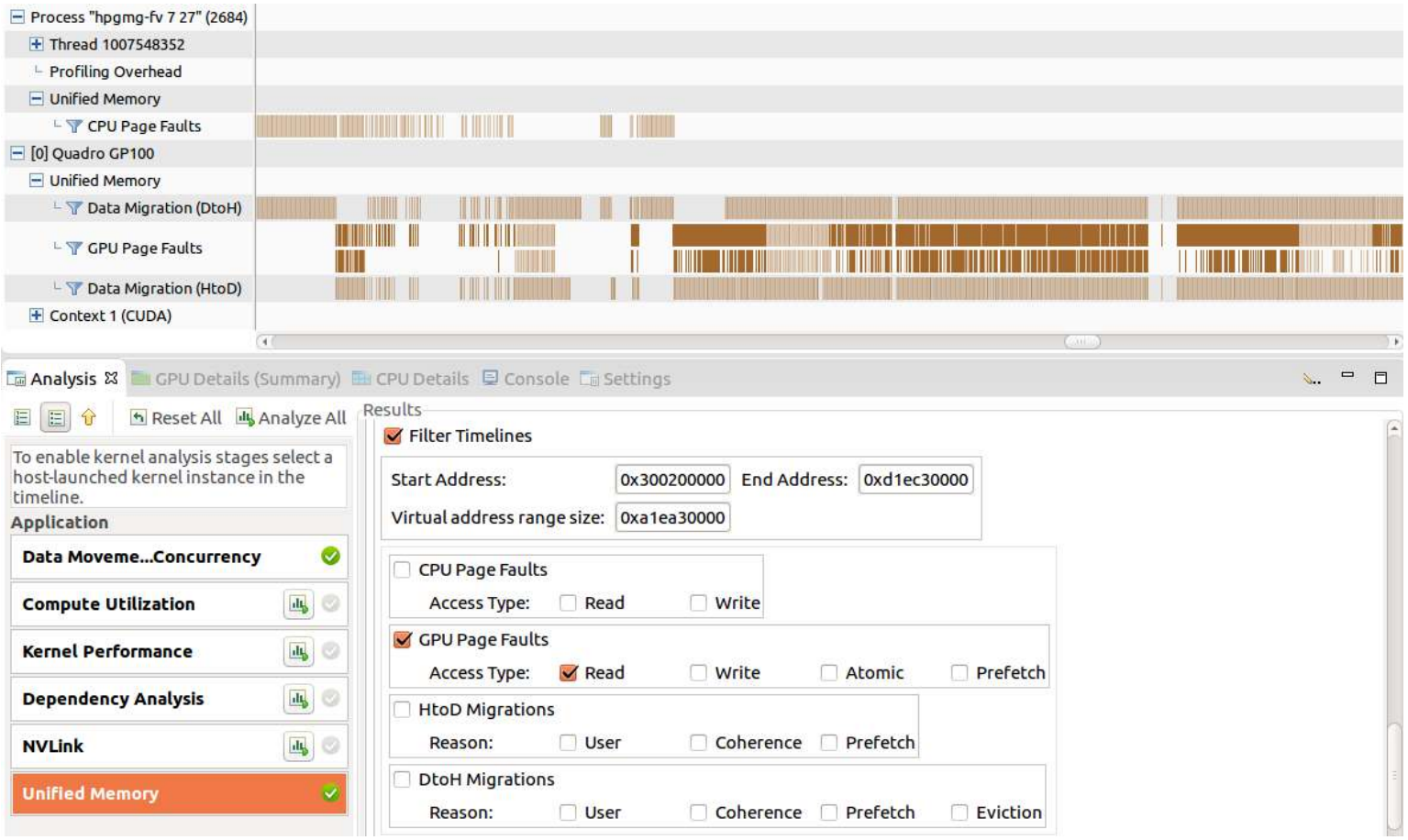
Data Migration (DtoH)

Start	170.845 s (170,844,511,496 ns)
End	170.845 s (170,844,671,976 ns)
Duration	160.48 µs
Size	2.097 MB
Migration Cause	Eviction
Throughput	13.068 GB/s
Virtual Address	0x4b2a00000
Memory operations	Browse for executable...
Process	2684

GPU Page Fault groups

Start	170.925 s (170,924,798,600 ns)
End	170.925 s (170,924,919,048 ns)
Duration	120.448 µs
Memory Acccess Type	Write
Virtual Address	0x34b680000
GPU Page Faults	2
Memory operations	Browse for executable...
Process	2684

PROFILER: FILTER



PROFILER: CORRELATE

*NewSession2

0.463 s 0.465 s

- Process "uvm_migration" (8021)
  - Thread 2816767808
    - Runtime API
    - Driver API
  - Profiling Overhead
  - Unified Memory
    - CPU Page Faults
- [0] Quadro GP100
  - Unified Memory
    - GPU Page Faults
    - Data Migration (HtoD)
  - Context 1 (CUDA)
    - Compute
      - 100.0% write(int*, u...
    - Streams
      - Default

Properties

Data Migration (HtoD)

Start	465.073 ms (465,072,848 ns)
End	465.15 ms (465,149,520 ns)
Duration	76.672 µs
Size	933.888 kB
Migration Cause	Prefetch
Throughput	12.18 GB/s
Virtual Address	0x50071c000
Memory operations	
Allocation	_Z13test_ondemandm@uvm_migration.cu:24
Free	_Z13test_ondemandm@uvm_migration.cu:40
Process	8021

uvm_migration.cu

```
  for(size_t tid = threadIdx.x + blockIdx.x * blockDim.x; tid < n;
    a[tid] = 2;
}

void test_ondemand(size_t s)
{
  int *a;
  cudaMallocManaged(&a, s);
  size_t n = s / sizeof(int);
  for (size_t i = 0; i < n; i++) a[i] = 1; // populate on the cpu

  struct timespec start, end;
  clock_gettime(CLOCK_MONOTONIC, &start);

  int block = 256;
  int grid = min((size_t)32 * 1024, (n+255)/block);
  write<<<grid, block>>>(a, n); // on-demand migration to the gpu
  cudaDeviceSynchronize();

  clock_gettime(CLOCK_MONOTONIC, &end);
  uint64_t t = get_time(start, end);
  printf("%.3f GB/s\n", (double)s/t);

  cudaFree(a);
}
```

More details tomorrow at 10:00 in Marriott Salon 3

USER HINTS

Why, When, and How to Use Them

If you know your application well you can optimize with hints

These are also useful to override some of the driver heuristics

`cudaMemPrefetchAsync``(ptr, size, processor, stream)`

Similar to `move_pages()` in Linux

`cudaMemAdvise``(ptr, size, advice, processor)`

Similar to `madvise()` in Linux

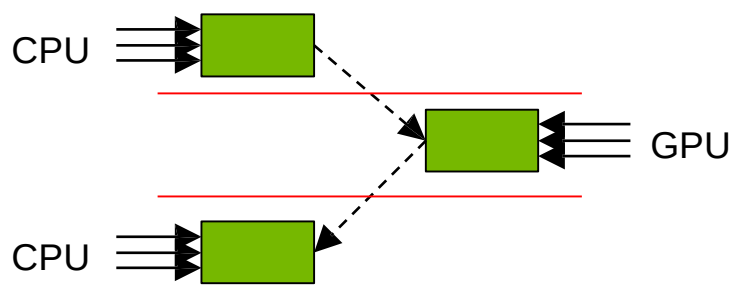
USER HINTS

Prefetching

```
char *data;
cudaMallocManaged(&data, N);

init_data(data, N);

cudaMemPrefetchAsync(data, N, myGpuId, s);
mykernel<<<..., s>>>(data, N);
cudaMemPrefetchAsync(data, N, cudaCpuDeviceId, s);
cudaStreamSynchronize(s);

use_data(data, N);

cudaFree(data);
```

Page faults can be expensive and they stall SM execution

Avoid faults by prefetching data to the accessing processor



USER HINTS

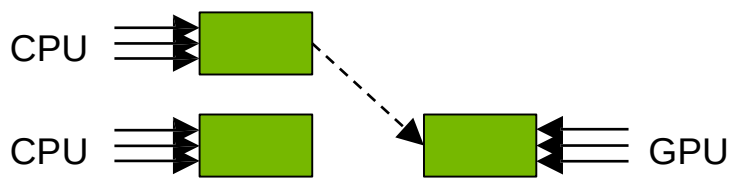
Read Mostly

```
char *data;
cudaMallocManaged(&data, N);

init_data(data, N);

cudaMemAdvise(data, N, ..SetReadMostly, myGpuId);
cudaMemPrefetchAsync(data, N, myGpuId, s);
mykernel<<<..., s>>>(data, N);


use_data(data, N);

cudaFree(data);
```

In this case prefetch creates a copy instead of moving data

Both processors can read data **simultaneously** without faults

Writes are allowed but they are expensive



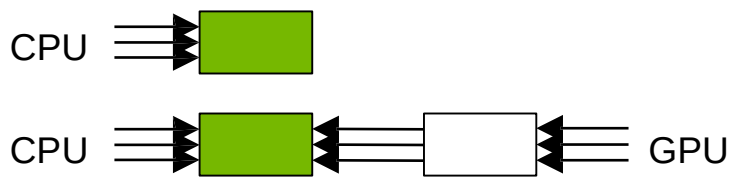
USER HINTS

Preferred Location

```
char *data;
cudaMallocManaged(&data, N);

init_data(data, N);

cudaMemAdvise(data, N, ..PreferredLocation, cudaCpuDeviceId);

mykernel<<<..., s>>>(data, N);


use_data(data, N);

cudaFree(data);
```

Here the kernel will *page fault* and generate direct mapping to data on the CPU

The driver will “resist” migrating data away from the preferred location



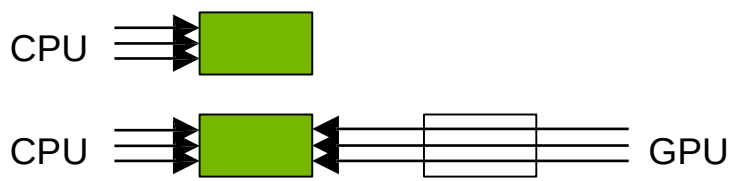
USER HINTS

Accessed By

```
char *data;
cudaMallocManaged(&data, N);

init_data(data, N);

cudaMemAdvise(data, N, ..SetAccessedBy, myGpuId);

mykernel<<<..., s>>>(data, N);


use_data(data, N);

cudaFree(data);
```

GPU will establish direct mapping of data in CPU memory, **no page faults** will be generated

Memory can move freely to other processors and mapping will carry over



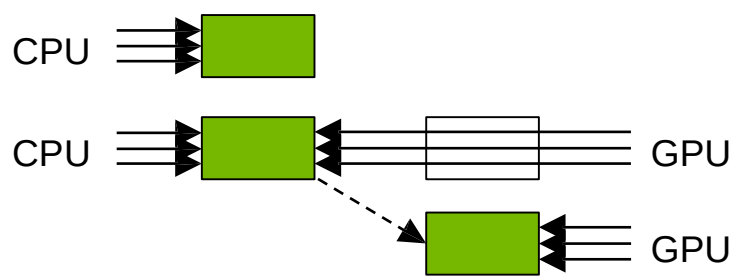
USER HINTS

Accessed By on Volta

```
char *data;
cudaMallocManaged(&data, N);

init_data(data, N);

cudaMemAdvise(data, N, ..SetAccessedBy, myGpuId);

mykernel<<<..., s>>>(data, N);


use_data(data, N);

cudaFree(data);
```

GPU will establish direct mapping of data in CPU memory, **no page faults** will be generated

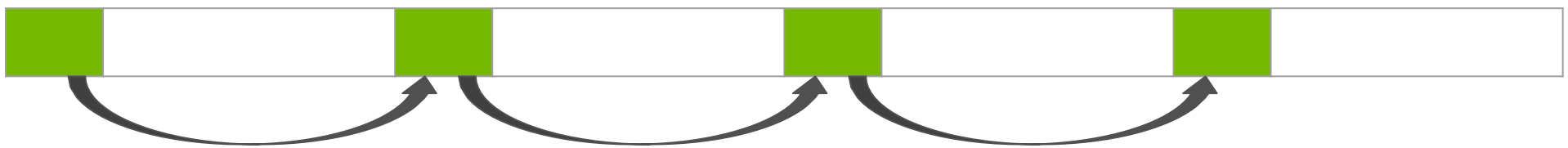
Access counters may eventually trigger migration of this memory to the GPU



PERFORMANCE

Page Fault Cost

How long does a page fault take to serve? - We can measure!

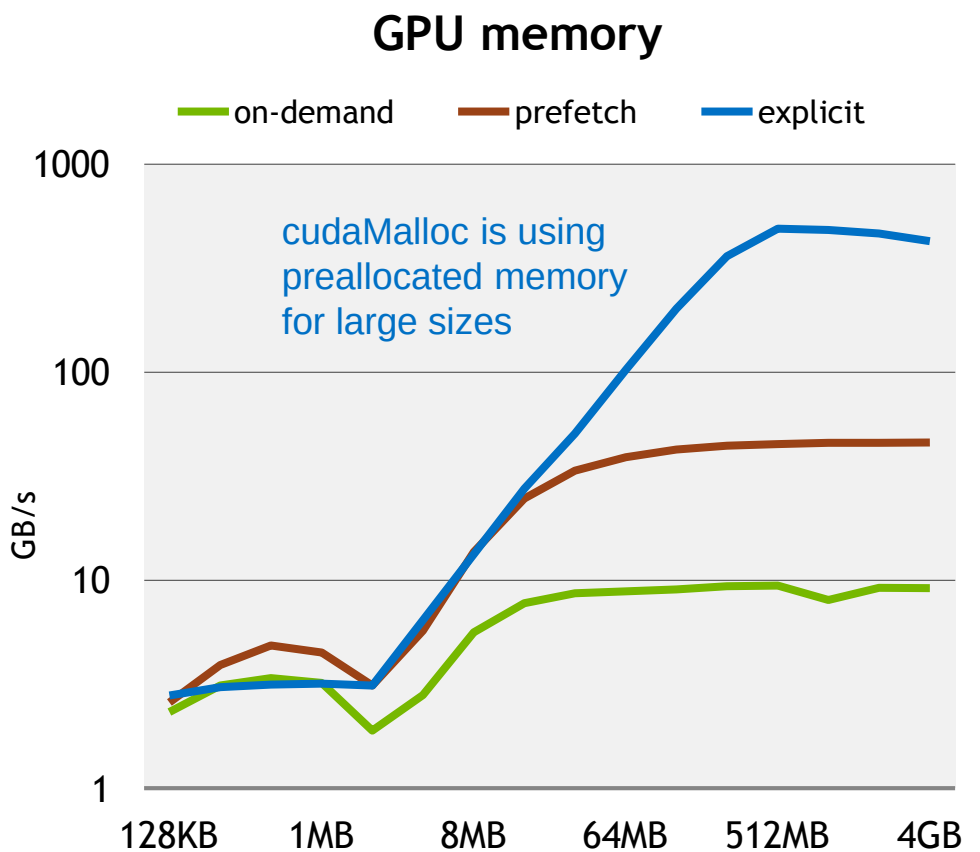


Linked list traversal with some large stride to avoid prefetching effects

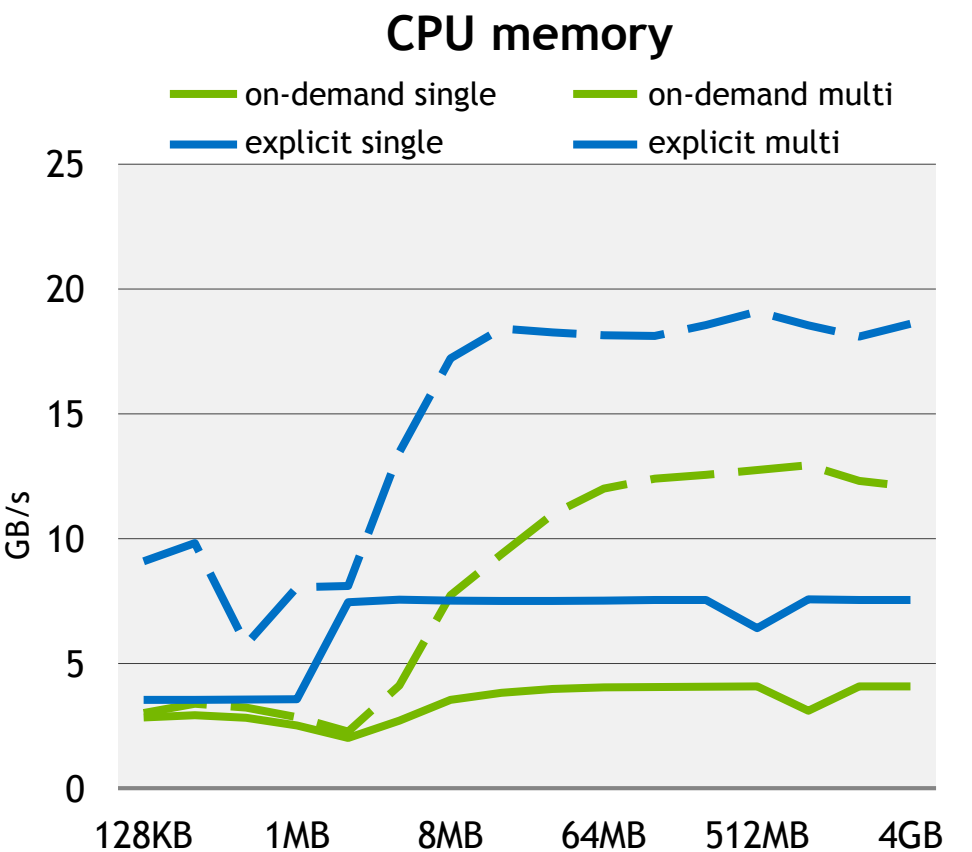
Page fault cost (us)	DtoH	HtoD
x86 + PCIe + GP100	20	30
P8 + NVLINK + GP100	20	20

PERFORMANCE

Page Allocation Throughput



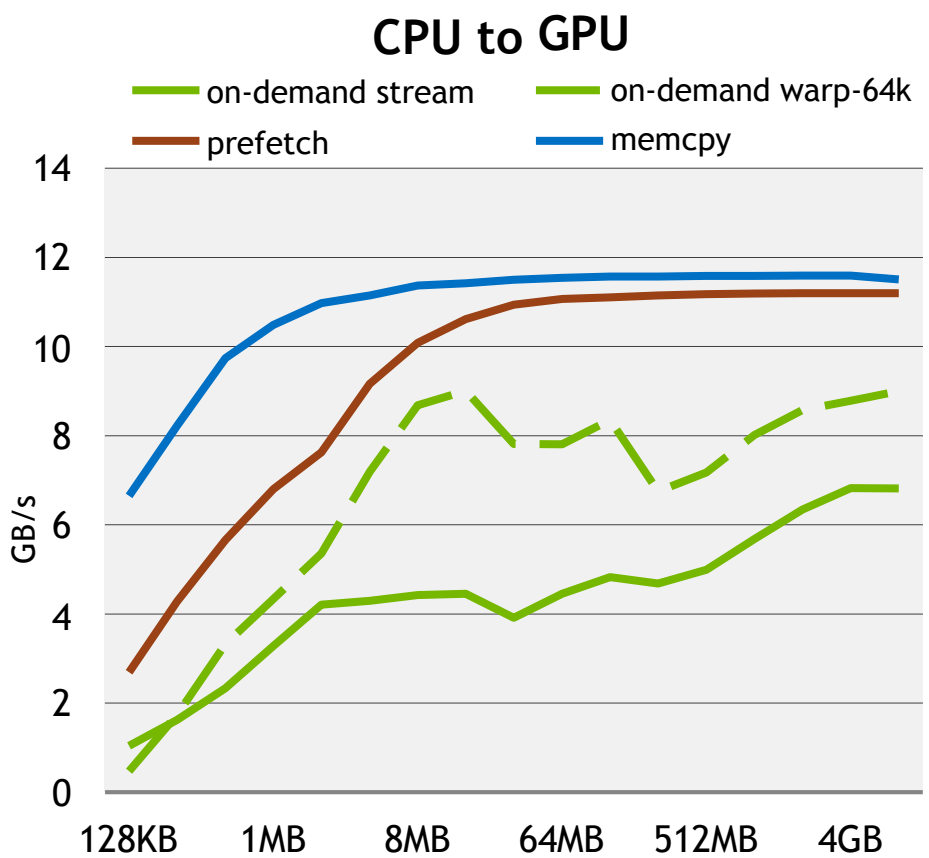
cudaMallocManaged/cudaMalloc + cudaMemset

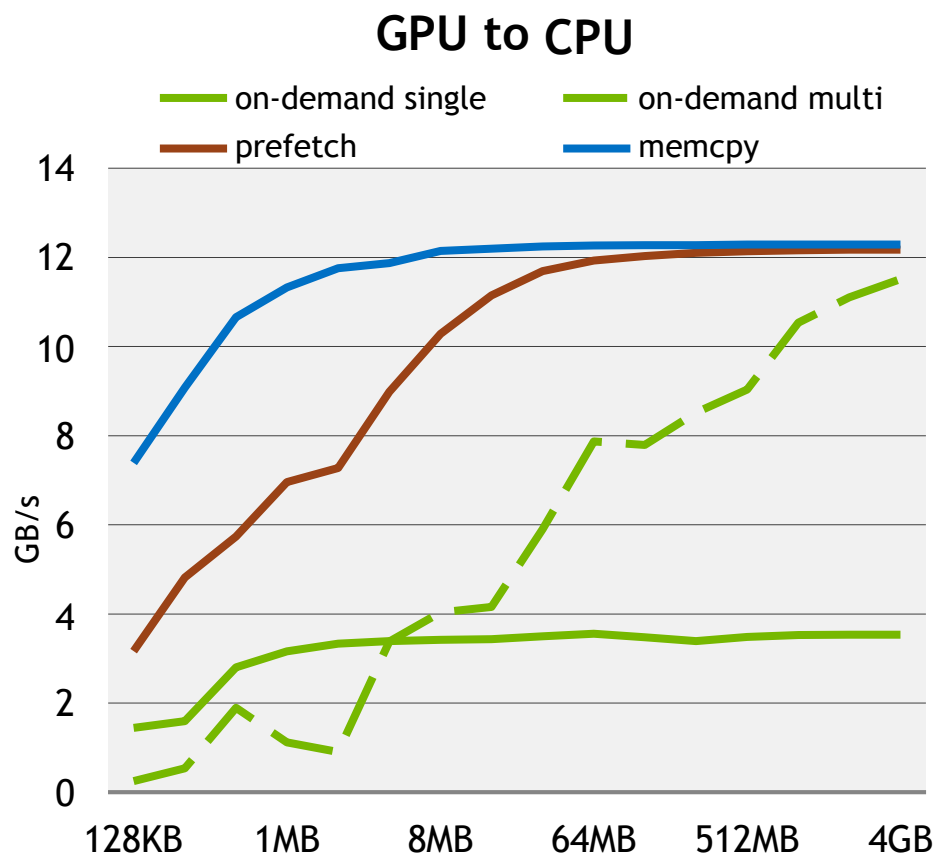


cudaMallocManaged/mmap + fill on the CPU

PERFORMANCE

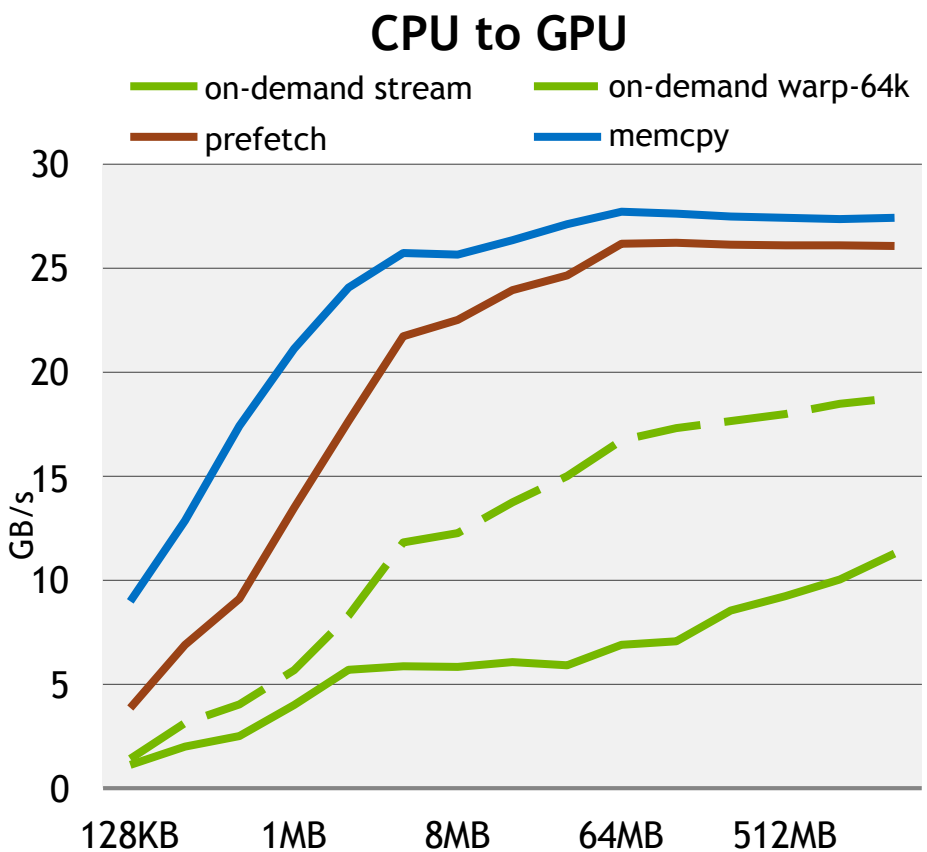
Page Migration Throughput (PCIe)

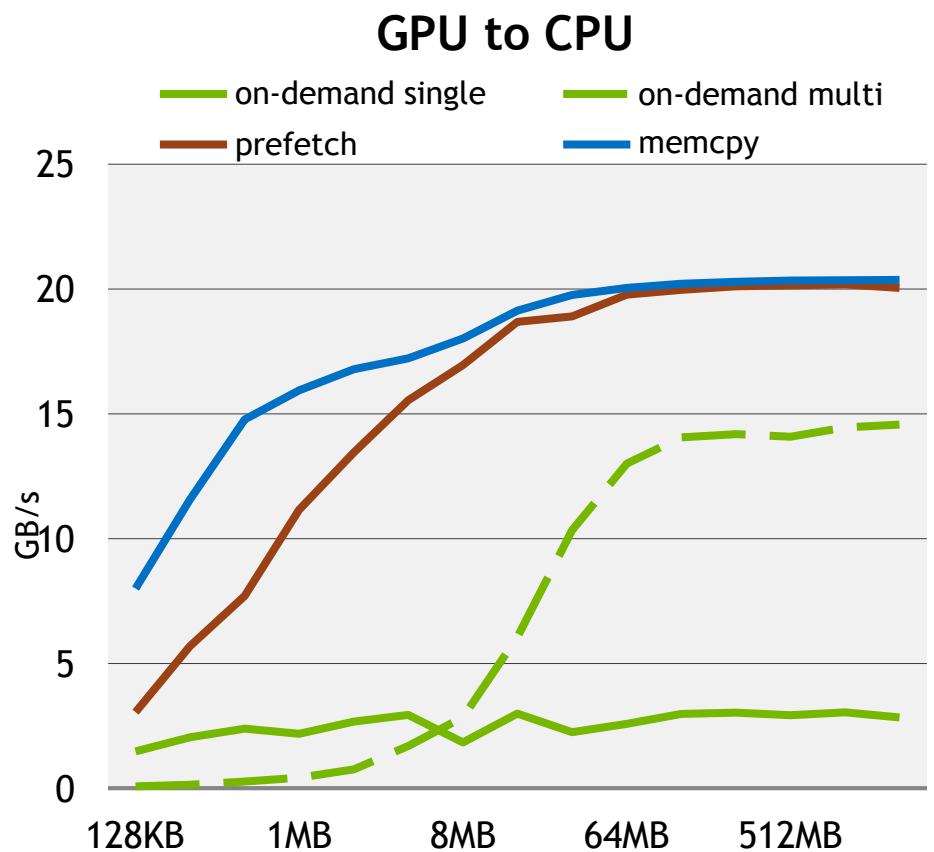




PERFORMANCE

Page Migration Throughput (2x NVLINK)





PERFORMANCE

Page Granularity Overhead

cudaMallocManaged *alignment*: 512B on Pascal/Volta, 4KB on Kepler/Maxwell

Too many small allocations will use up many pages

cudaMallocManaged memory is moved at *system page* granularity

For small allocations more data could be moved than necessary

Solution: use cached allocator or memory pools

AGENDA

Unified Memory Fundamentals

Under the Hood Details

Performance Analysis and Optimizations

Applications Deep Dive

HPC: HPGMG

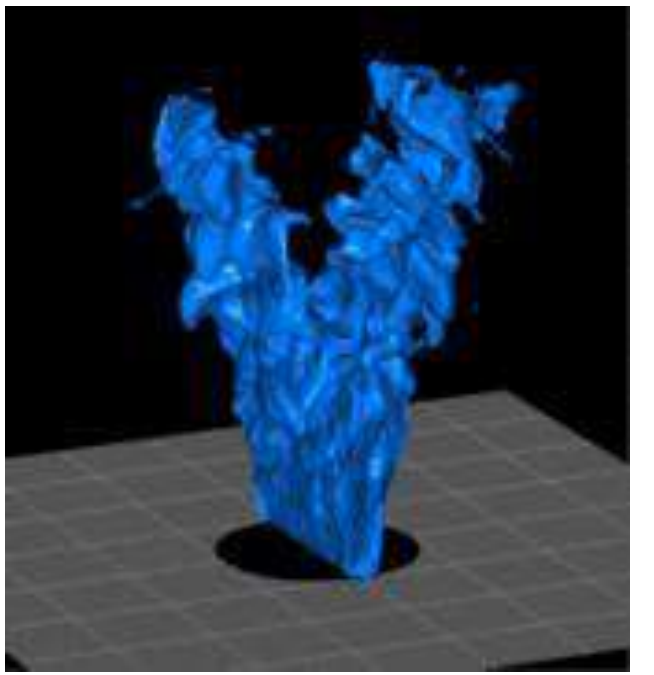
Combustion Simulation

High-Performance Geometric Multigrid

Proxy AMR and Low Mach Combustion codes

Used in Top500 benchmarking

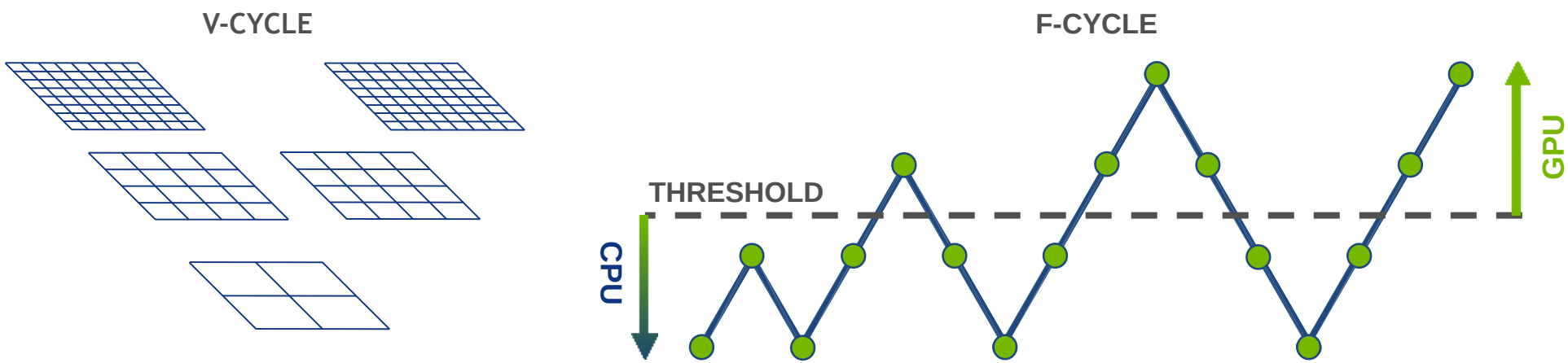
High memory usage requirements



http://crd.lbl.gov/departments/computer-science/PAR/research/hpgmg/

HPC: HPGMG

Taking Advantage of the CPU and the GPU

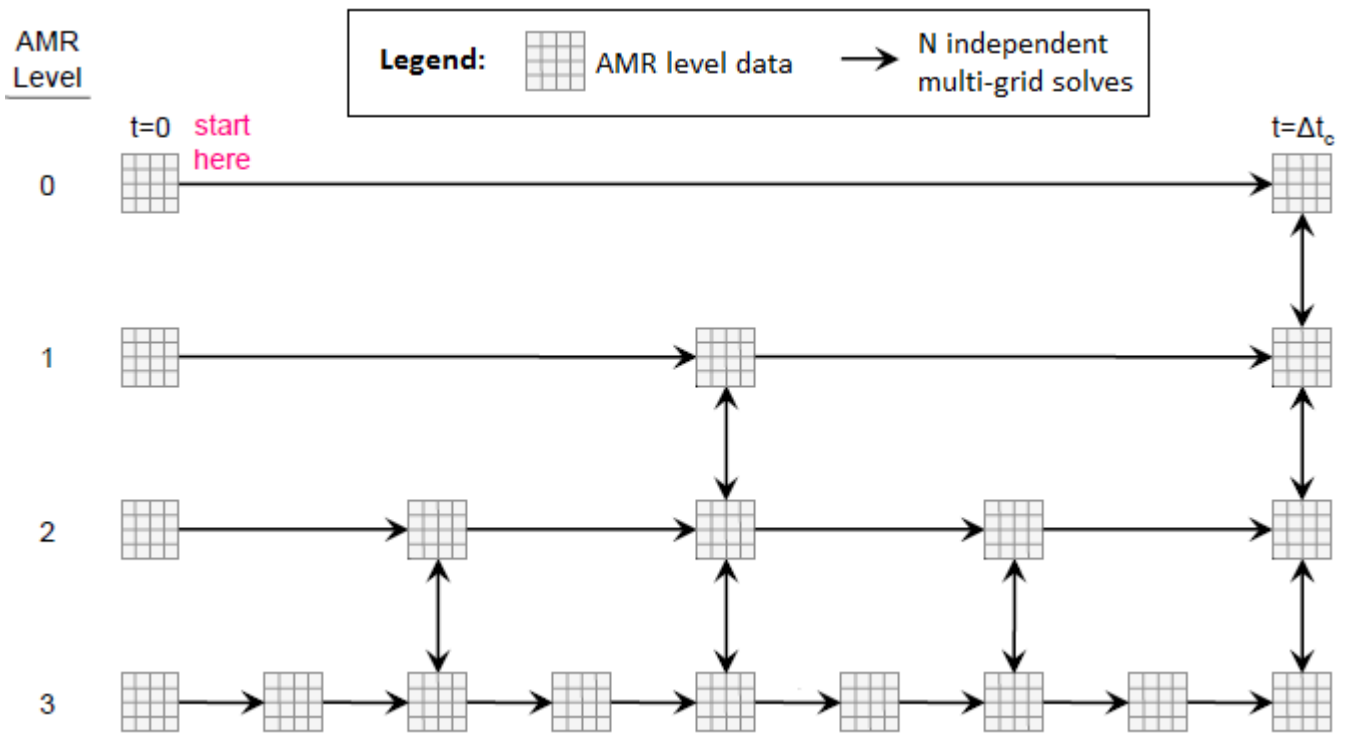


Hybrid implementation requires very careful memory management

Frequent data sharing when crossing the CPU-GPU threshold

HPGMG: AMR PROXY

Data Locality and Reuse of AMR Levels

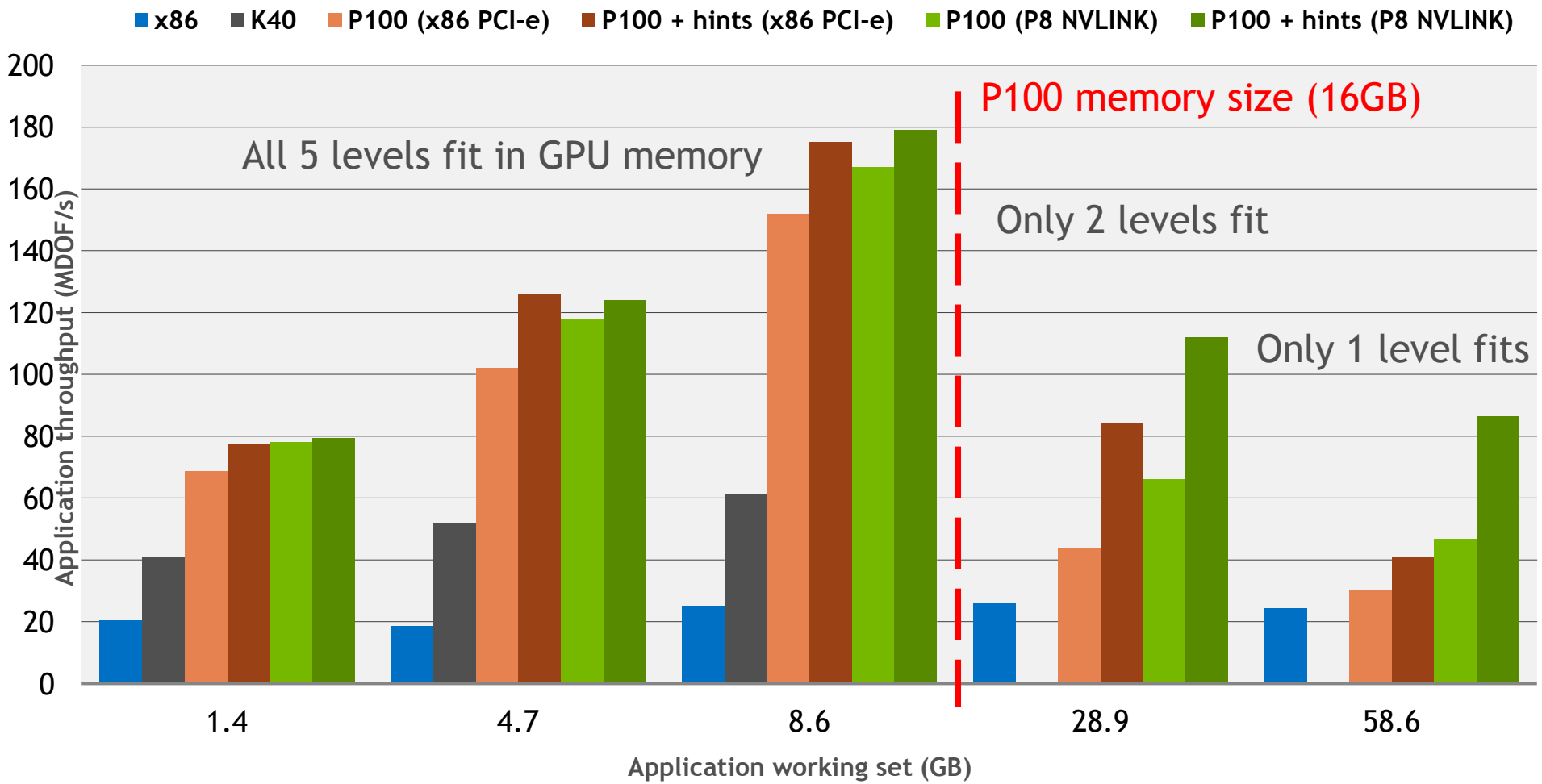


Optimization: prefetch the next AMR level while running computations on the current level

We can use a separate non-blocking CUDA stream to overlap with the default stream

https://devblogs.nvidia.com/parallelforall/beyond-gpu-memory-limits-unified-memory-pascal/

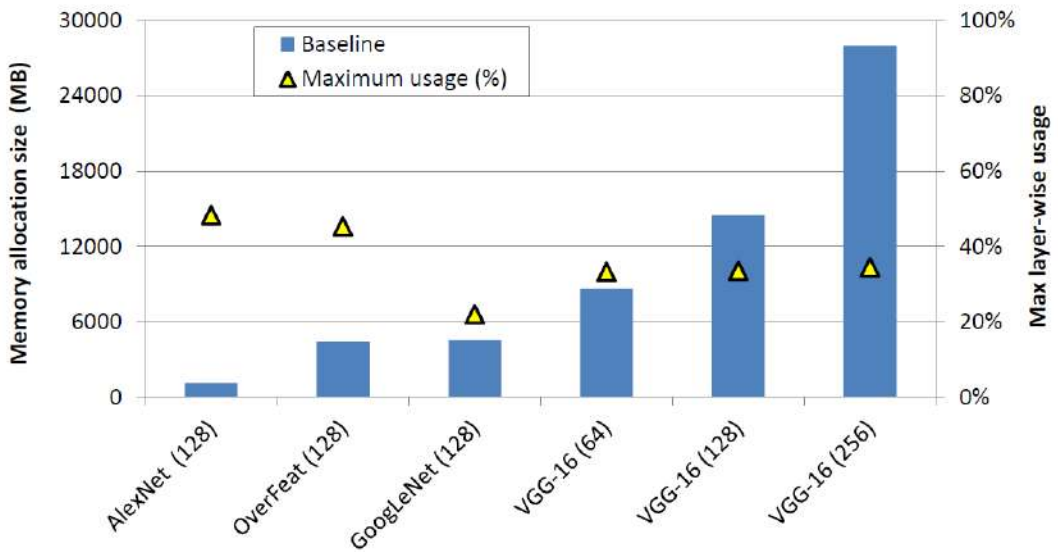
AMR PROXY OVERSUBSCRIPTION



***x86 CPU**: Intel E5-2630 v3, 2 sockets of 10 cores each with HT on (40 threads)*

DEEP LEARNING

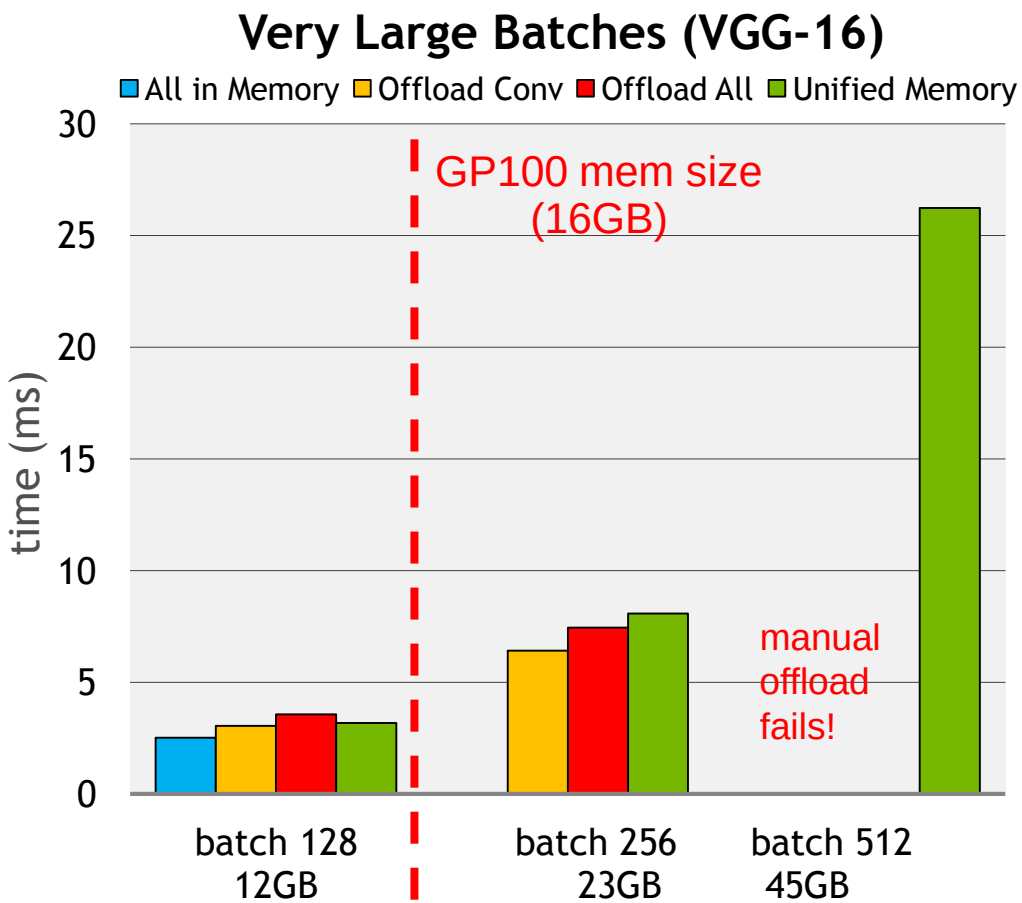
vDNN: Virtualized DNN for Scalable, Memory-Efficient Neural Network Design

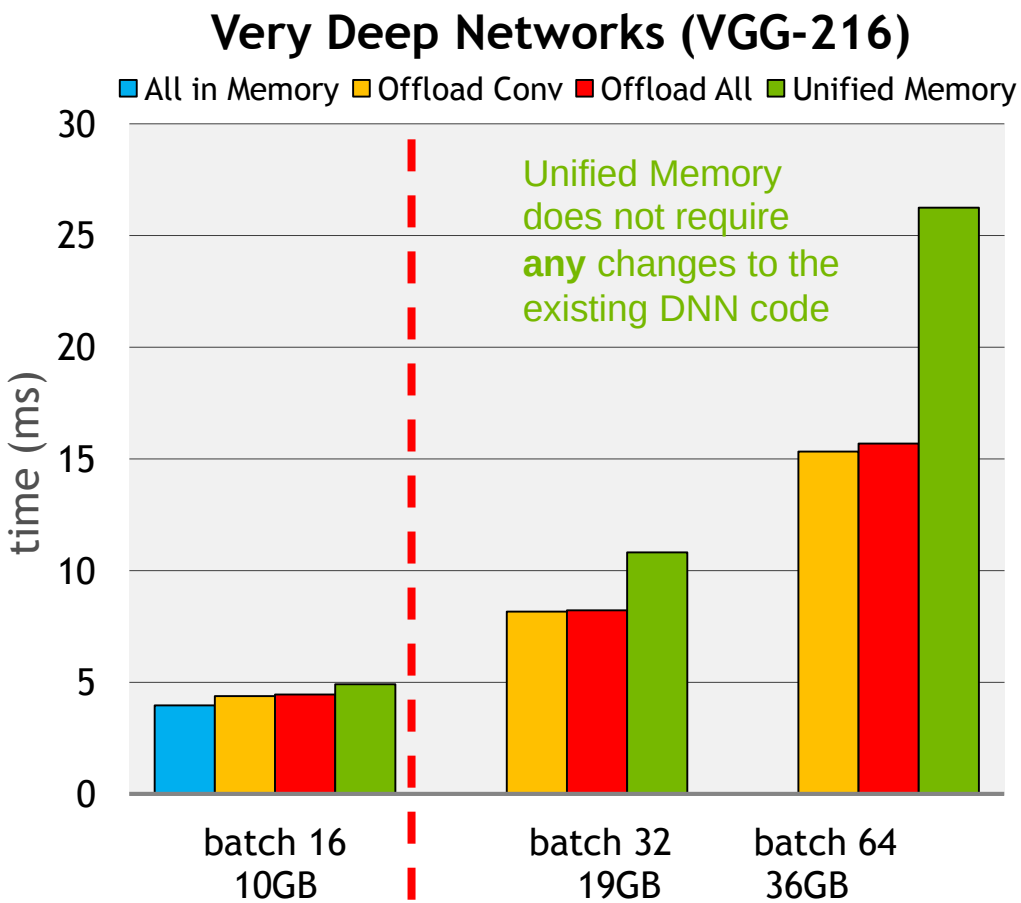


Original version implemented custom heuristics to prefetch and offload data

Unified Memory can automatically migrate memory as needed!

DEEP LEARNING OVERSUBSCRIPTION

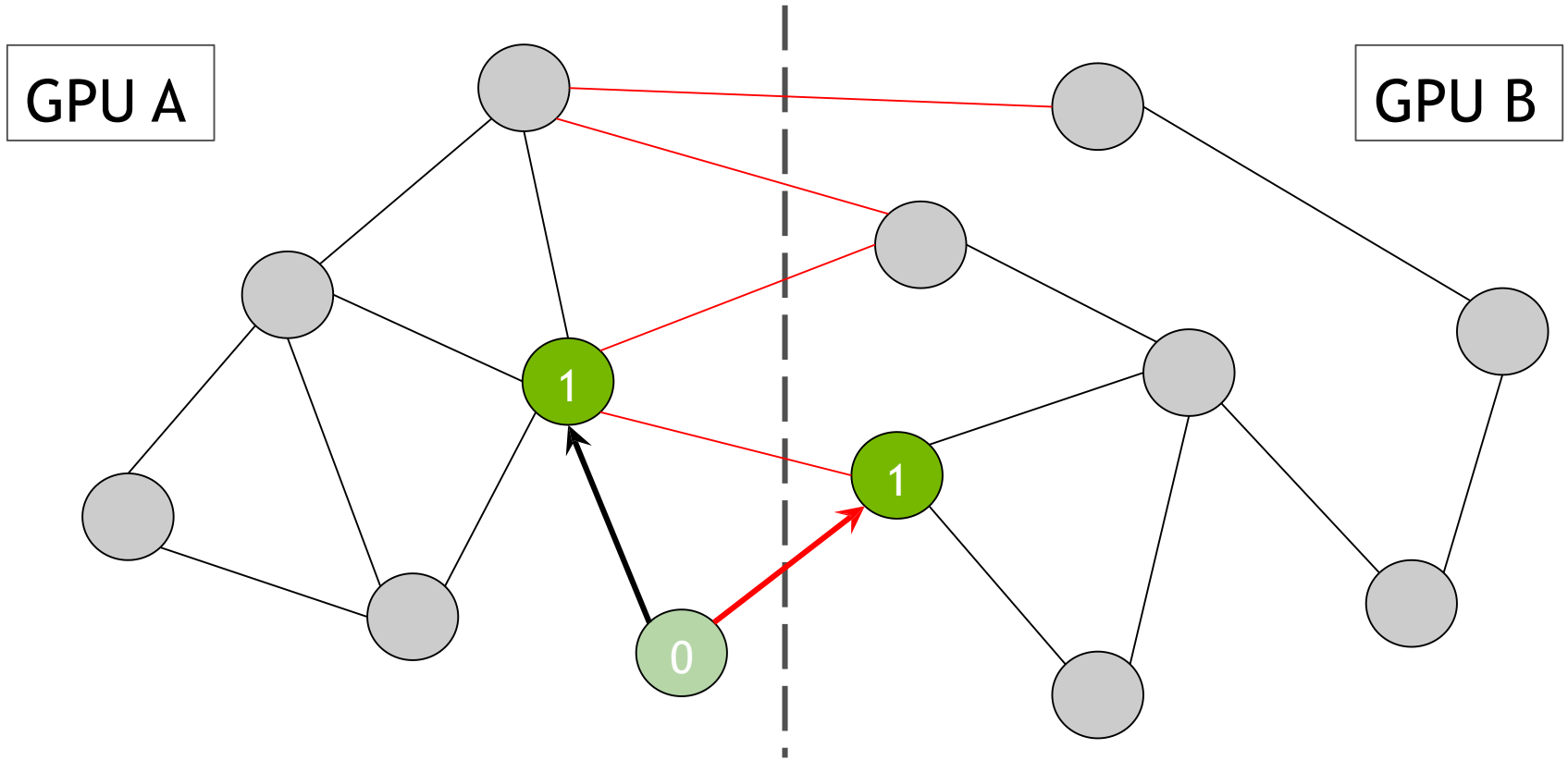




GPU: NVIDIA Quadro GP100; cuDNN 5.1, CUDA 9

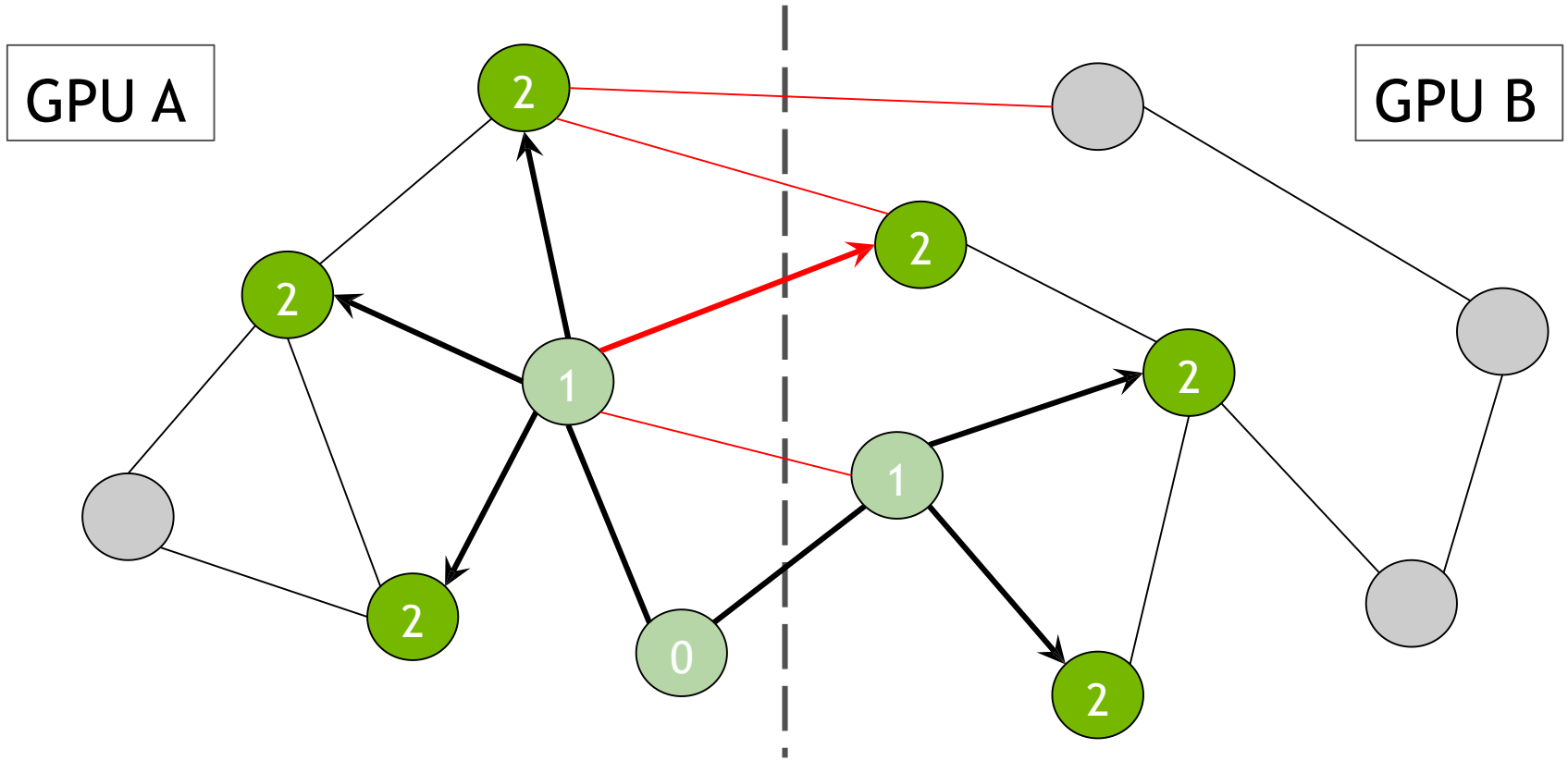
GRAPH ANALYTICS

BFS Traversal



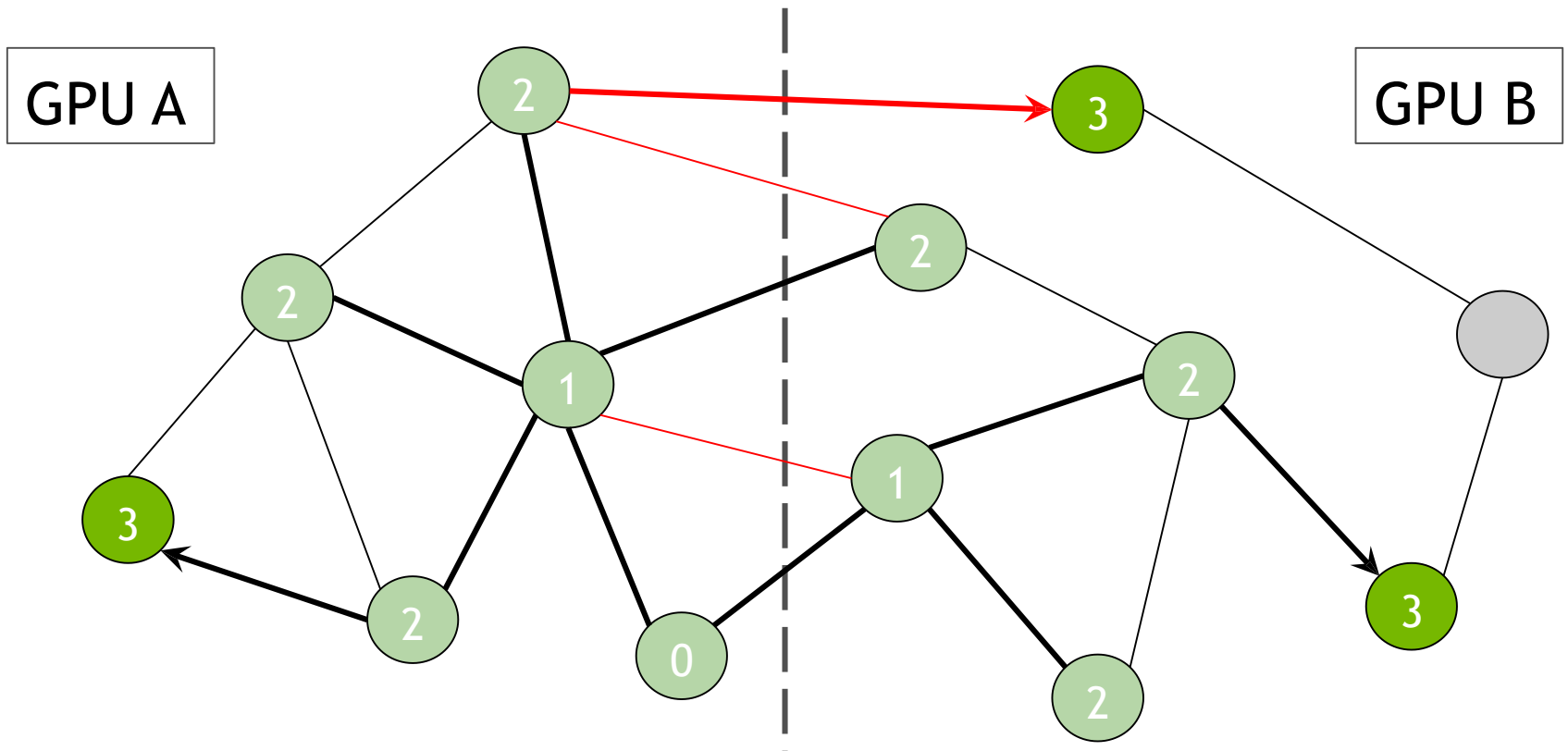
GRAPH ANALYTICS

BFS Traversal



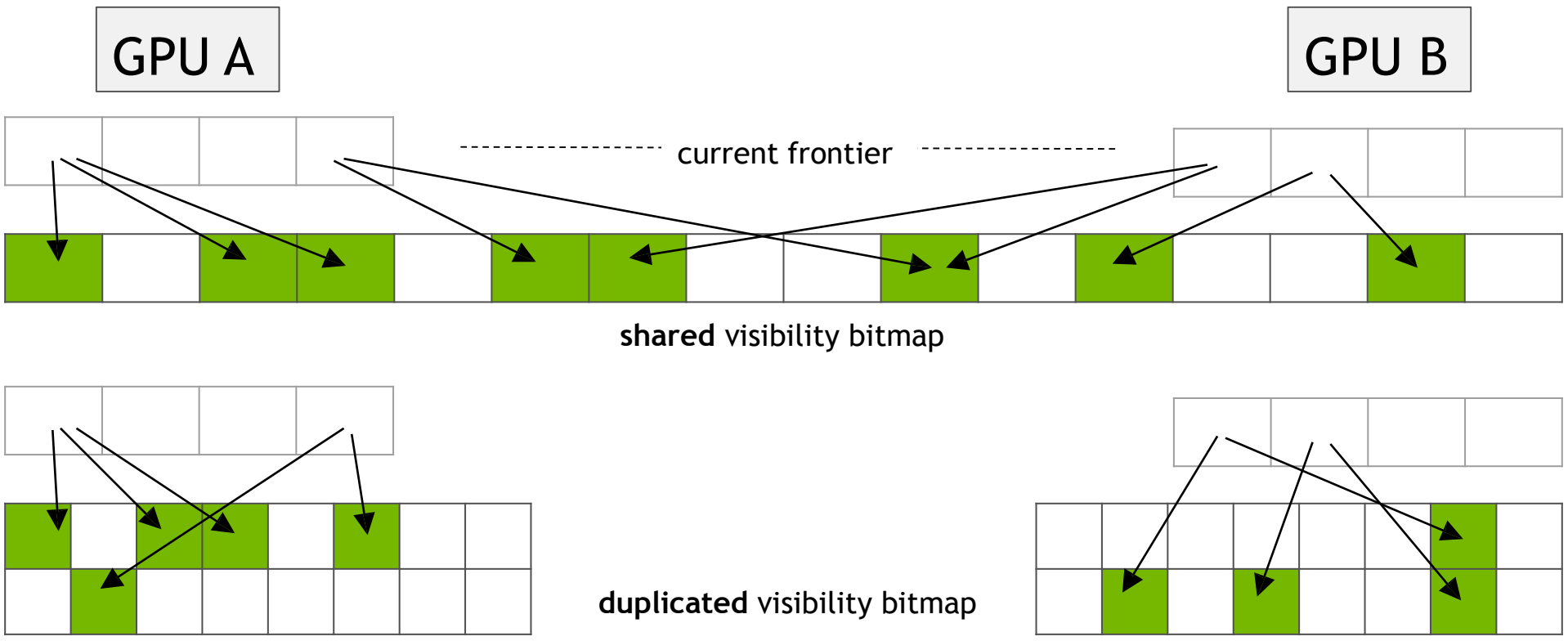
GRAPH ANALYTICS

BFS Traversal



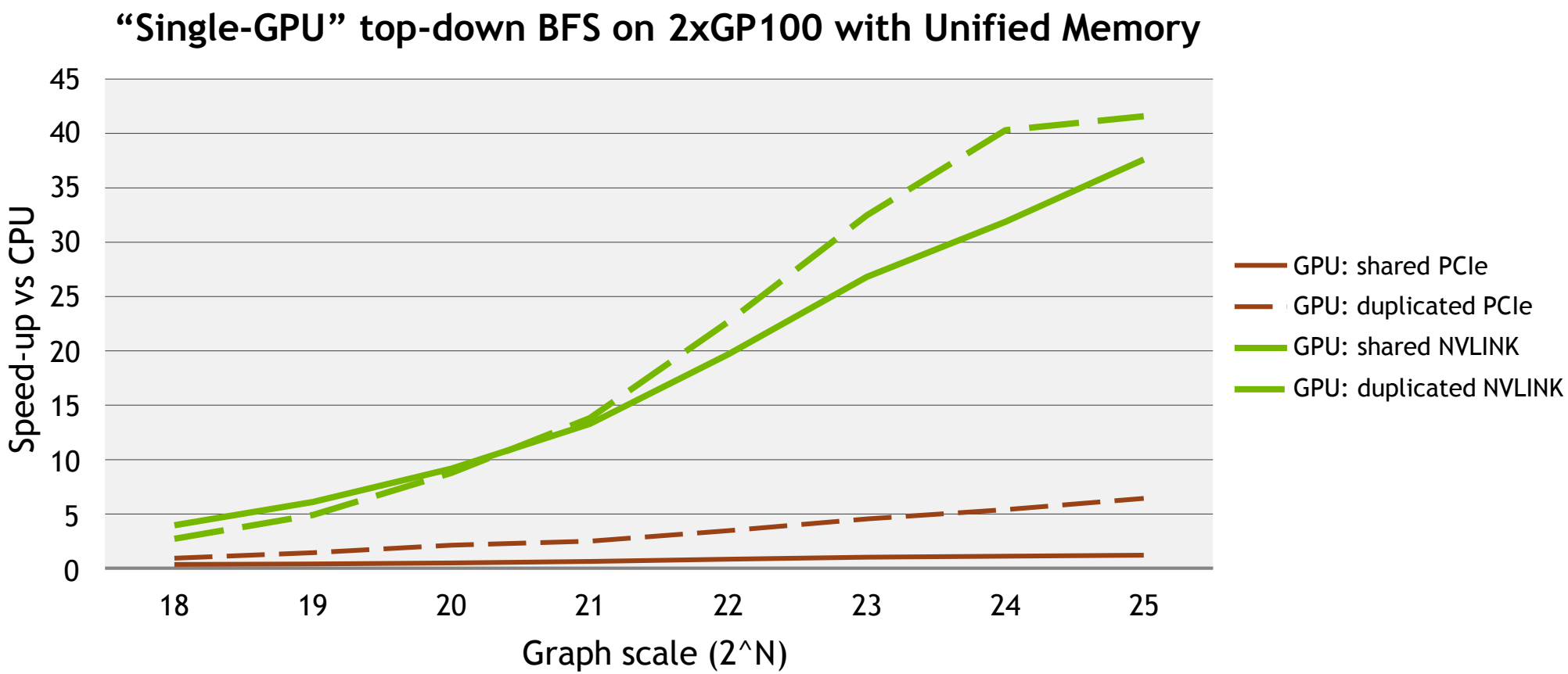
GRAPH ANALYTICS

Shared vs Duplicated Visibility Vector



GRAPH ANALYTICS

Software vs Hardware Atomics



CPU: *Intel Core i7-5930K @ 3.50GHz;* ***GPU:*** *NVIDIA Quadro GP100; edgefactor 16, harmonic mean over 64 random sources*